Block practice · Core maths skills for biology data

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Construct or complete a results or frequency table

Read or estimate a value from a graph, chart, table or image

Read two values from a graph, chart or table and find the difference

Express a part as a percentage of a total

Find a mean, median, mode or mean rate of change

Calculate and express a ratio in a required form

Draw a bar chart and justify its use

Plot a line graph from a table of results

Draw a line or curve of best fit through plotted points

Calculate a percentage change or percentage difference

Calculate an amount from a percentage, fraction or probability

Describe a trend or compare data sets and conclude

Judge whether data support a claim and spot their limits

Scale per-unit or reference quantities and combine them

Substitute into a given equation, rearranging and converting units

Practice workbook · Construct or complete a results or frequency table

Raw results, lists or tallies organised into a ruled table with headings and units, or a missing entry filled in.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-CONSTRUCT-TABLE

1. Question 4 · part a.i · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.i only.

4 A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

1.00 au = −7 %	1.50 au = −10 %	0.00 au (distilled water) = +8 %
	0.25 au = +4 %	0.50 au = −1 %

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

(4)

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P 7 8 7 6 9 A 0 8 2 0

1. Question 4 · part a.i · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.i only.

(ii) Determine the median value for sugar concentration.

(1)

median value = ...

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

(3)

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mass of cylinder = ... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

(4)

concentration of sugar solution = ... au

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1. Question 4 · part a.i · 4 marks · Working and answer

Working and answer

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2. Question 1 · part a.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

1 The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 cm^3 of water, and mix into a smooth paste
- put 10 cm^3 of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

room temperature = 5 heated to 70 °C = 0

kept at 3 °C = 1 acid added = 2 alkali added = 3

no hydrogen peroxide added = 0

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2. Question 1 · part a.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

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(a) (i) Give the student's results in the form of a suitable table.

(3)

(ii) State why this investigation is not reliable.

(1)

(iii) Give a reason why the results of this investigation are not accurate.

(1)

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

(3)

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2. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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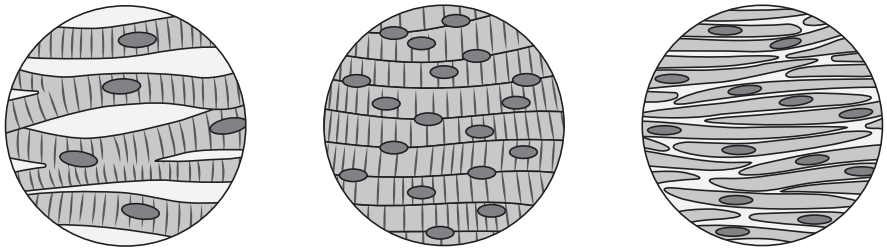
3. Question 6 · part b.ii · 3 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

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6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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3. Question 6 · part b.ii · 3 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

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(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

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3. Question 6 · part b.ii · 3 marks · Working and answer

Working and answer

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4. Question 4 · part a.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.i only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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P 7 4 4 9 5 A 0 1 2 2 8

4. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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5. Question 2 · part a · 3 marks

January 2020 Paper 2

Further practice · Attempt part a only.

2 A student records her pulse rate in beats per minute (bpm) after different activities.

These are the student's results.

- at rest 78 bpm
- walking 95 bpm
- jogging 138 bpm
- running 146 bpm

(a) Draw a table to show the student's results.

(3)

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5. Question 2 · part a · 3 marks

January 2020 Paper 2

Further practice · Attempt part a only.

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(b) Describe a method that the student could use to obtain these results.

(4)

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(c) State one factor that the student should control during the investigation.

(1)

(d) Name a body process, other than pulse rate, that increases during exercise.

(1)

(Total for Question 2 = 9 marks)

5. Question 2 · part a · 3 marks · Working and answer

Working and answer

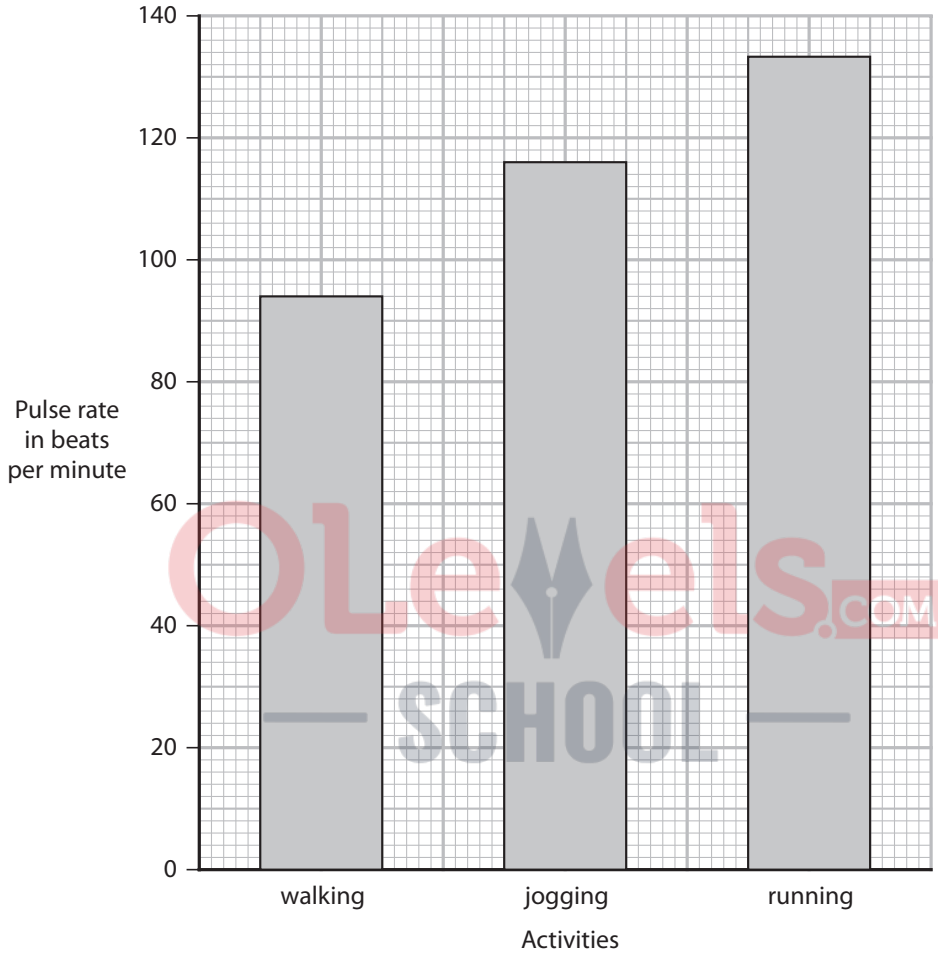
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6. Question 4 · part a.i · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a.i only.

4 (a) The bar chart shows the pulse rate of a student at the end of three different activities.



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6. Question 4 · part a.i · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a.i only.

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(i) Draw a table that the student could use to record these results.

(3)

(ii) Complete the table by adding the student's results.

(2)

(iii) Describe a method that the student could use to obtain the results shown in the bar chart.

(4)

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6. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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Practice workbook · Read or estimate a value from a graph, chart, table or image

Locating a required value, year or feature on a graph or chart, adding readings to a table, or estimating a proportion or size by comparing lengths in an image.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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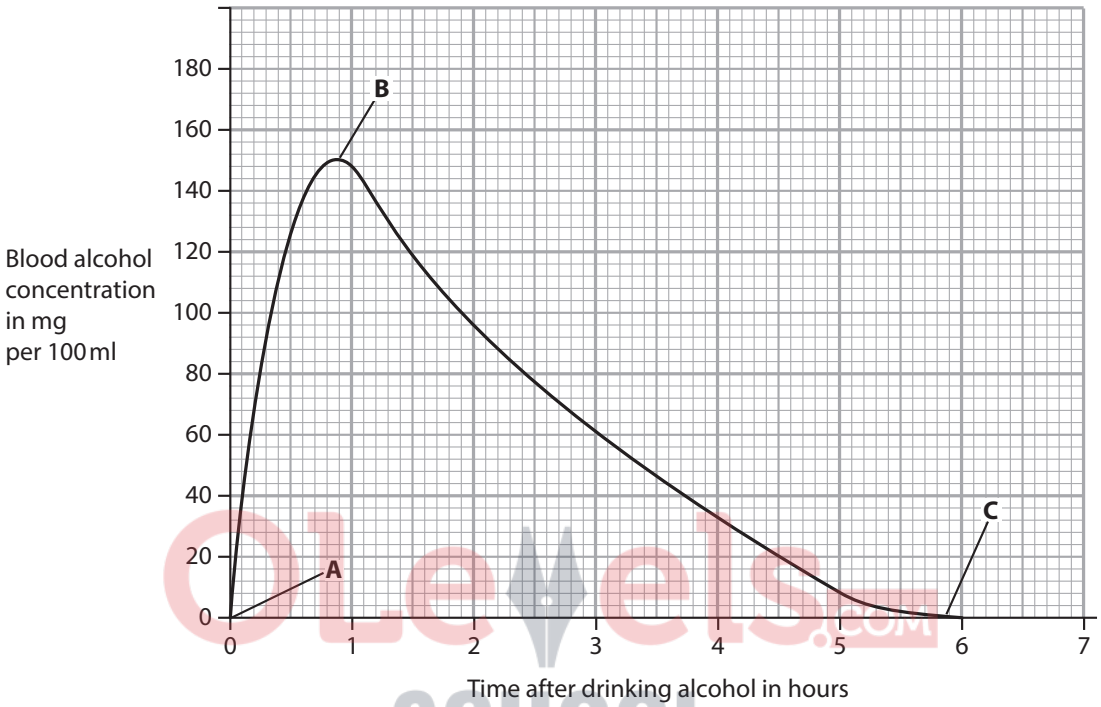
Type ID: HBI-MATH-READ-A-VALUE

1. Question 6 · part b.i · 1 mark

June 2025 Paper 1

Worked-example retry · Attempt part b.i only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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1. Question 6 · part b.i · 1 mark

June 2025 Paper 1

Worked-example retry · Attempt part b.i only.

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(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

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1. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

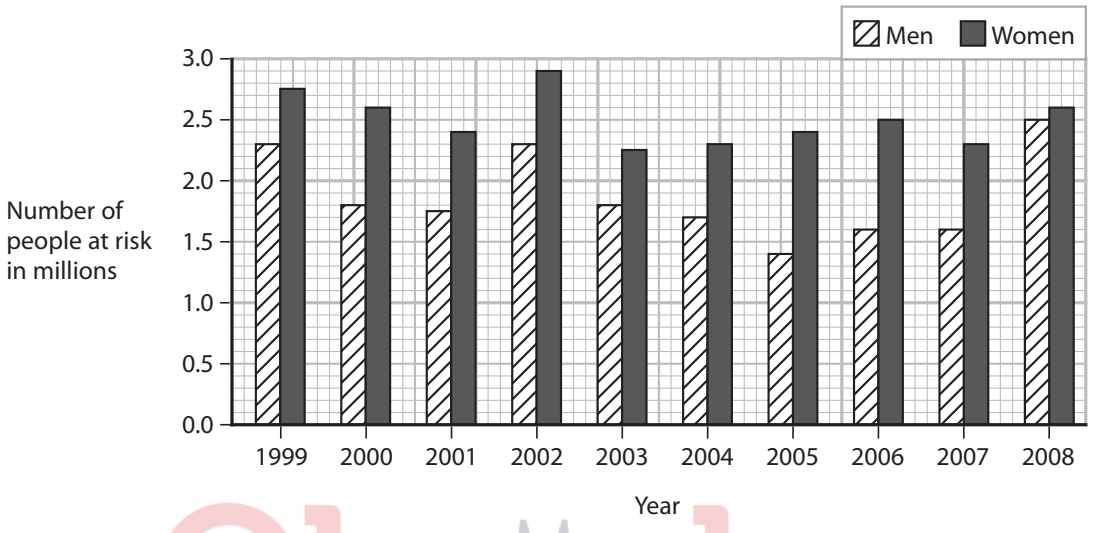
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2. Question 2 · part a.i · 1 mark

June 2022 Paper 2

Further practice · Attempt part a.i only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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2. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

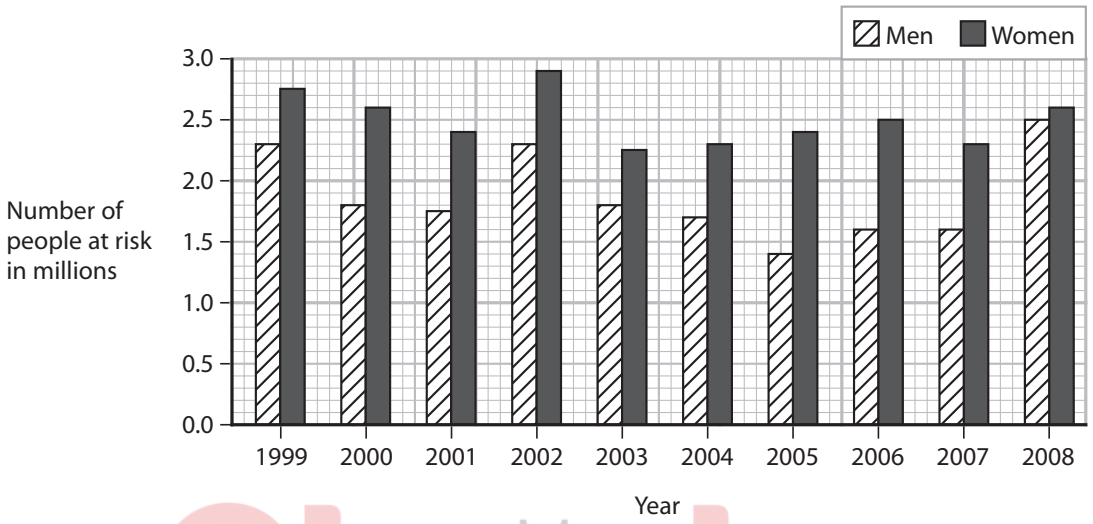
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3. Question 2 · part a.ii · 1 mark

June 2022 Paper 2

Further practice · Attempt part a.ii only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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3. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

8 (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

(1)

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

(3)

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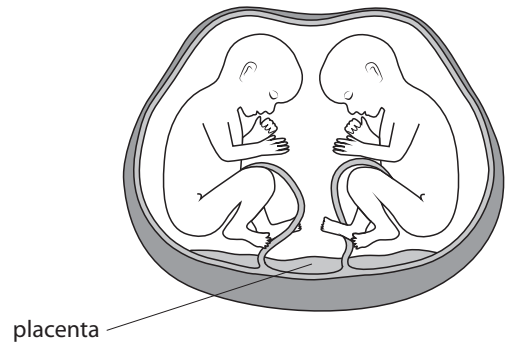
4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

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(b) Some identical twins share a placenta, as shown in the diagram.



(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

(3)

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(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

(2)

4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

(c) When babies are young, their growth is monitored using a percentile chart.

The diagram shows a percentile chart for baby girls.



(i) Determine the growth percentile of a six-week-old baby girl with a mass of 5 kg.

(1)

percentile = ...

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4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

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(ii) Describe how percentile charts are used to identify concerns about the growth of a baby.

(3)

(Total for Question 8 = 13 marks)

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Working and answer

5. Question 4 · part c.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part c.i only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

.....

.....

.....

.....

(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

.....

.....

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5. Question 4 · part c.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part c.i only.

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(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

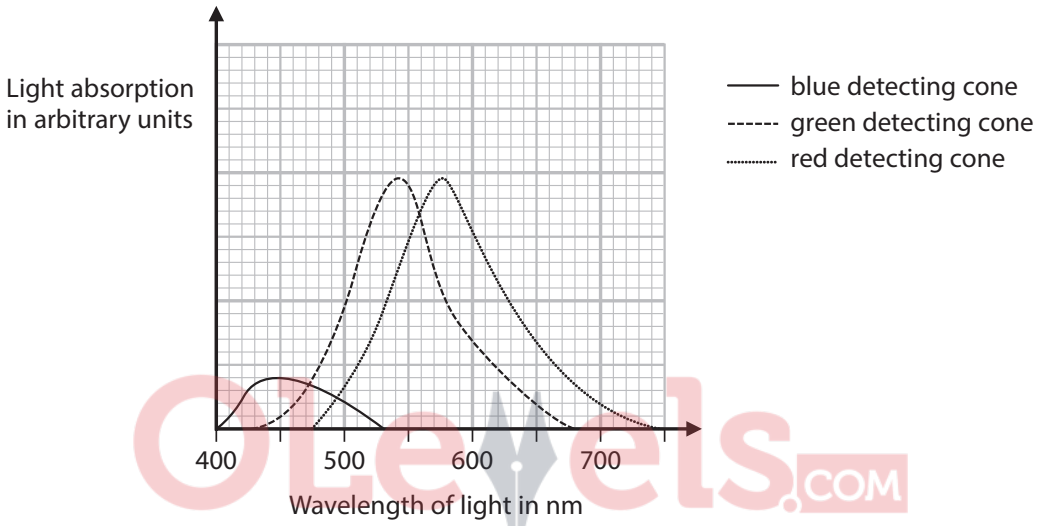
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5. Question 4 · part c.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part c.i only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.



(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

(1)

wavelength of light = ... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

(2)

difference in wavelengths = ... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

(2)

.....

.....

.....

.....

(Total for Question 4 = 14 marks)

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5. Question 4 · part c.i · 1 mark · Working and answer

Working and answer

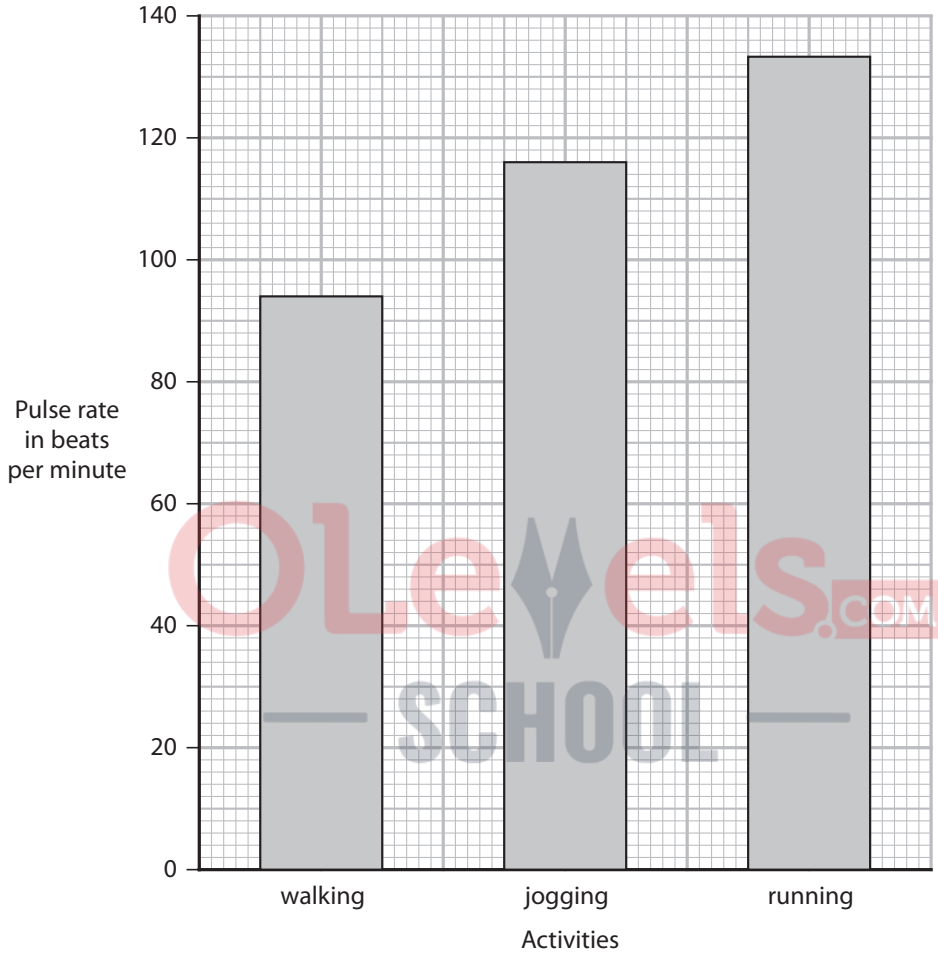
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6. Question 4 · part a.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

4 (a) The bar chart shows the pulse rate of a student at the end of three different activities.



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6. Question 4 · part a.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

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(i) Draw a table that the student could use to record these results.

(3)

(ii) Complete the table by adding the student's results.

(2)

(iii) Describe a method that the student could use to obtain the results shown in the bar chart.

(4)

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6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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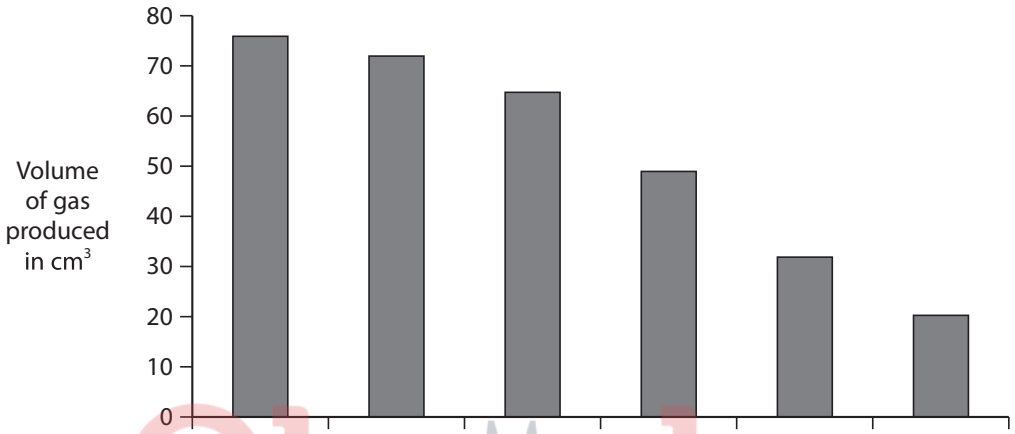
7. Question 3 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

.....

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.....

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.....

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7. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

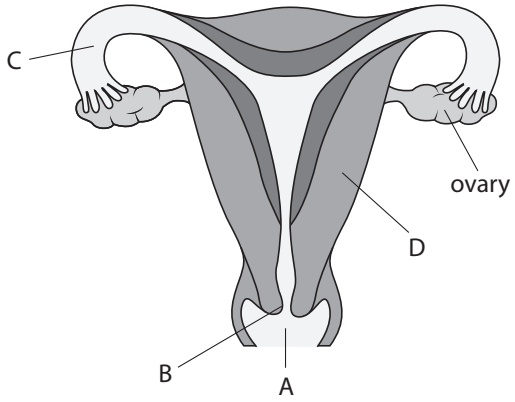
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8. Question 2 · part d.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part d.i only.

2 (a) The diagram shows some structures of the female reproductive system.



(i) Where are sperm deposited during sexual intercourse?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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8. Question 2 · part d.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part d.i only.

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(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

FSH	wall	fertilisation	oestrogen	ovulation
lining	ADH	follicles	menstruation	

Use words from the box to complete the passage about the ovaries.

(3)

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

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(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

(1)

.....

.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

(1)

.....

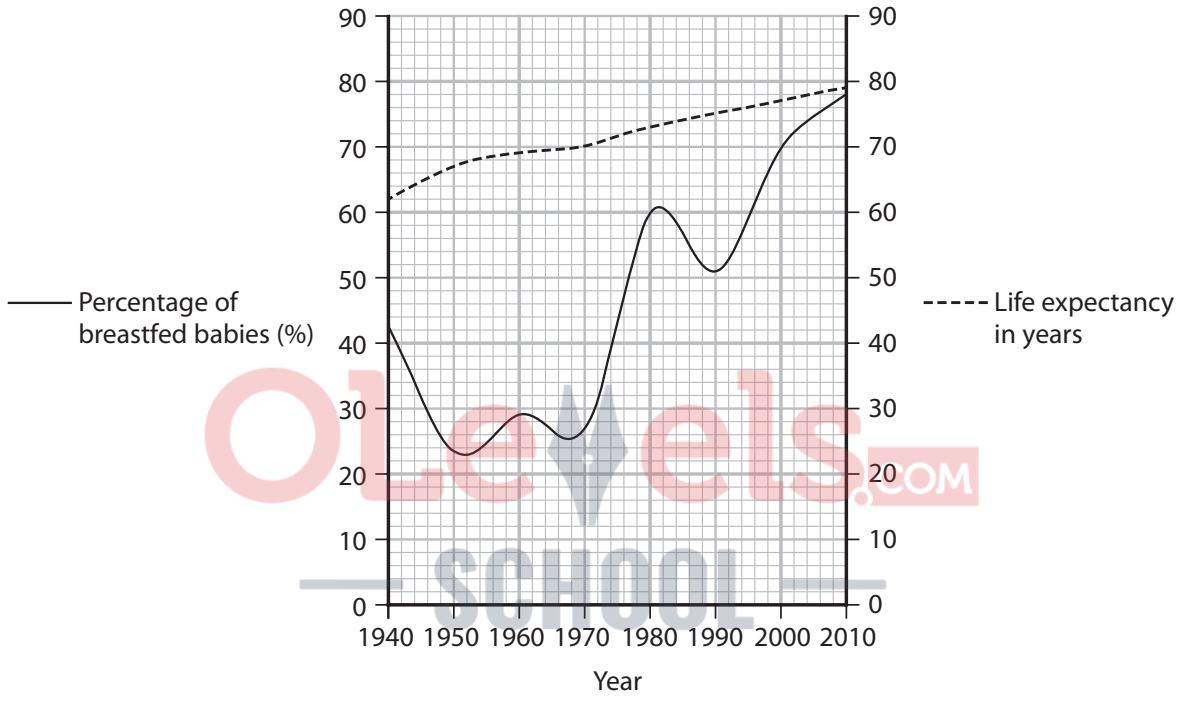
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8. Question 2 · part d.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part d.i only.

(d) The graph shows the percentage of babies that are breastfed and the life expectancy of breastfed babies from 1940 to 2010 in the UK.



(i) Compare the percentage of babies that were breastfed in 1940 with the percentage of babies that were breastfed in 2010.

(1)

(ii) The graph suggests that breastfeeding has no effect on life expectancy.

Describe how the graph shows this.

(2)

(Total for Question 2 = 10 marks)

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8. Question 2 · part d.i · 1 mark · Working and answer

Working and answer

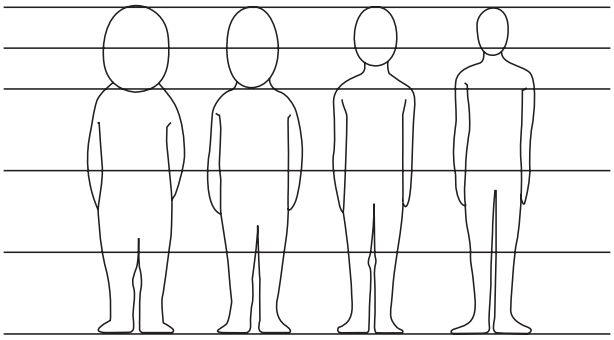
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9. Question 4 · part a.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a.i only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

proportion = ...

(b) Describe two main changes in the proportions of the body between birth and 18 years.

(2)

1 .....

.....

2 .....

.....

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9. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

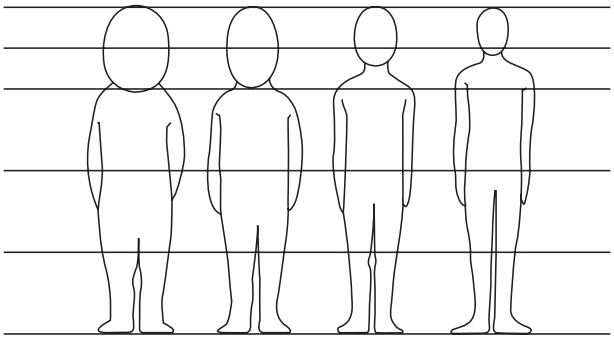
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10. Question 4 · part a.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

proportion = ...

(b) Describe two main changes in the proportions of the body between birth and 18 years.

(2)

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2 .....

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10. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

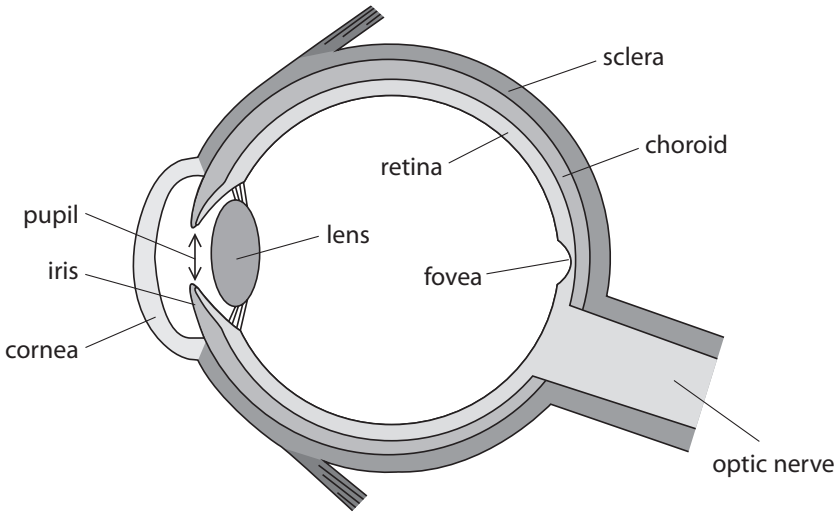
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11. Question 3 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye? **(1)**

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye. **(3)**

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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11. Question 3 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

(b) The photomicrograph gives the thickness of the two outer layers of the eye.



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Estimate the thickness of the retina.

(1)

thickness of the retina = ... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

(2)

.....

.....

.....

.....

(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

(2)

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(Total for Question 3 = 9 marks)

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11. Question 3 · part b · 1 mark · Working and answer

Working and answer

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Practice workbook · Read two values from a graph, chart or table and find the difference

Two points, bars, peaks and troughs, or ends of a range are located and the gap between them is the answer.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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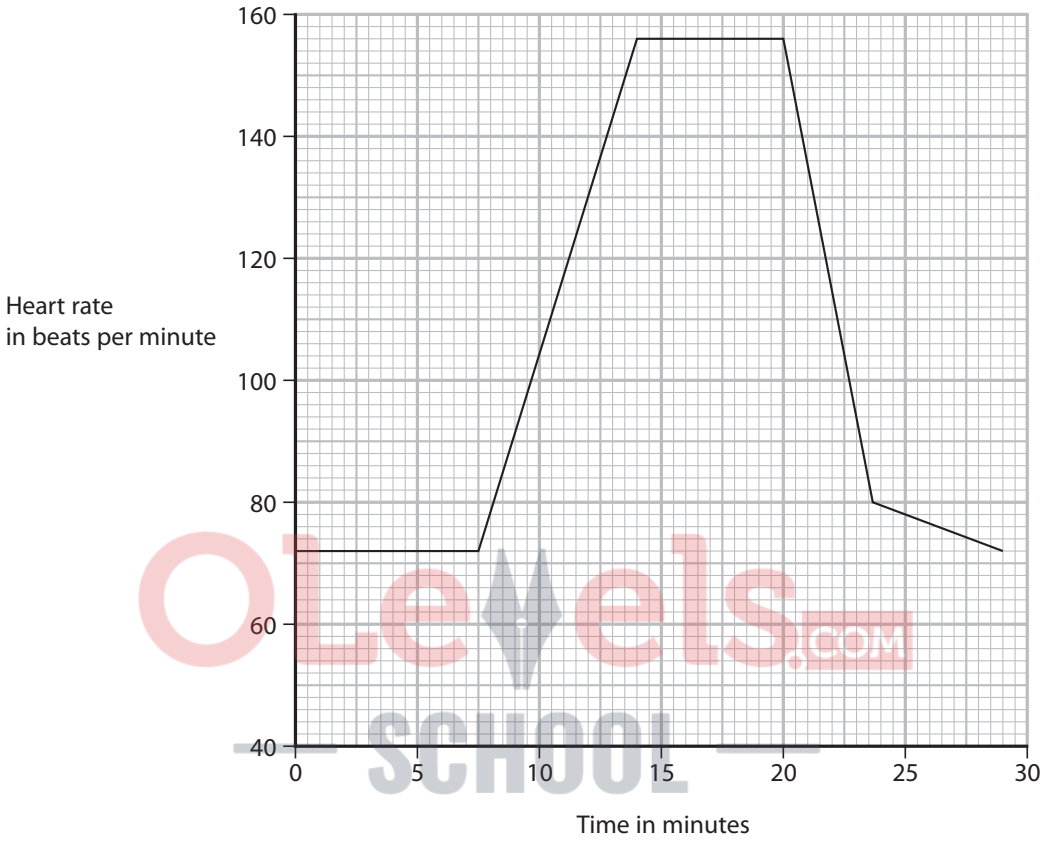
Type ID: HBI-MATH-READ-AND-SUBTRACT

1. Question 2 · part a.i · 2 marks

June 2022 Paper 1

Worked-example retry · Attempt part a.i only.

2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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1. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

5 A student investigates the effect of temperature on the volume of sweat produced.

This is the student's method.

- sit a person in a room at 20 °C
- measure the volume of sweat the person produces
- repeat with the room at different temperatures
- repeat again when the person is active

The table shows the student's results.

Temperature of room in °C	Volume of sweat produced in arbitrary units	
	Sitting	Active
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

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2. Question 5 · part b · 2 marks

June 2019 Paper 1R

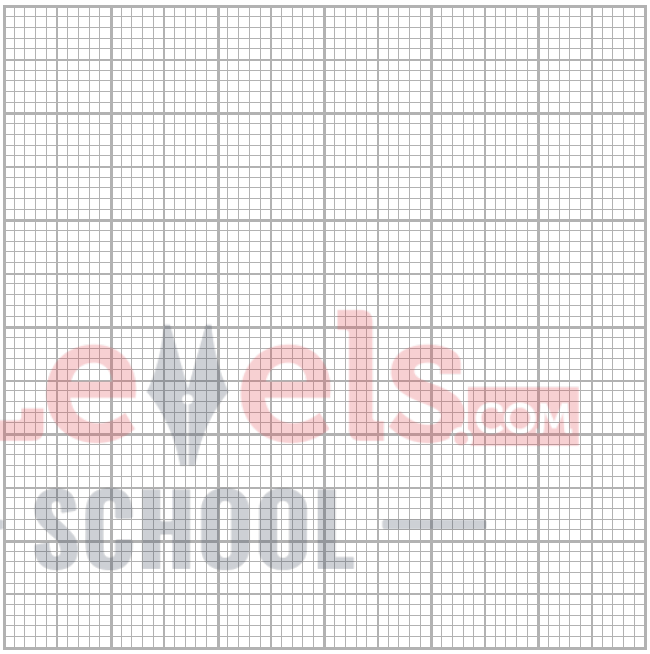
Further practice · Attempt part b only.

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(a) Plot the student's results on the grid.

Join the points with straight lines.

(5)



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(b) Use your graph to determine the difference in the volume of sweat produced at 27 °C when sitting and when active.

(2)

difference = ... arbitrary units

2. Question 5 · part b · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

(c) Explain the difference between the volume of sweat produced when sitting and when active.

(3)

(Total for Question 5 = 10 marks)

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Working and answer

3. Question 4 · part b.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

4 (a) State three methods of birth control.

(3)

1 .....

.....

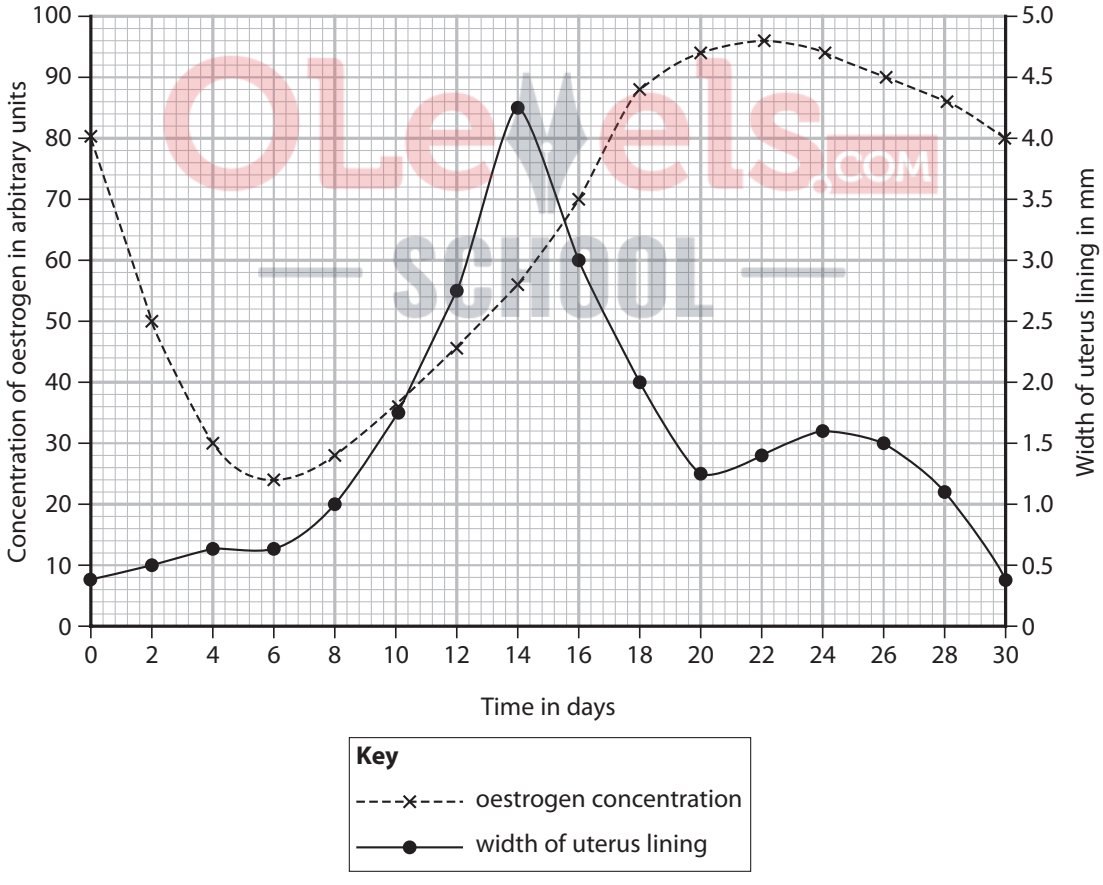
2 .....

.....

3 .....

.....

(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.



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3. Question 4 · part b.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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(i) Determine the change in width of the uterus lining between day 6 and day 14.

(2)

change in width = ... mm

(ii) Explain the importance of this change in width.

(3)

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(iii) Determine during which days menstruation occurred.

(1)

(iv) Ovulation takes place on day 15.

Describe how the two lines on the graph would be different if the ovum had been fertilised.

(3)

(Total for Question 4 = 12 marks)

3. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 6 · part a.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

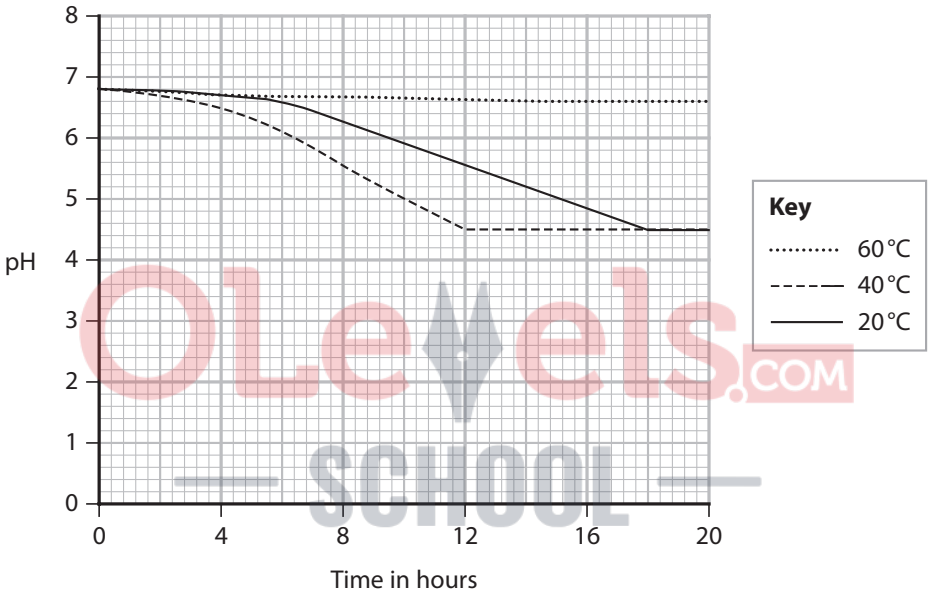
6 Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.



(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

(2)

time = ... hours

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4. Question 6 · part a.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

(ii) Give two variables that need to be controlled in this investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

(2)

.....

.....

.....

.....

.....

(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

(2)

boiling

.....

.....

cooling

.....

.....

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4. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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5. Question 6 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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5. Question 6 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

1 .....

.....

.....

2 .....

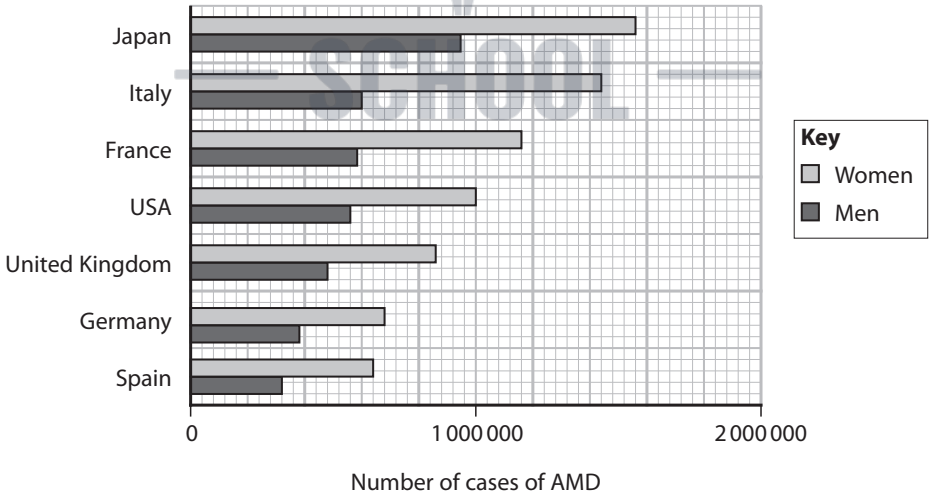
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.



(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

(2)

difference = ...

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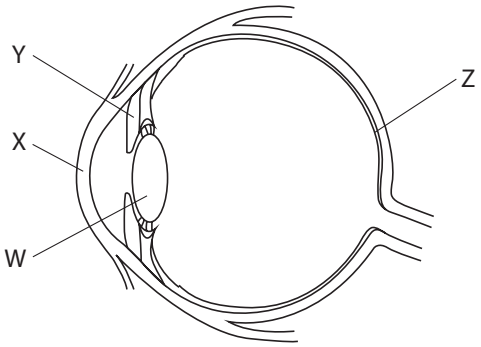
5. Question 6 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) The diagram shows a human eye.

AMD affects the retina of the eye.



Which of these structures is affected by AMD?

(1)

- ☒ **A** structure W
- ☒ **B** structure X
- ☒ **C** structure Y
- ☒ **D** structure Z

(c) A deficiency of vitamin A can also cause blindness.

A genetically engineered rice product can be used to prevent deficiency of vitamin A.

Describe how genetic engineering is used to produce this rice product.

(3)

(Total for Question 6 = 11 marks)

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5. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

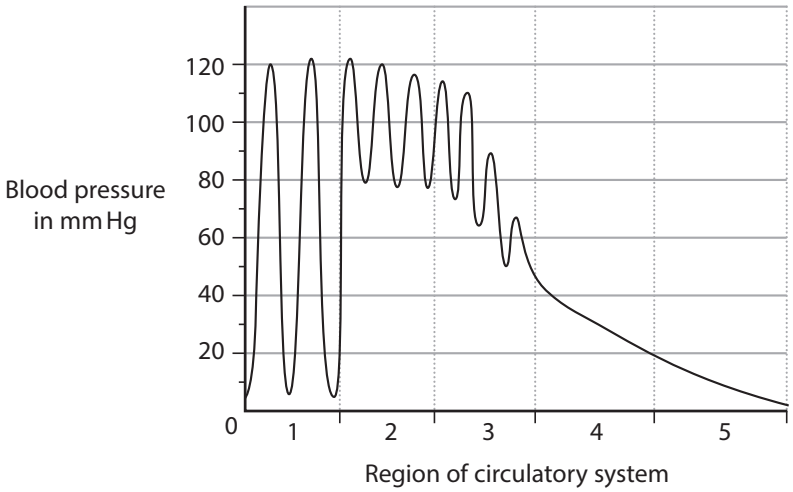
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6. Question 3 · part a.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part a.i only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1. **(2)**

difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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6. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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7. Question 4 · part c.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.ii only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

.....

.....

.....

.....

(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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7. Question 4 · part c.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.ii only.

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(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

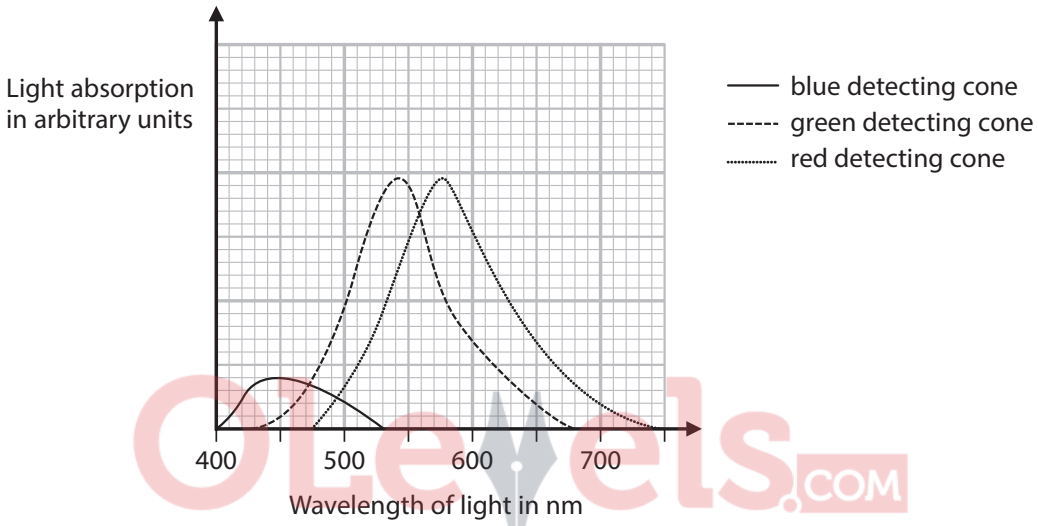
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7. Question 4 · part c.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.ii only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.



(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

(1)

wavelength of light = ... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

(2)

difference in wavelengths = ... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

(2)

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(Total for Question 4 = 14 marks)

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7. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

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8. Question 4 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

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4 (a) (i) Complete the word equation for one type of respiration.

(2)

oxygen + ... → ..... + ...

(ii) State the type of respiration shown by the equation.

(1)

.....

(iii) In which part of the cell does this type of respiration occur?

(1)

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

(1)

.....

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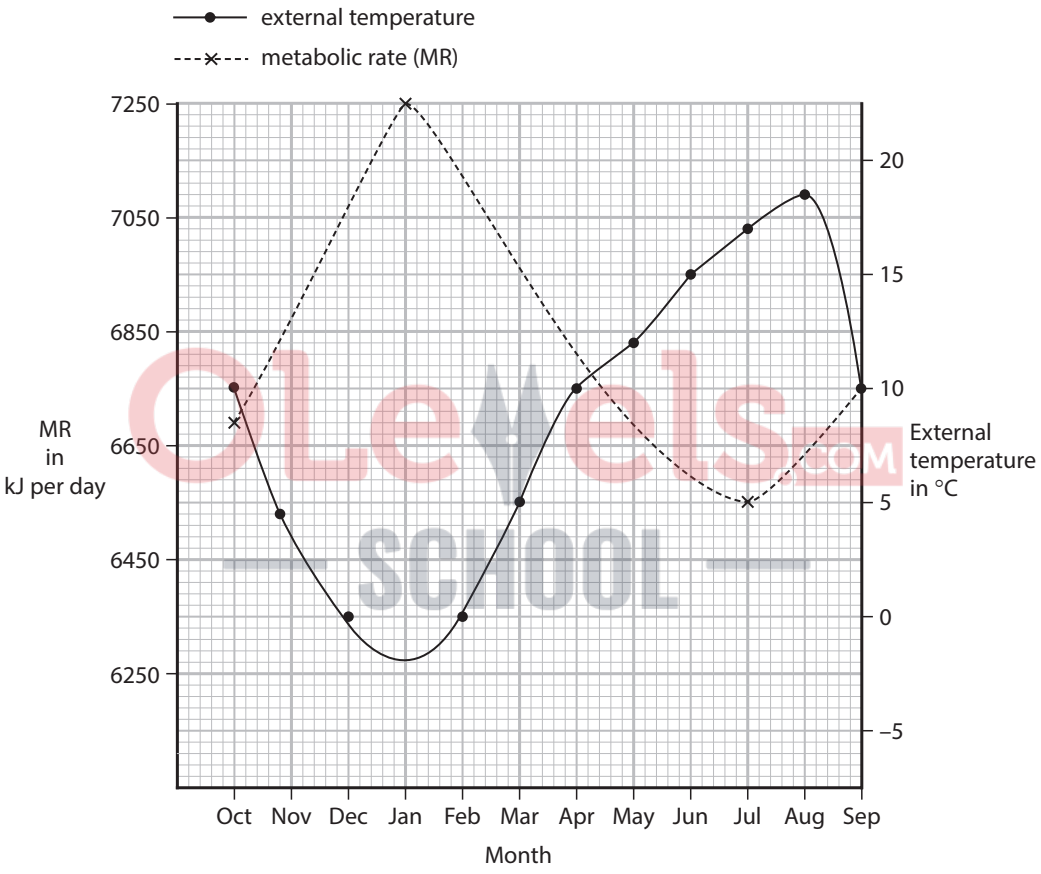
8. Question 4 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

(b) An investigation was carried out to see how a person's metabolic rate (MR) and the external temperature change at different times in one year.

The graph shows the results of the investigation for one person.



(i) Describe how the investigation could be made more reliable.

(2)

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8. Question 4 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Determine the difference between the maximum metabolic rate and minimum metabolic rate.

(2)

difference = ... kJ per day

(iii) Explain why the metabolic rate changes throughout the year.

(5)

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8. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 3 · part b.i · 1 mark

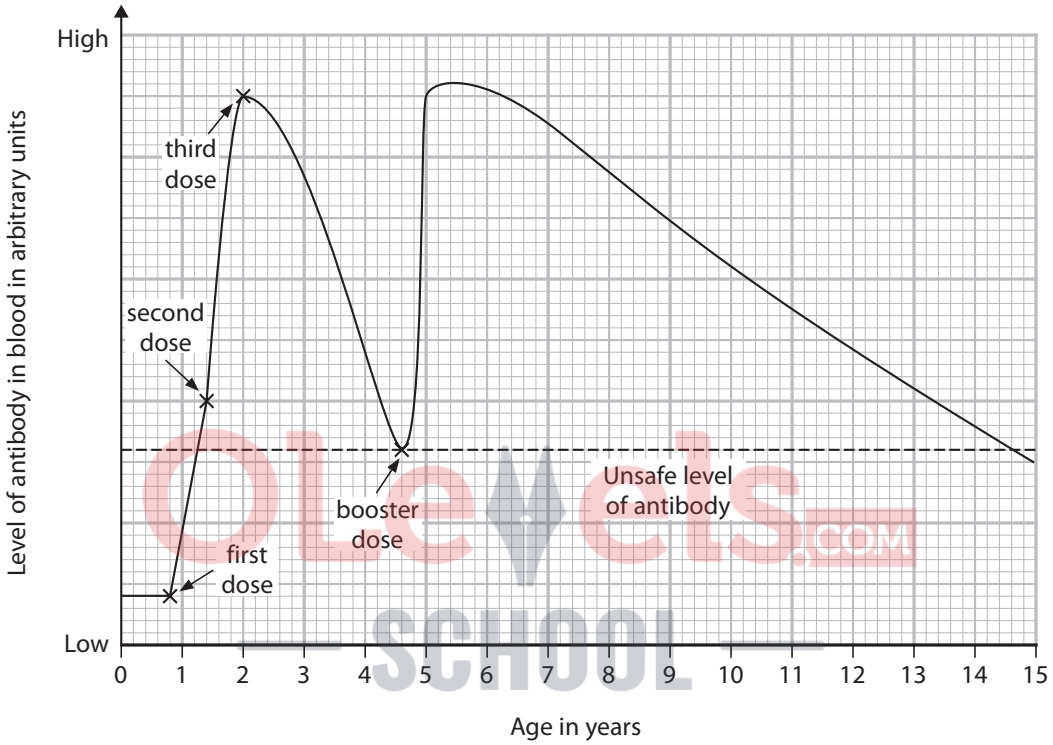
November 2023 Paper 2

Further practice · Attempt part b.i only.

3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

.....

.....

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9. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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10. Question 7 · part b.i · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

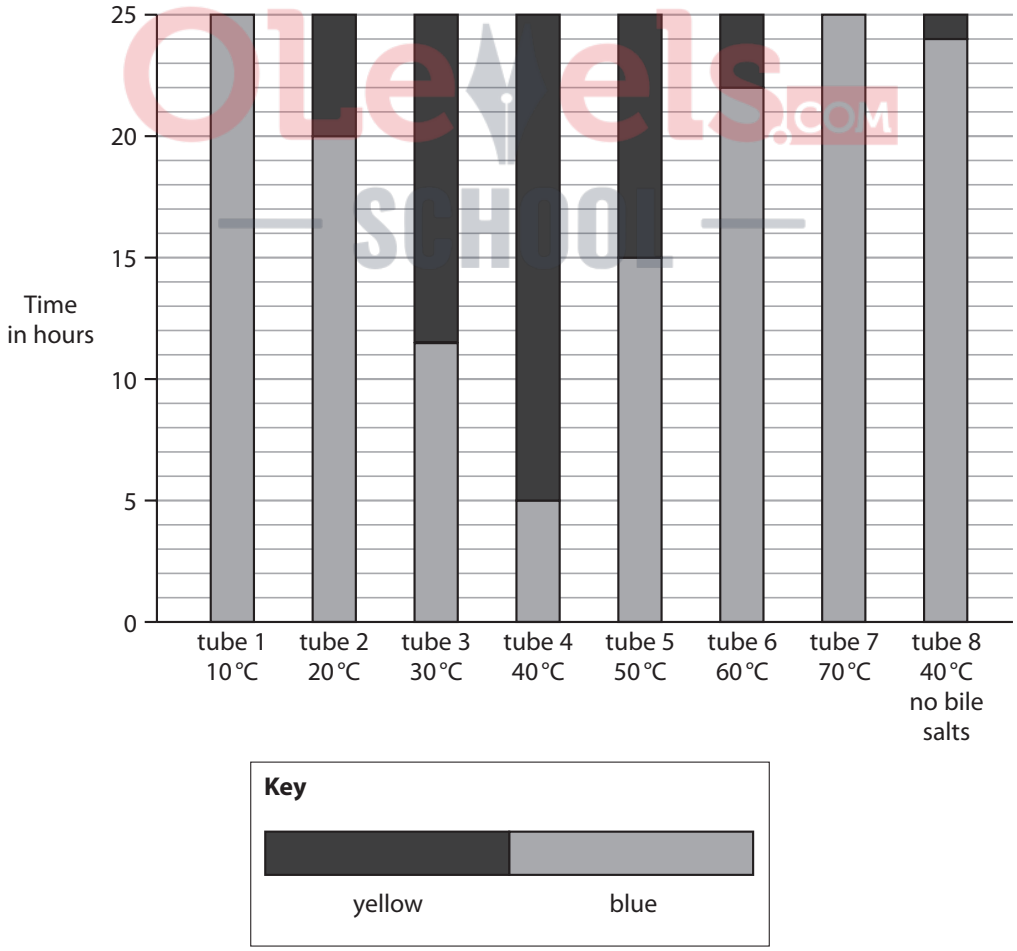
7 A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.



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10. Question 7 · part b.i · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

(3)

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

(1)

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

(3)

.....

.....

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

(2)

time = ... hours

(ii) Explain why the pH decreases during the 25 hours.

(3)

.....

.....

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.....

10. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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11. Question 1 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

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1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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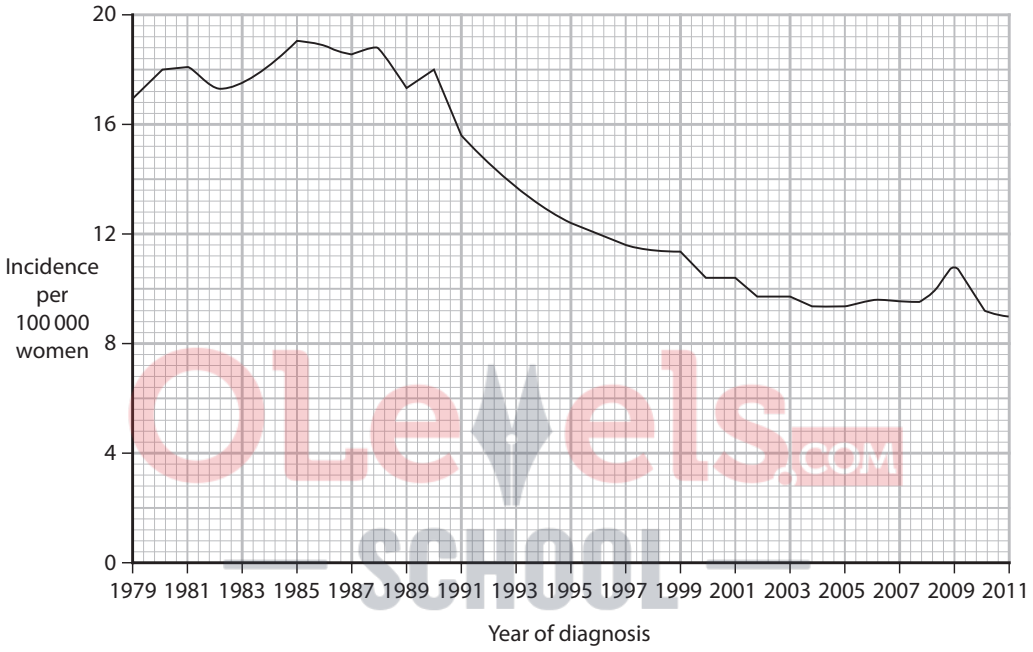
11. Question 1 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.



(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

(2)

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11. Question 1 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

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(ii) Calculate the difference in the incidence per 100 000 of women with cervical cancer between 1979 and 2011.

(2)

difference = ... per 100 000

(d) Cervical cancer that is caused by a virus can now be prevented by vaccination.

The table gives information about vaccination.

Which row gives correct information about vaccination?

(1)

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	Contents of vaccination	Response by the body	How the pathogen is destroyed
☒ **A**	antibodies	antigens produced	antigens destroy pathogen
☒ **B**	antibodies	white blood cells produced	white blood cells destroy pathogen
☒ **C**	antigens	antibodies produced	antibodies destroy pathogen
☒ **D**	antigens	platelets produced	platelets destroy pathogen

(Total for Question 1 = 12 marks)

Working and answer

12. Question 2 · part a.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

2 To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

Time in minutes	Blood glucose levels in mg per 100 cm^3	
	person without diabetes	person with diabetes
0	65	110
30	120	215
60	130	230
90	95	200
120	80	190
150	65	180
180	65	170

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12. Question 2 · part a.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

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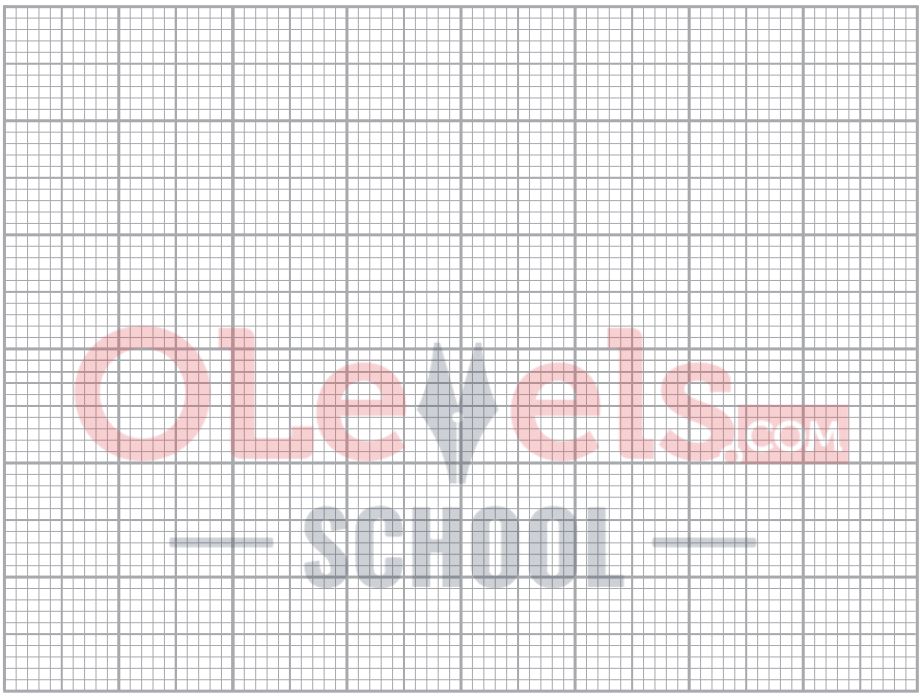
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(a) (i) Plot the results on the grid.

Join the points with straight lines.

(6)



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

(2)

without diabetes = ... mg per 100 cm^3

with diabetes = ... mg per 100 cm^3

12. Question 2 · part a.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

(iii) Give **three** differences between the glucose levels for the two people during the three hours.

(3)

1 .....

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2 .....

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3 .....

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(b) One symptom of diabetes is glucose in the urine.

Describe how a sample of urine could be tested safely for glucose.

(5)

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(Total for Question 2 = 16 marks)

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12. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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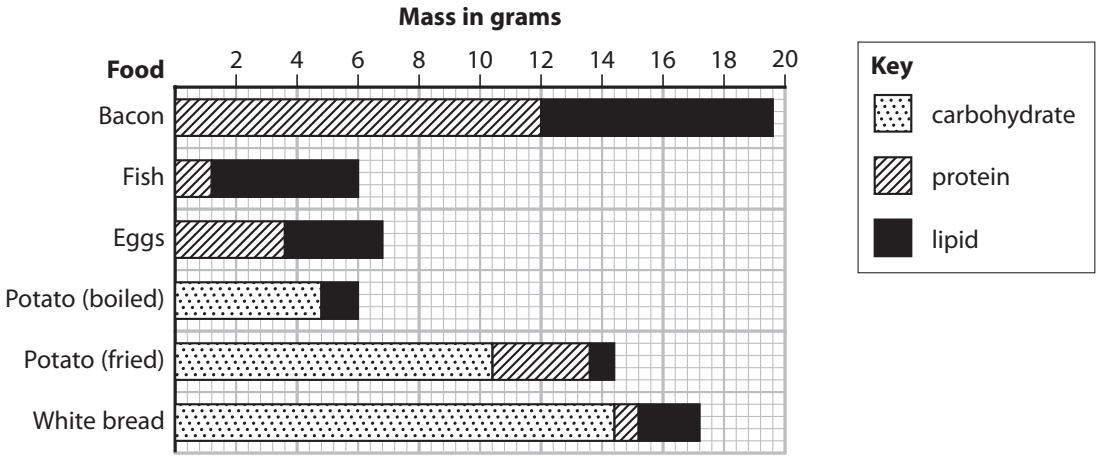
13. Question 7 · part a.iii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

7 Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.



(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

(1)

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

(1)

(iii) Deduce the mass of lipid in 26 g of bacon.

(2)

mass of lipid = ... g

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P 6 7 0 5 9 A 0 2 0 2 4

13. Question 7 · part a.iii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

(iv) The energy in one gram of protein is 17 kJ and the energy in one gram of lipid is 37 kJ.

Calculate the total energy in 26 g of fish.

(3)

total energy = ... kJ

(b) Describe how to test for the presence of lipid in the fish.

(3)

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(Total for Question 7 = 10 marks)

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13. Question 7 · part a.iii · 2 marks · Working and answer

Working and answer

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Practice workbook · Express a part as a percentage of a total

One subgroup or component is expressed as a percentage of its whole, sometimes after totalling a table or to support a comparison between groups.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PERCENTAGE-OF-TOTAL

1. Question 7 · part b.ii · 2 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.ii only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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1. Question 7 · part b.ii · 2 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.ii only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

(2)

percentage = ... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

(2)

1 .....

.....

2 .....

.....

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1. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a.iv · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a.iv only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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2. Question 4 · part a.iv · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a.iv only.

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(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

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.....

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.....

(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

OLevels.COM — SCHOOL —

percentage = ... %

2. Question 4 · part a.iv · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a.iv only.

(b) Kidney failure can be treated by a kidney transplant or by dialysis.

Discuss the advantages and disadvantages of kidney transplants.

(5)

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(Total for Question 4 = 12 marks)

2. Question 4 · part a.iv · 2 marks · Working and answer

Working and answer

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3. Question 3 · part a.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

3 A woman who smokes when she is pregnant risks harming her developing fetus.

The table lists the birth masses of 310 children born to non-smoking mothers or to mothers who smoke.

	Birth masses of children in kg					
	Less than 2.4	**2.4 – 3.1**	**3.2 – 3.5**	**3.6 – 3.9**	**More than 3.9**	**Total number of births in sample**
Number of children born to non-smoking mothers	4	42	70	60	34	210
Number of children born to mothers who smoke	3	24	44	23	6	100

(a) (i) What is the most common birth mass in kg?

(1)

- ☒ **A** 2.4 – 3.1
- ☒ **B** 3.2 – 3.5
- ☒ **C** 3.6 – 3.9
- ☒ **D** More than 3.9

(ii) Calculate the percentage of children with a birth mass of 3.6 kg or more in the sample of non-smoking mothers.

(2)

percentage = ...

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8

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3. Question 3 · part a.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

(b) Use data from the table to deduce what can be concluded about the effect of smoking on birth mass.

(2)

(c) Suggest why smoking affects birth mass.

(5)

(Total for Question 3 = 10 marks)

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3. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part b.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.iii only.

4 (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

(2)

fingers	carbohydrate	bread	ankles	protein	fat

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

Year	Number of cases of kwashiorkor
2010	32
2011	21
2012	23
2013	19

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4. Question 4 · part b.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.iii only.

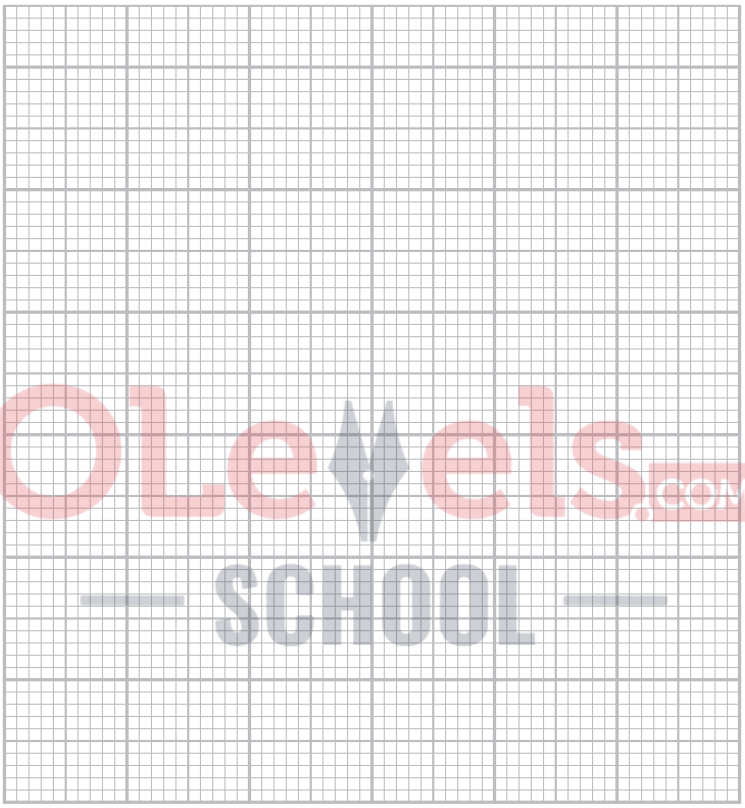
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(i) Plot a bar chart of the data on the grid.

(3)



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(ii) Suggest a reason for the difference in the number of recorded cases of kwashiorkor in 2010 compared with 2013.

(1)

.....

.....

(iii) In a hospital, 95 out of a total of 1978 children with malnutrition were found to have kwashiorkor.

Calculate the percentage of children with kwashiorkor.

(2)

percentage of children = ... %

(Total for Question 4 = 8 marks)

4. Question 4 · part b.iii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part a.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part a.ii only.

4 The table shows the results of a survey about the number of teeth per student in a class of young students.

Number of teeth	Number of students
23	1
24	2
25	3
26	4
27	7
28	13

(a) (i) Draw a bar chart to show the results of the survey.

(4)



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5. Question 4 · part a.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part a.ii only.

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(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

(3)

percentage = ... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

(2)

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5. Question 4 · part a.ii · 3 marks · Working and answer

Working and answer

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6. Question 4 · part a.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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7. Question 8 · part b.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

8 *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

(3)

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

Region of the world	Q	R	S	T	U
Total number of deaths from malaria	191 000	800	3800	14 400	1200
Number of deaths caused by *P. vivax*	1000	500	1400	4900	700

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

(3)

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P 5 8 5 6 8 A 0 2 2 2 8

7. Question 8 · part b.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

(1)

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

(3)

(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

(3)

(Total for Question 8 = 13 marks)

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7. Question 8 · part b.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Find a mean, median, mode or mean rate of change

A set of repeat readings or survey values needs an average, or a change over a time interval needs a mean rate per unit time.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-MEAN-MEDIAN-MODE-RATE

1. Question 5 · part a.i · 2 marks

June 2019 Paper 1

Worked-example retry · Attempt part a.i only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

.....

2 .....

.....

(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

.....

2 .....

.....

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1. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 7 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

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(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

2. Question 7 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

Number of bubbles of CO_2	Time taken for chymosin to turn milk protein solid, in seconds				
	Test 1	Test 2	Test 3	Test 4	Mean
100	253	257	250	260	255
150	238	232	241	229	
200	216	214	219	211	215
250	208	202	212	198	205
300	210	200	199	311	203

(i) State three factors that the student should control.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

(2)

mean time taken = ...

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P 6 2 0 5 4 A 0 2 0 2 4

2. Question 7 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) In test 4 there is an anomalous result.

State how the student deals with this result.

(1)

.....

.....

(iv) State why recording the number of bubbles may produce inaccurate results.

(1)

.....

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(Total for Question 7 = 14 marks)

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Working and answer

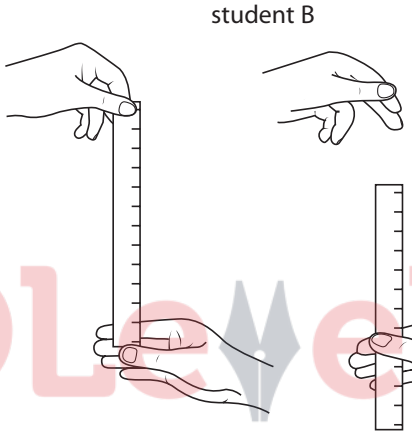
3. Question 5 · part a · 2 marks

November 2021 Paper 2

Further practice · Attempt part a only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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3. Question 5 · part a · 2 marks

November 2021 Paper 2

Further practice · Attempt part a only.

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The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

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(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

3. Question 5 · part a · 2 marks · Working and answer

Working and answer

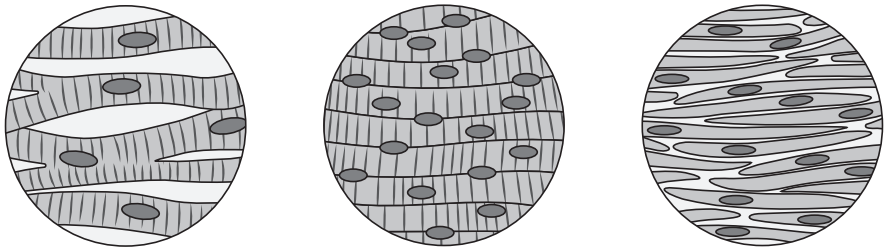
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4. Question 6 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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4. Question 6 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

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4. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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5. Question 7 · part c.ii · 2 marks

June 2022 Paper 1

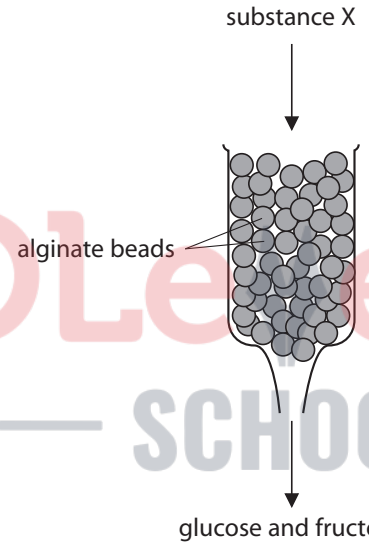
Further practice · Attempt part c.ii only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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20

5. Question 7 · part c.ii · 2 marks

June 2022 Paper 1

Further practice · Attempt part c.ii only.

(b) Which process is involved in the conversion of glucose into glycogen in the human body?

(1)

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

(2)

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number of people = ... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

38	62	51	96	49	78	82	91	65

Determine the median number of people with diabetes.

(2)

median = ...

(Total for Question 7 = 10 marks)

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5. Question 7 · part c.ii · 2 marks · Working and answer

Working and answer

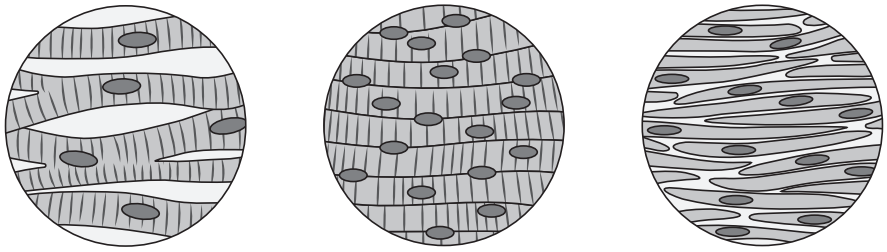
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6. Question 6 · part b.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.iii only.

6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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6. Question 6 · part b.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.iii only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

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(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

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6. Question 6 · part b.iii · 1 mark

November 2024 Paper 1

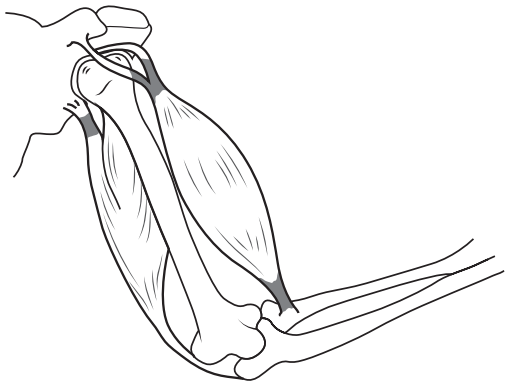
Further practice · Attempt part b.iii only.

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(c) The diagram shows two muscles in the arm.



Explain how these muscles help to raise the forearm.

(4)

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(Total for Question 6 = 17 marks)

6. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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7. Question 4 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

4 A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

1.00 au = −7 %	1.50 au = −10 %	0.00 au (distilled water) = +8 %
	0.25 au = +4 %	0.50 au = −1 %

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

(4)

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P 7 8 7 6 9 A 0 8 2 0

7. Question 4 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Determine the median value for sugar concentration.

(1)

median value = ...

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

(3)

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mass of cylinder = ... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

(4)

concentration of sugar solution = ... au

7. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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8. Question 3 · part a.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a.i only.

3 A woman who smokes when she is pregnant risks harming her developing fetus.

The table lists the birth masses of 310 children born to non-smoking mothers or to mothers who smoke.

	Birth masses of children in kg					
	Less than 2.4	**2.4 – 3.1**	**3.2 – 3.5**	**3.6 – 3.9**	**More than 3.9**	**Total number of births in sample**
Number of children born to non-smoking mothers	4	42	70	60	34	210
Number of children born to mothers who smoke	3	24	44	23	6	100

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(a) (i) What is the most common birth mass in kg? **(1)**

- ☒ **A** 2.4 – 3.1
- ☒ **B** 3.2 – 3.5
- ☒ **C** 3.6 – 3.9
- ☒ **D** More than 3.9

(ii) Calculate the percentage of children with a birth mass of 3.6 kg or more in the sample of non-smoking mothers. **(2)**

percentage = ...

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8. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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9. Question 7 · part b.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

(3)

1 .....

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2 .....

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3 .....

.....

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9. Question 7 · part b.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

(ii) State the dependent variable in the student's investigation.

(1)

.....

(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

(2)

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9. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

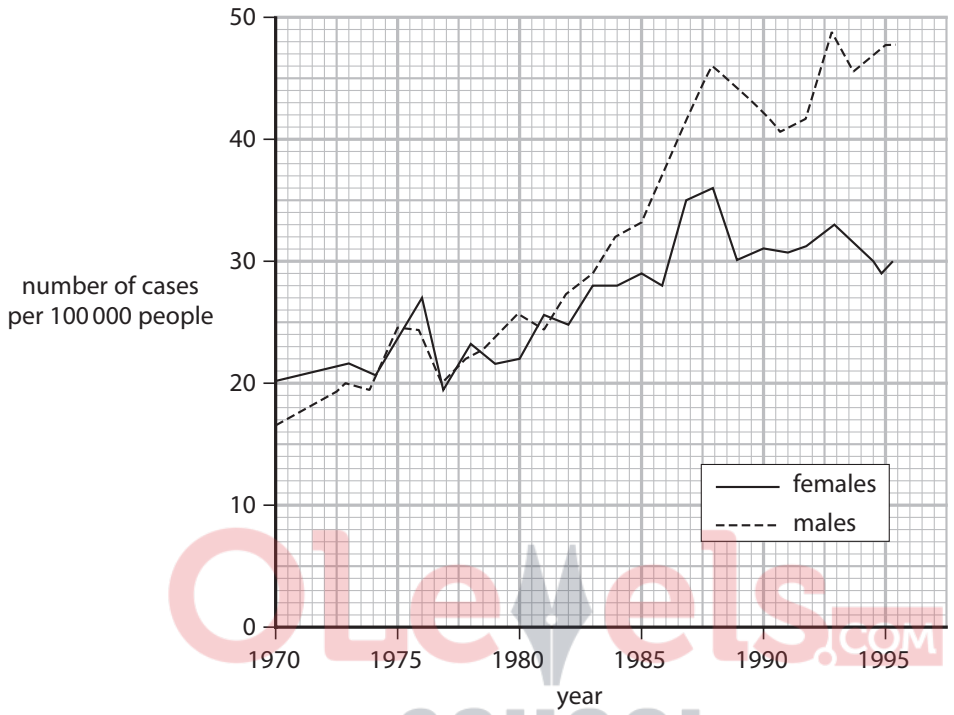
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10. Question 6 · part b.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.i only.

6 A form of skin cancer called melanoma has been increasing in both males and females.



(a) (i) Describe the trends shown by the graph.

(3)

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10. Question 6 · part b.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.i only.

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(ii) Suggest why cancer can occur in the skin.

(2)

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

(3)

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mean increase per 100 000 males per year = ...

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(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

(3)

number of females with melanoma = ...

(Total for Question 6 = 11 marks)

10. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

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11. Question 7 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease?

(1)

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics.

(2)

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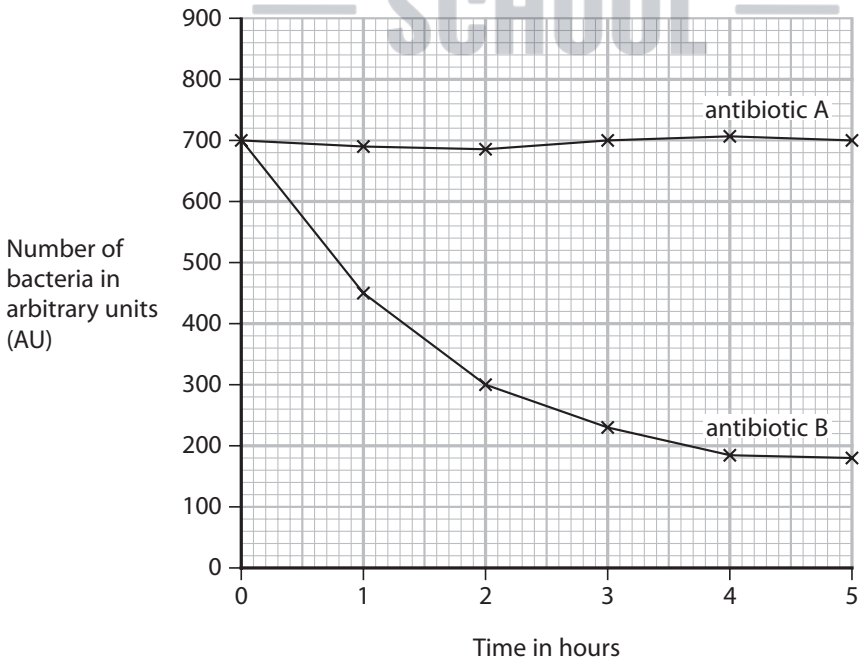
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



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11. Question 7 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

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(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

(3)

mean rate of change = ... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

(2)

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11. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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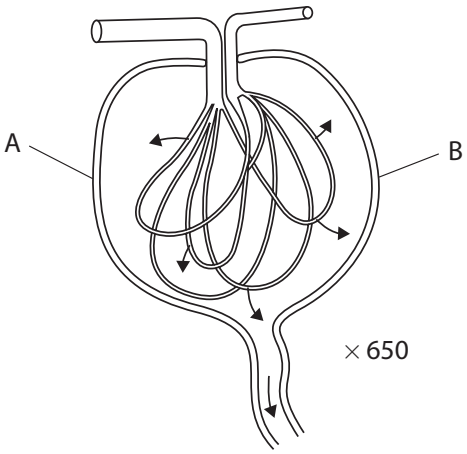
12. Question 1 · part b.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

12. Question 1 · part b.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
0.5	70	50
1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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12. Question 1 · part b.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

(i) State two factors that should be controlled in the investigation.

(2)

1 .....

.....

2 .....

.....

(ii) The mean rate of urine production on day 1 is 276 cm^3 per hour.

Calculate the mean rate of urine production on day 2 during the 4 hour time period.

(2)

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mean rate of urine production on day 2 = ... cm^3 per hour

(iii) Explain the difference between the mean rate on day 1 and the mean rate on day 2.

(3)

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(Total for Question 1 = 16 marks)

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12. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Calculate and express a ratio in a required form

Two values, counted, read from a graph or table, are compared as a ratio in simplest form, 1 : n, n : 1, or as how many times greater.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-RATIO

1. Question 5 · part a.i · 2 marks

November 2024 Paper 1

Worked-example retry · Attempt part a.i only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

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.....

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.....

(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

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1. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 7 · part a.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

.....

2 .....

.....

(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

.....

2 .....

.....

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2. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	MRSA	*S. aureus*	MRSA
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

1 .....

.....

.....

2 .....

.....

.....

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3. Question 6 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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3. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

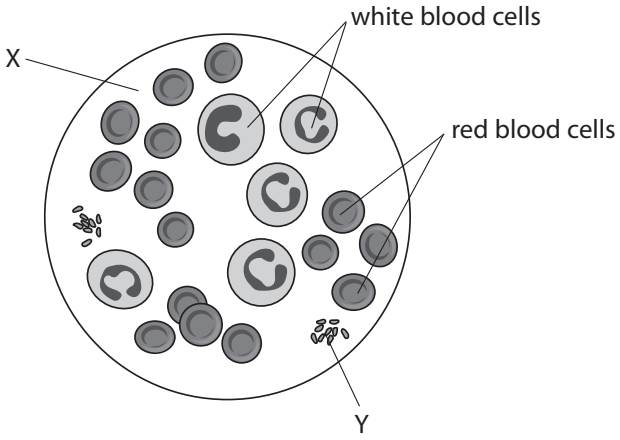
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4. Question 1 · part c.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.i only.

1 The diagram shows a sample of human blood.



(a) State the function of red blood cells.

(1)

.....

.....

(b) Give the name of X and Y.

(2)

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

(2)

ratio = ...

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4. Question 1 · part c.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.i only.

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(ii) Suggest why this ratio will change if a person has a bacterial infection.

(2)

(Total for Question 1 = 7 marks)

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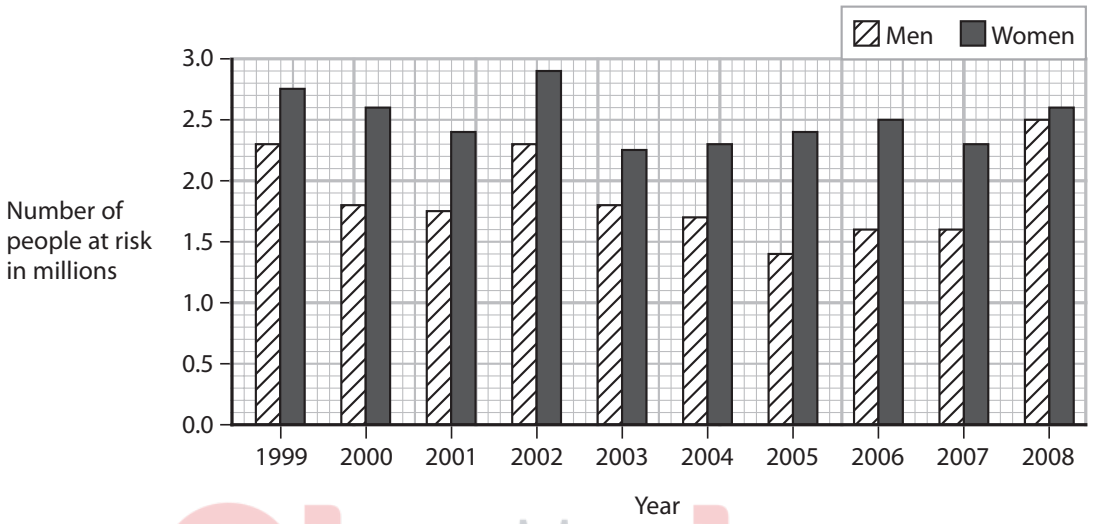
Working and answer

5. Question 2 · part a.iii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.iii only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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5. Question 2 · part a.iii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.iii only.

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(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

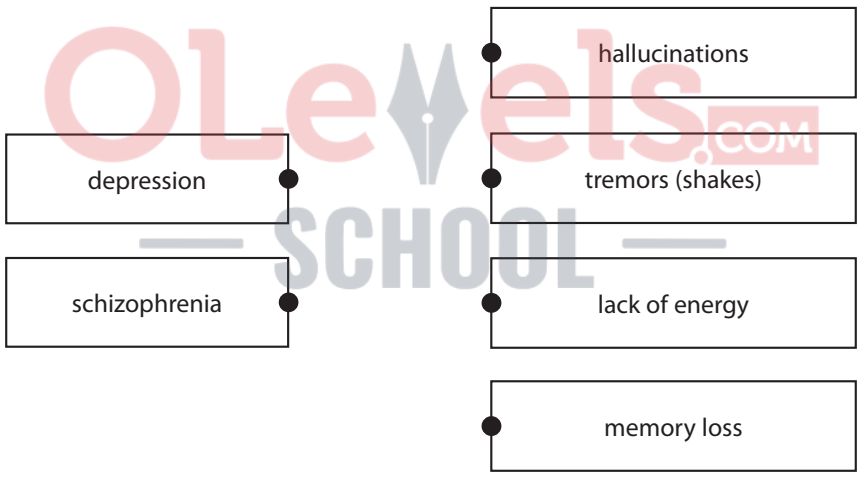
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(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



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(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

5. Question 2 · part a.iii · 2 marks · Working and answer

Working and answer

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6. Question 4 · part c · 2 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

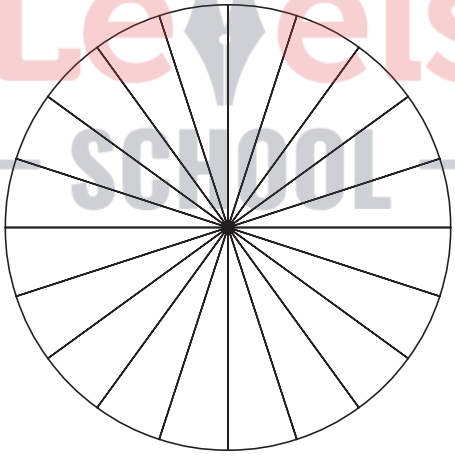
4 The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

Component	Percentage (%) composition of diet	
	Area A	Area B
cereals	75.0	15.0
fruit and vegetables	15.0	37.5
eggs, fish and meat	2.5	27.5
milk, cheese and butter	7.5	20.0

(a) Complete the pie chart to show the diet of the girl in Area B.

(4)



(b) Explain what is meant by a balanced diet.

(2)

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6. Question 4 · part c · 2 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

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(c) Calculate the ratio of the percentage composition of cereals in the diet of the girl in Area A compared with the diet of the girl in Area B.

Give your answer in the form n : 1

(2)

ratio = ...

(d) Discuss the effects that the percentage composition given for eggs, fish and meat in the diet may have on the growth of the two girls.

(5)

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(Total for Question 4 = 13 marks)

6. Question 4 · part c · 2 marks · Working and answer

Working and answer

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7. Question 3 · part a.i · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.i only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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7. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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8. Question 4 · part c.i · 1 mark

November 2023 Paper 2

Further practice · Attempt part c.i only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

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8. Question 4 · part c.i · 1 mark

November 2023 Paper 2

Further practice · Attempt part c.i only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

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(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

8. Question 4 · part c.i · 1 mark

November 2023 Paper 2

Further practice · Attempt part c.i only.

(ii) Discuss the results of the study using information from the table.

(4)

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(Total for Question 4 = 14 marks)

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Working and answer

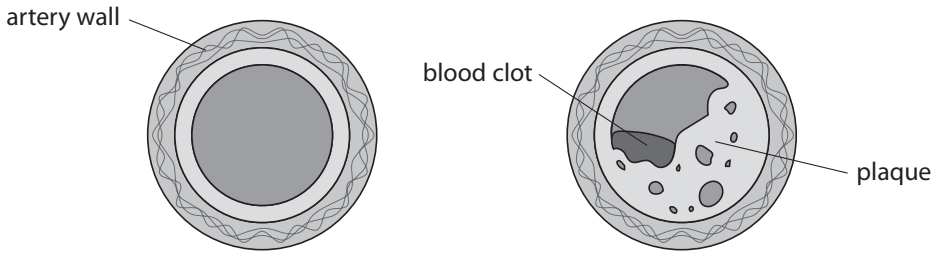
9. Question 6 · part b.ii · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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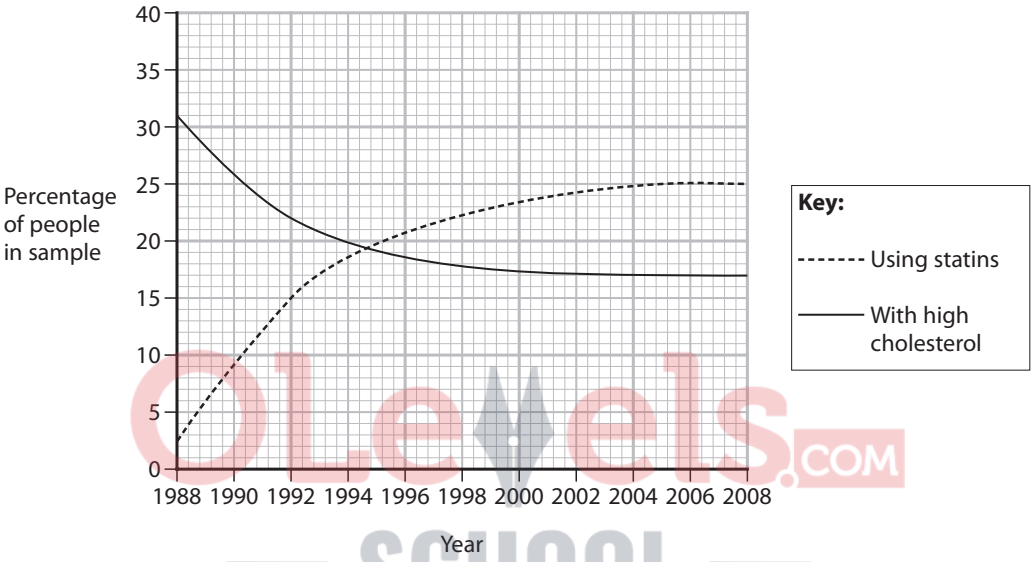
9. Question 6 · part b.ii · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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P 7 4 4 9 5 A 0 1 8 2 8

9. Question 6 · part b.ii · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

(iii) Describe a method that scientists could use to obtain the data shown in the graph.

(5)

(iv) Give two factors the scientists need to consider to make sure that their investigation is valid.

(2)

1 .....

.....

2 .....

.....

(Total for Question 6 = 15 marks)

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9. Question 6 · part b.ii · 3 marks · Working and answer

Working and answer

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10. Question 4 · part c.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.iii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

.....

.....

malaria

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10. Question 4 · part c.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.iii only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

10. Question 4 · part c.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.iii only.

(iii) Calculate the ratio of the number of people with non-infectious diseases in a higher income country to the number of people with non-infectious diseases in a lower income country (lines 20 and 21).

Give your answer in the form n : 1

(3)

ratio = ... : .....

(iv) Suggest why there are more cases of dementia and cancers in higher income countries (lines 21 and 22).

(2)

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(Total for Question 4 = 15 marks)

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10. Question 4 · part c.iii · 3 marks · Working and answer

Working and answer

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11. Question 4 · part a.ii · 2 marks

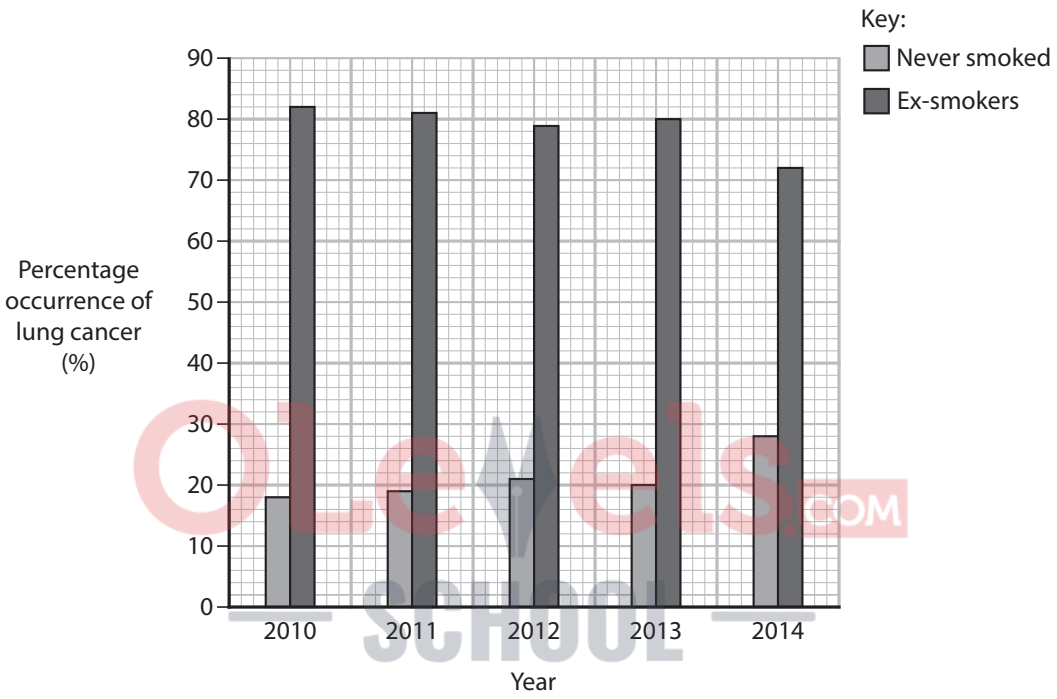
June 2025 Paper 2R

Further practice · Attempt part a.ii only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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11. Question 4 · part a.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

.....

.....

.....

.....

(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

.....

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11. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Draw a bar chart and justify its use

Data for separate categories or grouped classes are shown as bars; may include adding an axis scale or saying why bars suit the data.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-BAR-CHART

1. Question 5 · part b.i · 5 marks

January 2022 Paper 1

Worked-example retry · Attempt part b.i only.

5 (a) Describe the possible effects of a DNA mutation.

(3)

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

Age of mother in years	15–19	20–24	25–29	30–34	35–39	40–44
Probability of chromosome aberration	0.0010	0.0010	0.0010	0.0015	0.0045	0.0140

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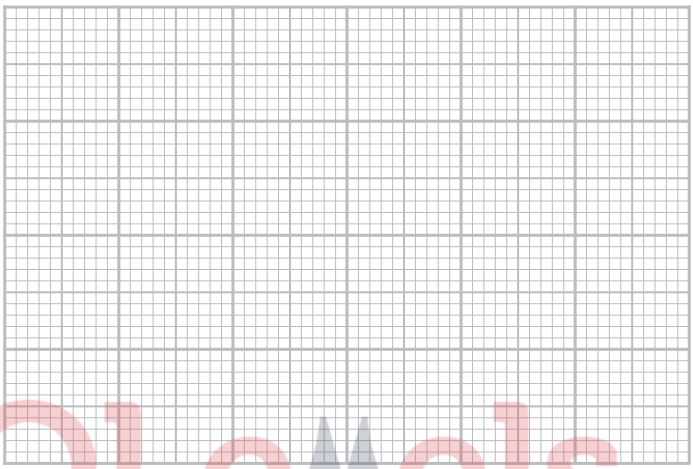
1. Question 5 · part b.i · 5 marks

January 2022 Paper 1

Worked-example retry · Attempt part b.i only.

(i) Plot a bar chart of the results.

(5)



(ii) In one year, 43 000 children are born to mothers aged 30–34.

Calculate the probable number of these children who will have a chromosome aberration.

Give your answer to two significant figures.

(3)

number of children = ...

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1. Question 5 · part b.i · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 4 · part b.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b.i only.

4 (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

(2)

fingers	carbohydrate	bread	ankles	protein	fat

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

Year	Number of cases of kwashiorkor
2010	32
2011	21
2012	23
2013	19

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2. Question 4 · part b.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b.i only.

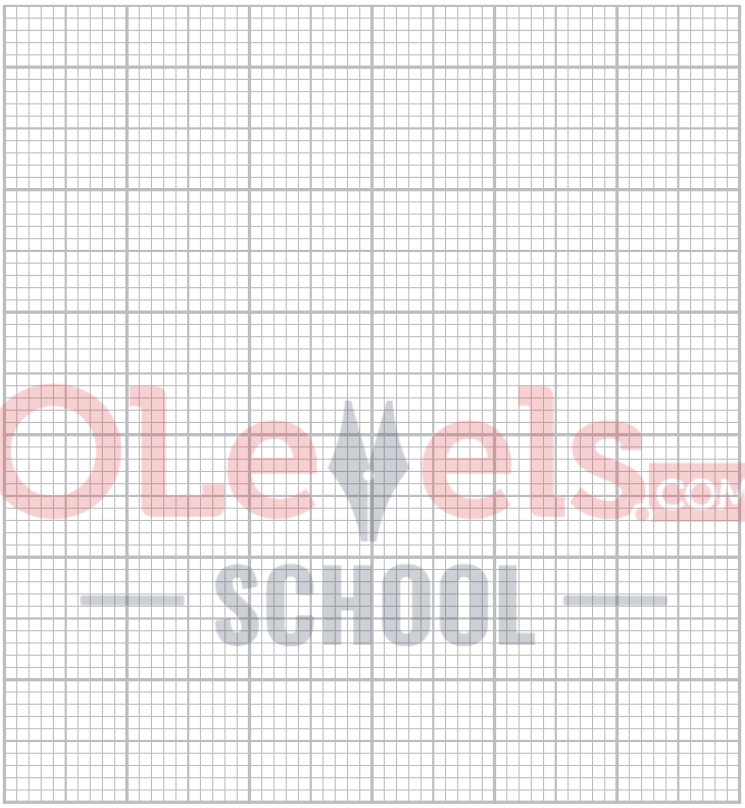
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(i) Plot a bar chart of the data on the grid.

(3)



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(ii) Suggest a reason for the difference in the number of recorded cases of kwashiorkor in 2010 compared with 2013.

(1)

.....

.....

(iii) In a hospital, 95 out of a total of 1978 children with malnutrition were found to have kwashiorkor.

Calculate the percentage of children with kwashiorkor.

(2)

percentage of children = ... %

(Total for Question 4 = 8 marks)

2. Question 4 · part b.i · 3 marks · Working and answer

Working and answer

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3. Question 3 · part a.i · 5 marks

November 2020 Paper 1

Further practice · Attempt part a.i only.

3 There are many factors that affect the height of a human.

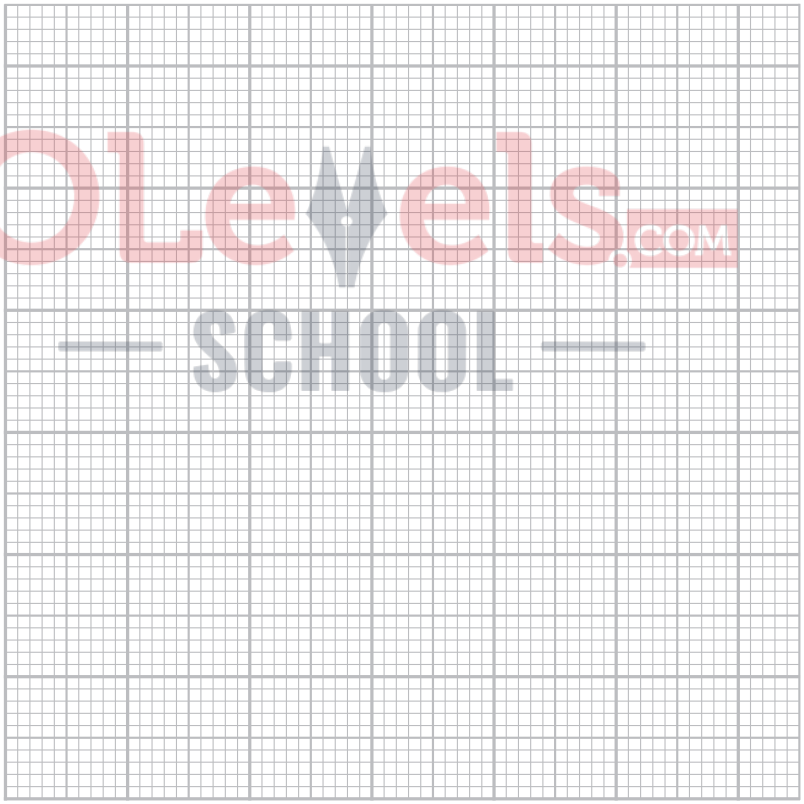
(a) The heights of a group of students are measured.

The table shows the results.

Height in cm	150–151	152–153	154–155	156–157	158–159	> 160
Number of students	4	14	25	20	10	4

(i) Plot a bar chart of the results.

(5)



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3. Question 3 · part a.i · 5 marks

November 2020 Paper 1

Further practice · Attempt part a.i only.

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(ii) Describe the trend shown in this bar chart.

(2)

.....

.....

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.....

(b) State three factors that could affect the heights of the students.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(Total for Question 3 = 10 marks)

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3. Question 3 · part a.i · 5 marks · Working and answer

Working and answer

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4. Question 4 · part a.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part a.i only.

4 The table shows the results of a survey about the number of teeth per student in a class of young students.

Number of teeth	Number of students
23	1
24	2
25	3
26	4
27	7
28	13

(a) (i) Draw a bar chart to show the results of the survey.

(4)



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4. Question 4 · part a.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part a.i only.

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(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

(3)

percentage = ... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

(2)

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4. Question 4 · part a.i · 4 marks · Working and answer

Working and answer

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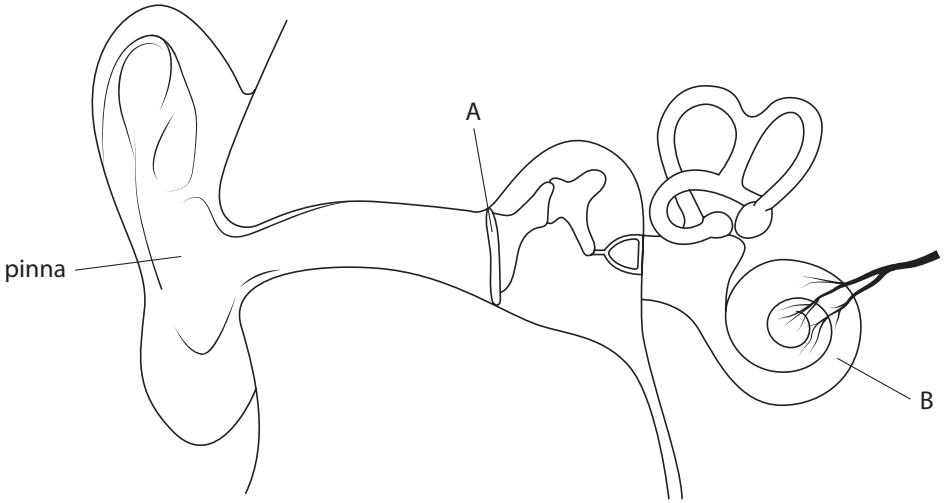
5. Question 5 · part b.i · 4 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

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5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

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5. Question 5 · part b.i · 4 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)

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5. Question 5 · part b.i · 4 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

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(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

(3)

percentage change = ... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

(2)

1 .....

.....

2 .....

.....

(ii) Describe how the reliability of the results could be improved.

(2)

.....

.....

.....

.....

(Total for Question 5 = 15 marks)

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5. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 2 · part a.i · 4 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

2 (a) The table gives information about different methods of food preservation.

It also gives information on the length of time in days that each method keeps foods fresh.

Method of food preservation	Time food stays fresh in days
refrigeration	5
freezing	90
pasteurisation	35
vacuum-packed food	14

(i) Draw a bar chart to show the information in the table.

(4)

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6. Question 2 · part a.i · 4 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

(ii) State how many of the methods shown in the bar chart use heat treatment to preserve food.

(1)

.....

(b) Bacteria can multiply rapidly in food that is not preserved.

Use words from the box to complete the passage about the growth of bacteria.

(6)

exponential	more than	stationary	equal to	lag
dying	death	less than	live	growth

The first stage in the growth of bacterial colonies is known as the ..... phase. During this phase, bacteria are not dividing.

The second stage of bacterial growth is known as the ..... phase. During this phase the bacteria are multiplying rapidly.

In the third stage, nutrients are running out and the number of dying cells is ..... the number of dividing cells.

Finally, during the ..... phase the number of dividing cells is ..... the number of ..... cells.

(Total for Question 2 = 11 marks)

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6. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

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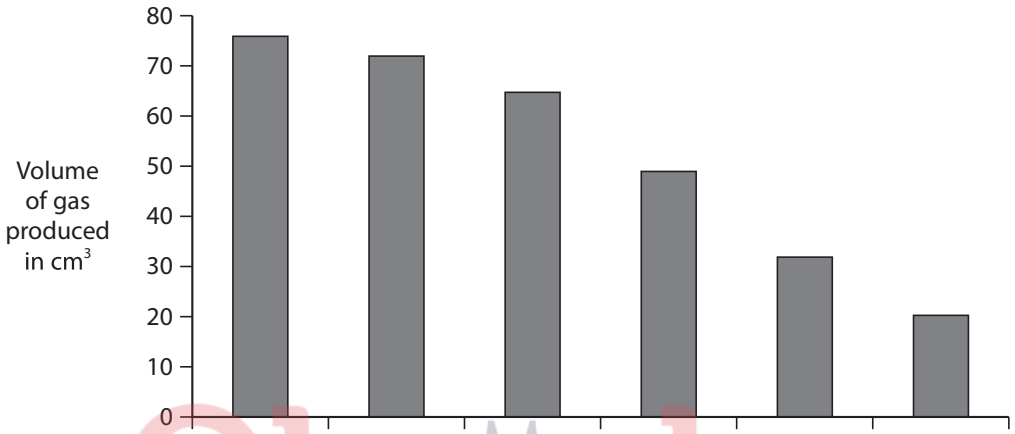
7. Question 3 · part a.i · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.i only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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P 7 2 5 8 5 A 0 6 2 0

7. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

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(ii) Describe the function of a motor neurone.

(2)

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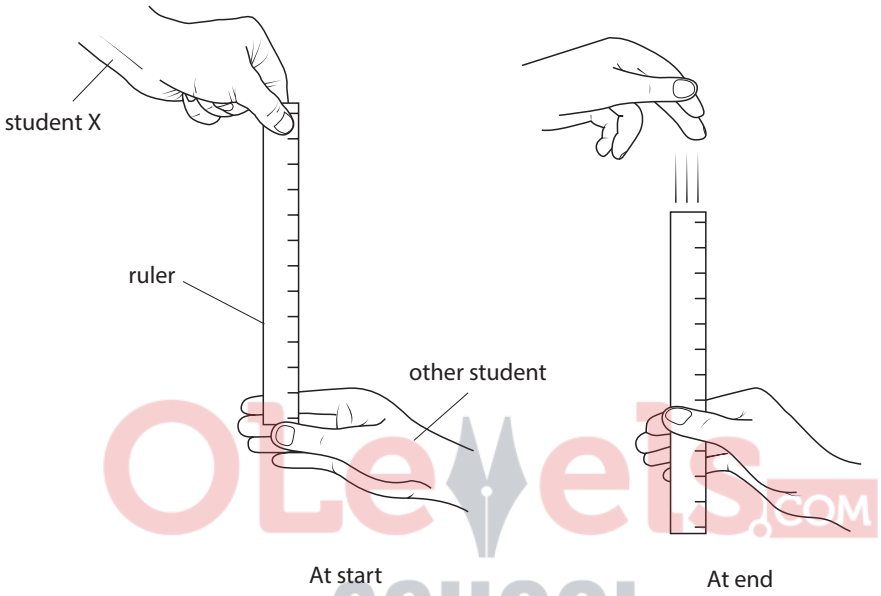
8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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(b) Student X uses this equipment to investigate the reaction time of another student.



Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

(6)

(ii) Name the dependent variable in this investigation.

(1)

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

(1)

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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(iv) The reading on the ruler for one of the students is 0.15 m.

Calculate the reaction time of this student.

Use this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

(2)

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reaction time = ... s

(Total for Question 3 = 16 marks)

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Working and answer

Practice workbook · Plot a line graph from a table of results

Continuous data in a table are plotted on a grid, for one or several series, joined with ruled lines or a smooth curve as instructed.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PLOT-LINE-GRAPH

1. Question 2 · part a.i · 6 marks

November 2023 Paper 1

Worked-example retry · Attempt part a.i only.

2 To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

Time in minutes	Blood glucose levels in mg per 100 cm^3	
	person without diabetes	person with diabetes
0	65	110
30	120	215
60	130	230
90	95	200
120	80	190
150	65	180
180	65	170

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1. Question 2 · part a.i · 6 marks

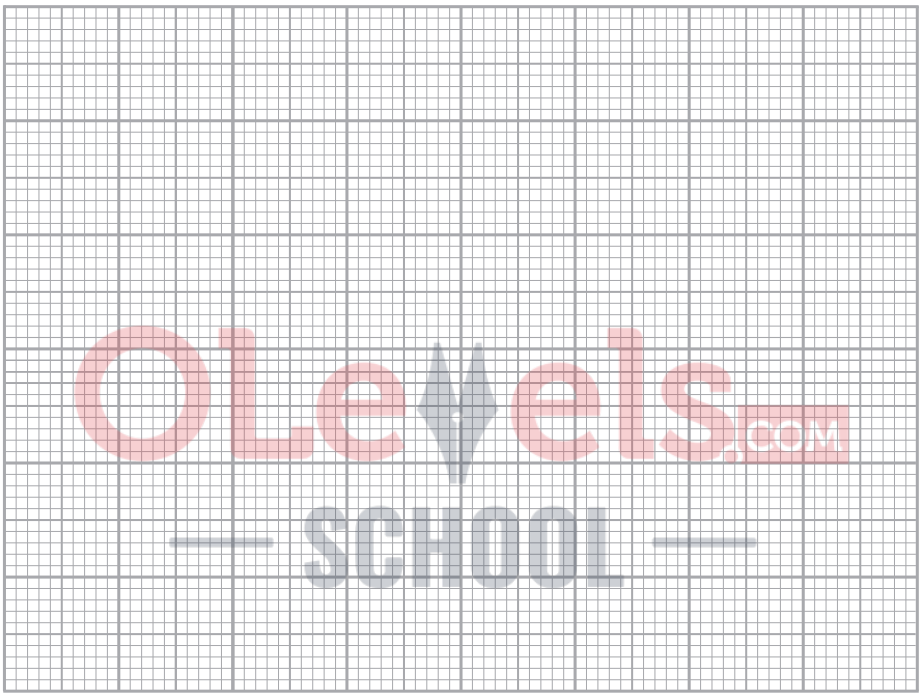
November 2023 Paper 1

Worked-example retry · Attempt part a.i only.

(a) (i) Plot the results on the grid.

Join the points with straight lines.

(6)



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

(2)

without diabetes = ... mg per 100 cm^3

with diabetes = ... mg per 100 cm^3

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1. Question 2 · part a.i · 6 marks · Working and answer

Working and answer

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2. Question 5 · part a · 5 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

5 A student investigates the effect of temperature on the volume of sweat produced.

This is the student's method.

- sit a person in a room at 20 °C
- measure the volume of sweat the person produces
- repeat with the room at different temperatures
- repeat again when the person is active

The table shows the student's results.

Temperature of room in °C	Volume of sweat produced in arbitrary units	
	Sitting	Active
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

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2. Question 5 · part a · 5 marks

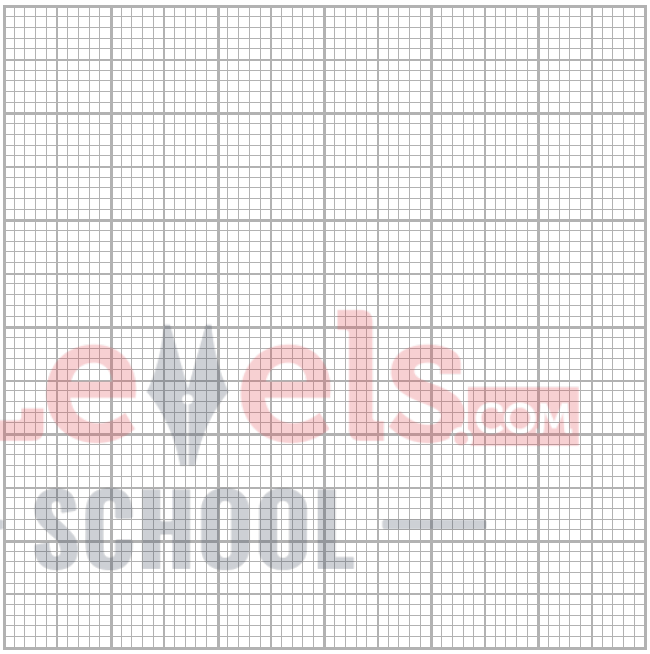
June 2019 Paper 1R

Further practice · Attempt part a only.

(a) Plot the student's results on the grid.

Join the points with straight lines.

(5)



(b) Use your graph to determine the difference in the volume of sweat produced at 27 °C when sitting and when active.

(2)

difference = ... arbitrary units

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2. Question 5 · part a · 5 marks · Working and answer

Working and answer

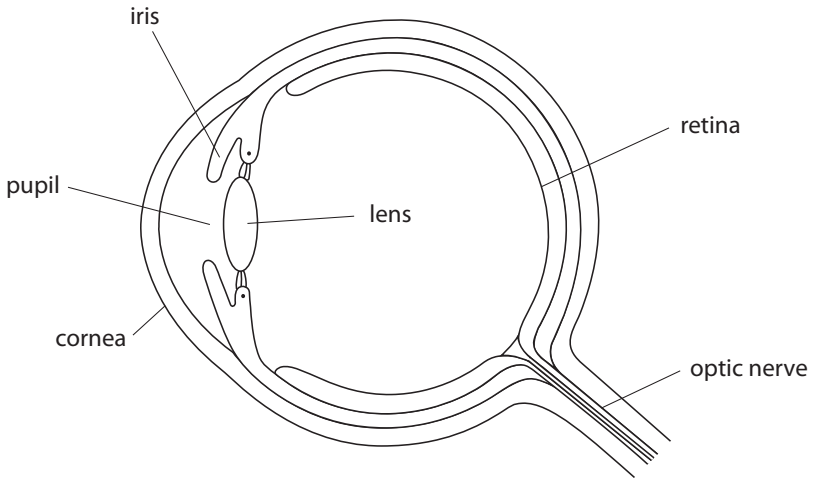
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3. Question 3 · part b.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.i only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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3. Question 3 · part b.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.i only.

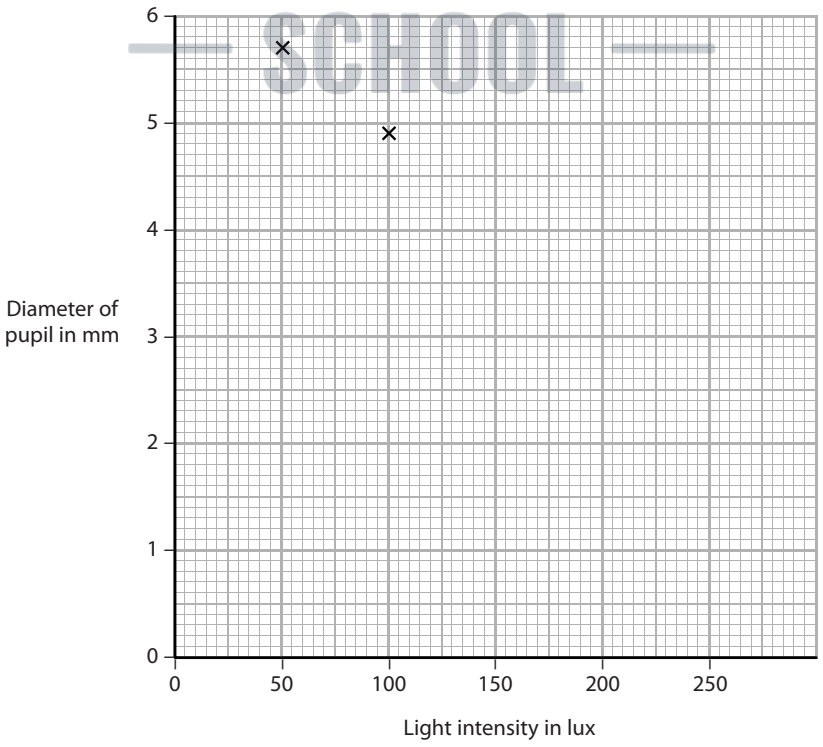
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 5 · part a.i · 6 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
atropine injected → 9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

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4. Question 5 · part a.i · 6 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

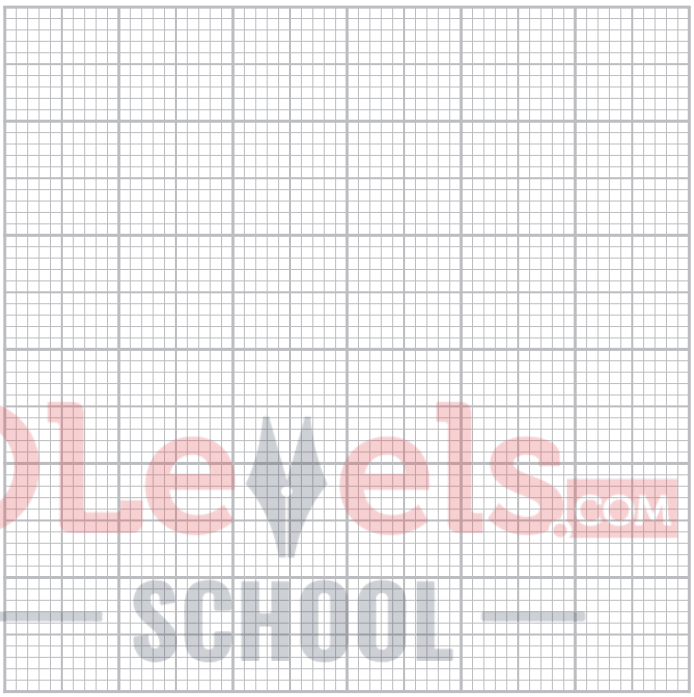
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

.....

.....

.....

.....

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

4. Question 5 · part a.i · 6 marks · Working and answer

Working and answer

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5. Question 4 · part a.i · 5 marks

November 2020 Paper 2

Further practice · Attempt part a.i only.

4 Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

Beaker	pH of contents of beaker					
	at start	1 minute	2 minutes	3 minutes	4 minutes	5 minutes
A – aspirin	7.6	7.1	6.6	6.2	5.8	5.4
B – aspirin and sucrose	7.2	6.7	6.3	5.7	5.2	4.7
C – aspirin and starch	7.1	7.2	7.3	7.3	7.4	7.4

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5. Question 4 · part a.i · 5 marks

November 2020 Paper 2

Further practice · Attempt part a.i only.

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(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

(5)



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(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

(1)

.....

5. Question 4 · part a.i · 5 marks · Working and answer

Working and answer

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6. Question 3 · part b.i · 5 marks

June 2021 Paper 1

Further practice · Attempt part b.i only.

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3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

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6. Question 3 · part b.i · 5 marks

June 2021 Paper 1

Further practice · Attempt part b.i only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

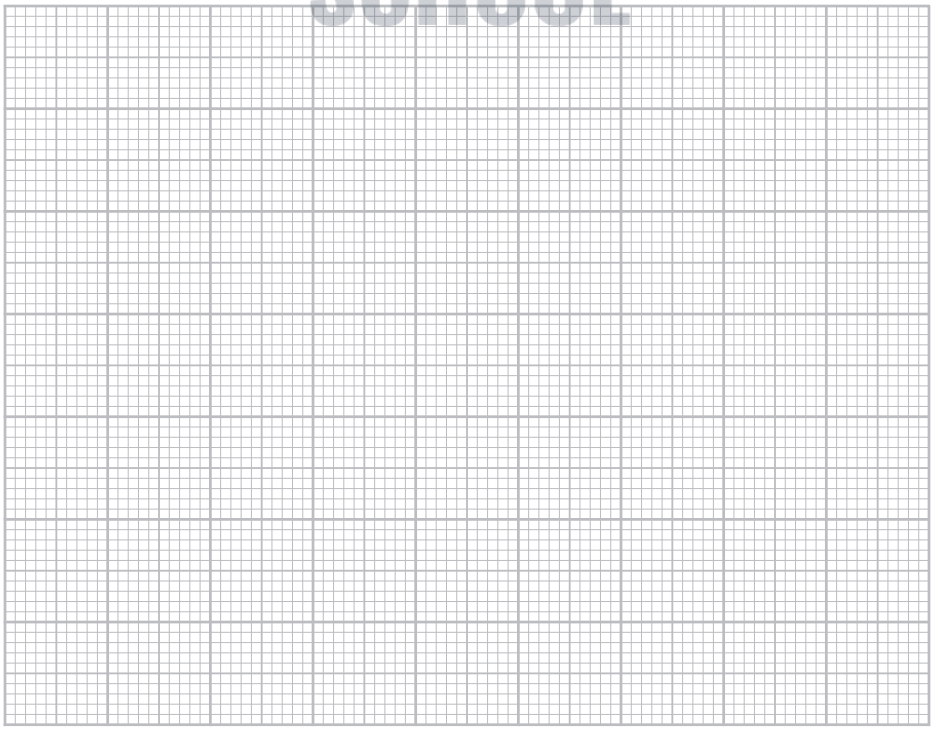
The results of the scientist's investigation are shown in the table.

Temperature in °C	Rate of protein digestion in arbitrary units	
	Enzyme A	Enzyme B
10	0	0
20	5	10
30	5	40
40	20	80
50	60	140
60	80	200
70	20	280
80	0	320
90	0	80

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

(5)



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8

P 6 6 4 3 2 A 0 8 2 4

6. Question 3 · part b.i · 5 marks

June 2021 Paper 1

Further practice · Attempt part b.i only.

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(ii) Describe the effects on enzyme B of increasing the temperature from 80°C to 90°C.

(3)

(iii) Describe the differences between the rate of protein digestion shown by enzyme A and enzyme B.

Refer to the graph in your answer.

(3)

(Total for Question 3 = 15 marks)

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6. Question 3 · part b.i · 5 marks · Working and answer

Working and answer

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7. Question 2 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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2 A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 cm^3 of each solution into a different test tube
- add 1.0 cm^3 of a competitive inhibitor and 1.0 cm^3 of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation.

(2)

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(ii) State three variables that should be kept constant in this investigation.

(3)

1 .....

2 .....

3 .....

7. Question 2 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

(b) The table shows the student's results.

Concentration of phenolphthalein phosphate solution in arbitrary units	Intensity of pink colour in arbitrary units
0.05	0.20
0.10	0.30
0.20	0.47
0.40	0.65
0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**

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P 7 0 9 5 0 A 0 6 2 4

7. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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8. Question 5 · part b.i · 4 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

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5 Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

(2)

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

(2)

.....

.....

.....

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8. Question 5 · part b.i · 4 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

Period of time in hours	Percentage change in mass of haemoglobin
0	0
100	0.7
200	2.4
300	4.1
400	5.9
500	6.5
600	7.2

(i) Plot the results on the grid. **(4)**

(ii) Draw the curve of best fit. **(1)**

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P 7 2 6 0 2 A 0 1 4 2 8

8. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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9. Question 5 · part b.i · 6 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

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5 A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 cm^2 area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

(2)

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(ii) State three variables that should be controlled.

(3)

1 .....

2 .....

3 .....

9. Question 5 · part b.i · 6 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

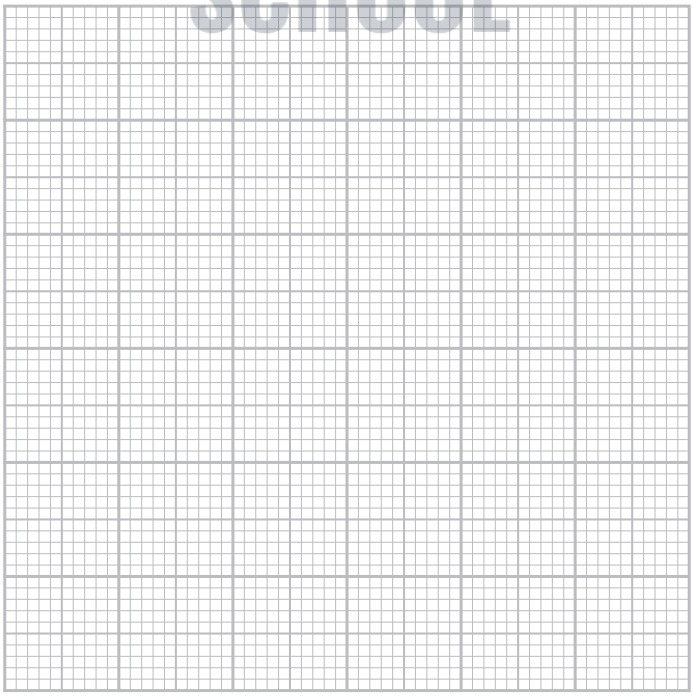
(b) The table shows the student's results.

Room temperature in °C	Amount of sweat produced by 25 cm^2 area of skin in arbitrary units	
	While inactive	During exercise
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

(i) Plot line graphs of the student's results.

Join the points with straight lines.

(6)



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P 7 2 5 8 6 A 0 1 2 2 0

9. Question 5 · part b.i · 6 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

(ii) Explain the differences between the two sets of results.

(5)

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(Total for Question 5 = 16 marks)

9. Question 5 · part b.i · 6 marks · Working and answer

Working and answer

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10. Question 7 · part a.i · 4 marks

June 2023 Paper 1

Further practice · Attempt part a.i only.

7 A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 cm^3 per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GFR in cm^3 per minute	80	75	79	79	67	60	50	45	35

(a) (i) Plot a line graph of the data on the grid.

(4)



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10. Question 7 · part a.i · 4 marks

June 2023 Paper 1

Further practice · Attempt part a.i only.

(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

(2)

(iii) A person needs dialysis if the GFR falls to 15 cm^3 per minute.

Use the data to estimate when the person will probably need to start dialysis.

(2)

(iv) Discuss the accuracy of this estimate.

(3)

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10. Question 7 · part a.i · 4 marks · Working and answer

Working and answer

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11. Question 4 · part a.i · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a.i only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

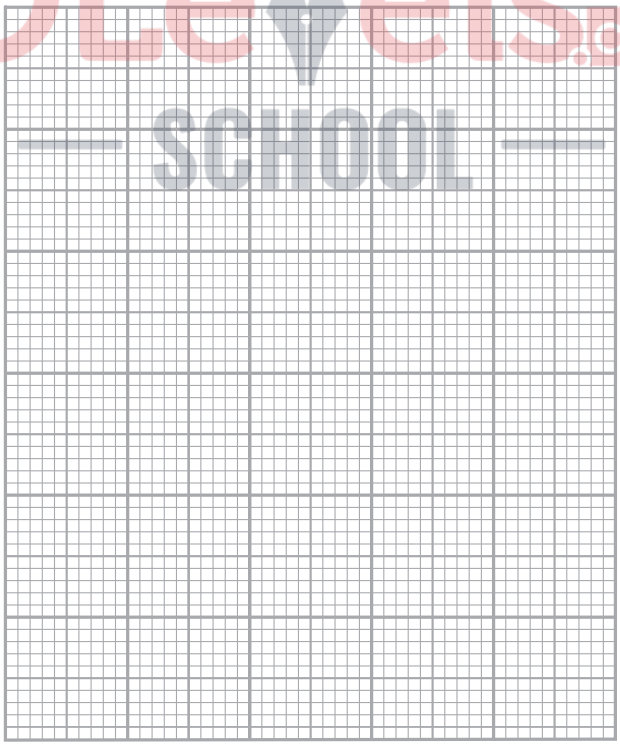
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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11. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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12. Question 2 · part a.i · 3 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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12. Question 2 · part a.i · 3 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

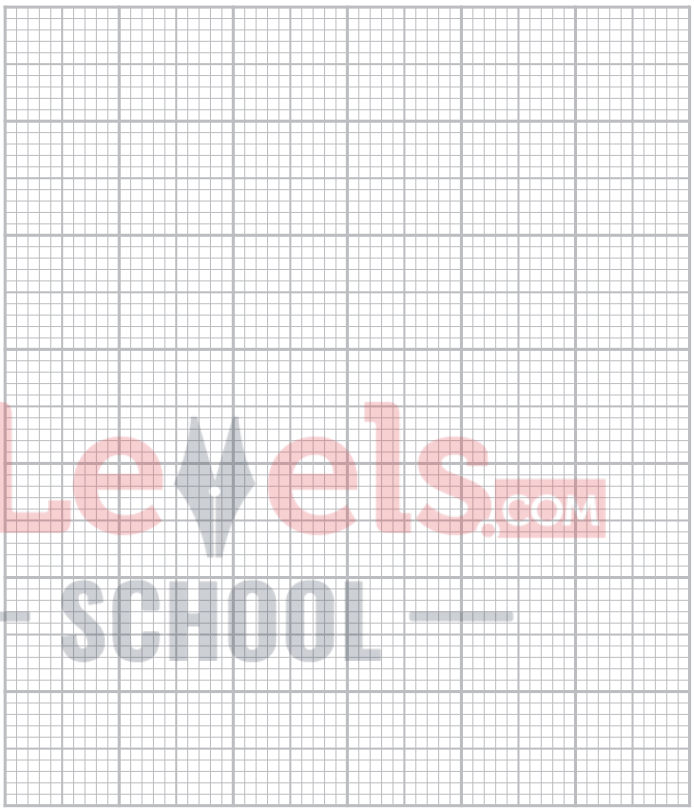
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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

.....

.....

12. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

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13. Question 4 · part a.i · 4 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

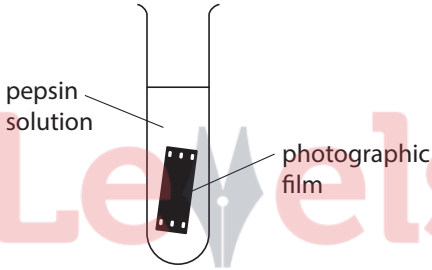
4 Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin



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The table shows the student's results.

Percentage pepsin concentration (%)	Time taken for film to become clear in seconds
1	112
2	102
3	93
4	84
5	84
6	84

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13. Question 4 · part a.i · 4 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

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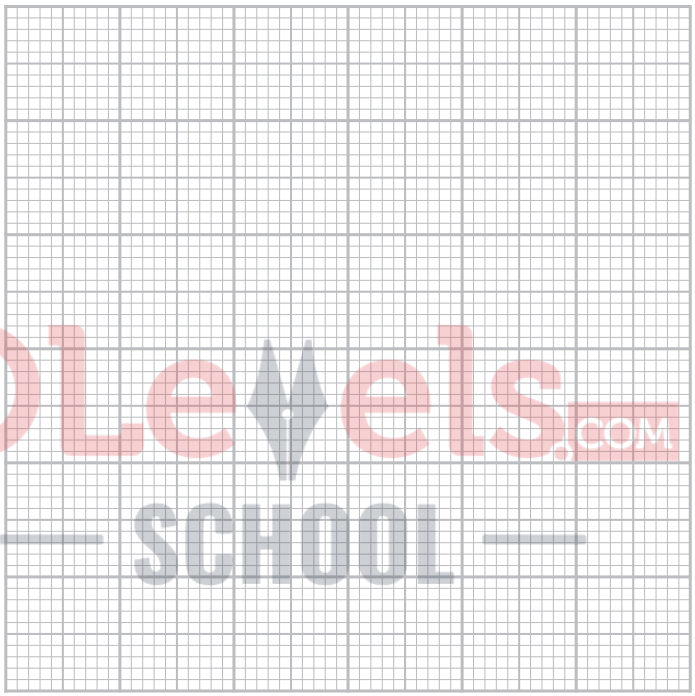
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(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



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(b) (i) Explain why the photographic film becomes clear.

(3)

13. Question 4 · part a.i · 4 marks · Working and answer

Working and answer

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14. Question 4 · part c.i · 5 marks

June 2025 Paper 1

Further practice · Attempt part c.i only.

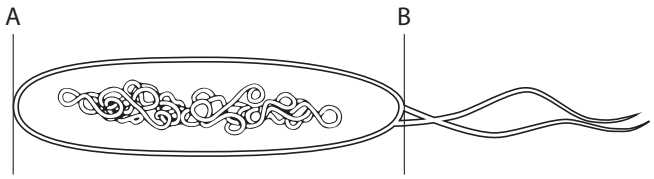
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4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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14. Question 4 · part c.i · 5 marks

June 2025 Paper 1

Further practice · Attempt part c.i only.

(c) The effectiveness of a vaccine in preventing disease was studied over six months.

Two different groups of people were used. One group of people were aged between 45 and 64 years and the second group were aged over 65 years.

The results are shown in the table.

	Percentage effectiveness against disease (%)	
Time since vaccination in months	**45 to 64 years**	**over 65 years**
1	90	80
2	85	78
3	80	75
4	68	58
5	60	50
6	50	40

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14. Question 4 · part c.i · 5 marks

June 2025 Paper 1

Further practice · Attempt part c.i only.

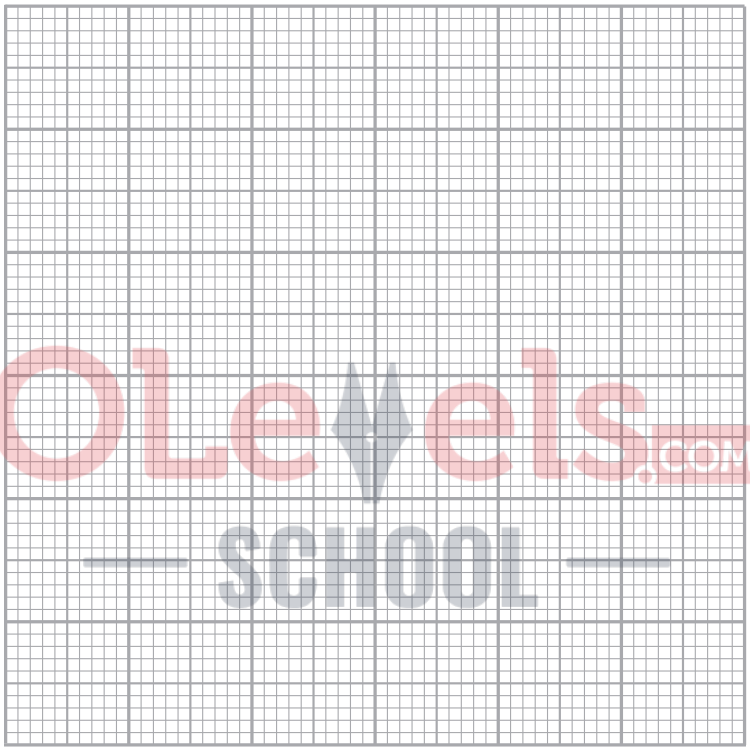
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(i) Plot a graph of the results. Join the points with straight lines.

(5)



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(ii) Describe the trends shown by this graph.

(3)

(Total for Question 4 = 15 marks)

14. Question 4 · part c.i · 5 marks · Working and answer

Working and answer

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Practice workbook · Draw a line or curve of best fit through plotted points

Points are already plotted and a single straight line or smooth curve showing the trend is requested.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-LINE-OF-BEST-FIT

1. Question 2 · part a.ii · 1 mark

November 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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1. Question 2 · part a.ii · 1 mark

November 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

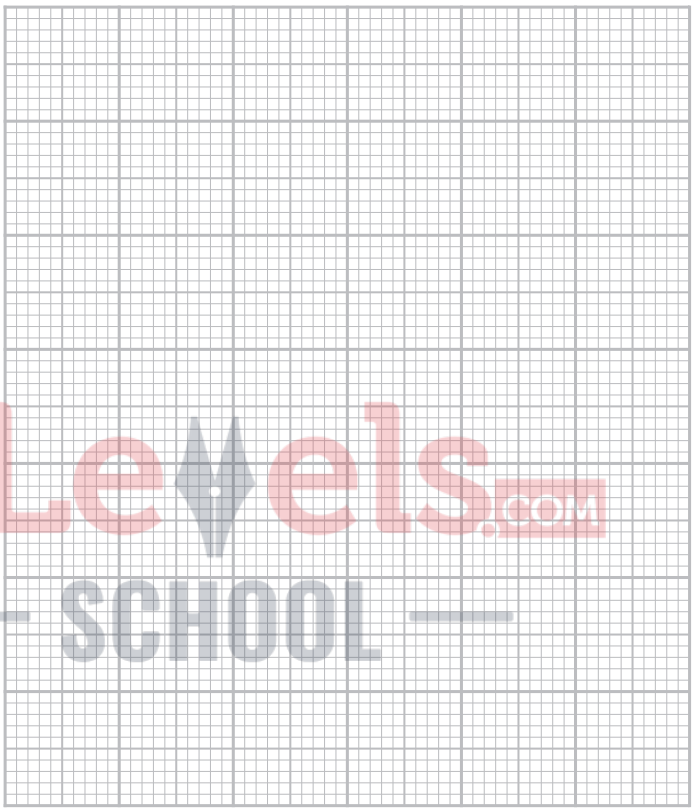
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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

.....

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1. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

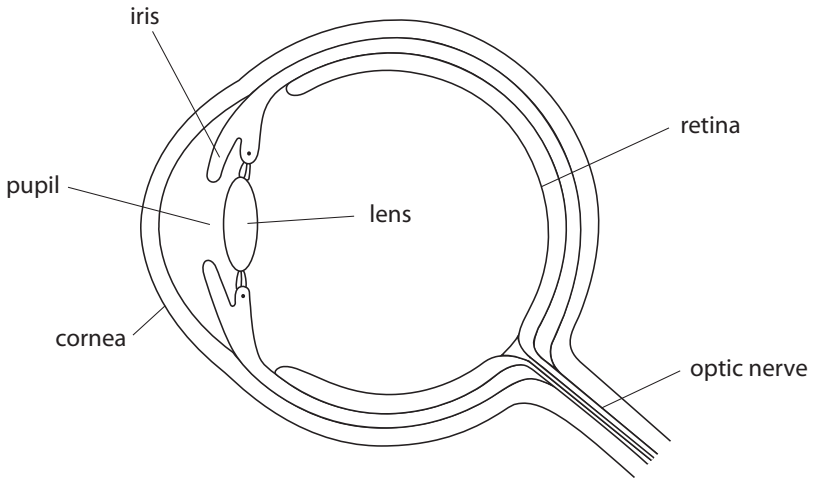
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2. Question 3 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

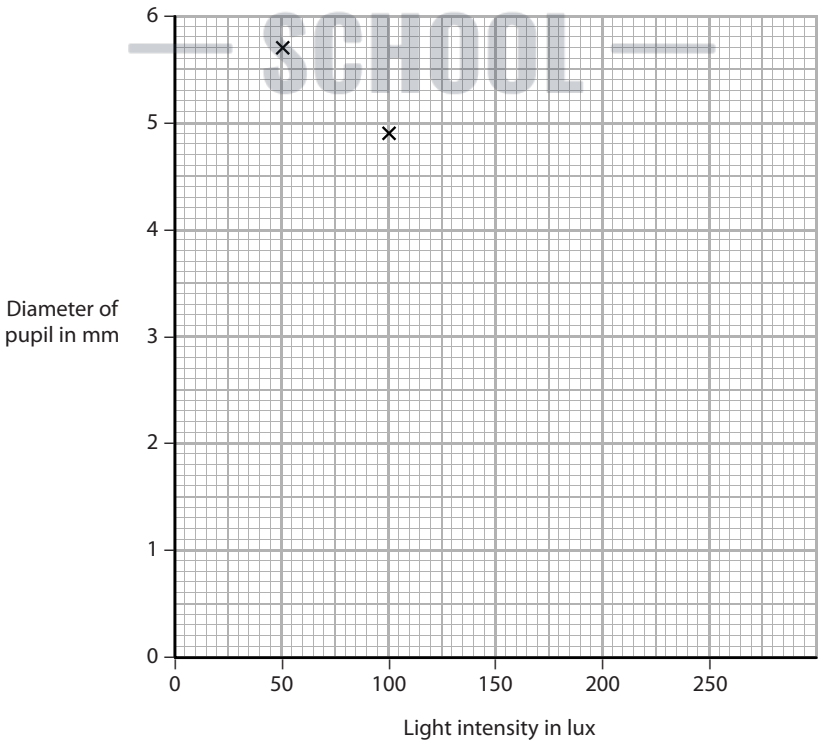
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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2. Question 3 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

(iii) Explain which result is anomalous.

(2)

(iv) State the effect of light intensity on the diameter of the pupil.

(1)

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

(2)

(Total for Question 3 = 8 marks)

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Working and answer

3. Question 2 · part b.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

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2 A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 cm^3 of each solution into a different test tube
- add 1.0 cm^3 of a competitive inhibitor and 1.0 cm^3 of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

2 .....

3 .....

3. Question 2 · part b.ii · 1 mark

June 2022 Paper 2R

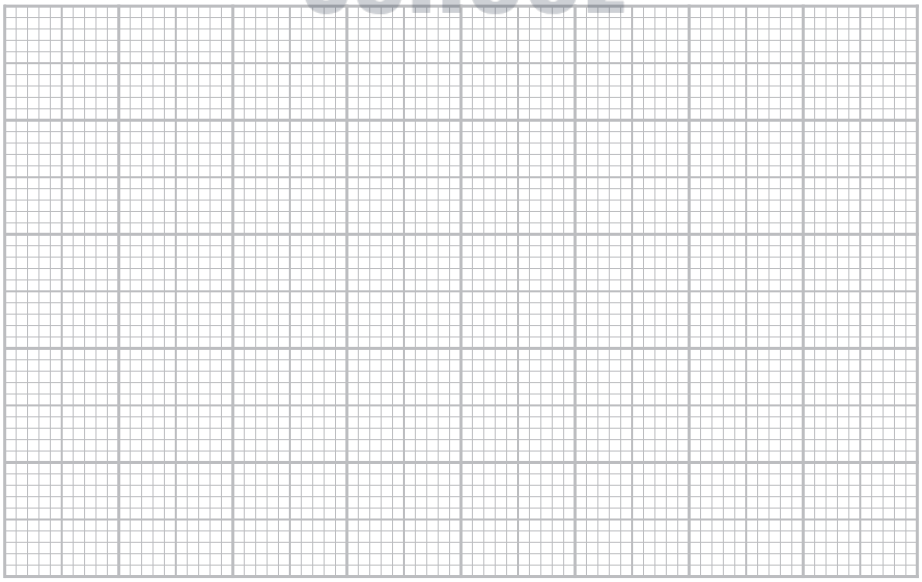
Further practice · Attempt part b.ii only.

(b) The table shows the student's results.

Concentration of phenolphthalein phosphate solution in arbitrary units	Intensity of pink colour in arbitrary units
0.05	0.20
0.10	0.30
0.20	0.47
0.40	0.65
0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**



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3. Question 2 · part b.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Add a second curve to show the results that would be obtained if the competitive inhibitor had not been added.

(2)

(c) Describe how the results of the investigation would be different if a non-competitive inhibitor had been used instead of a competitive inhibitor.

(2)

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.....

(Total for Question 2 = 13 marks)

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Working and answer

4. Question 5 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

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5 Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

(2)

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

(2)

.....

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4. Question 5 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

Period of time in hours	Percentage change in mass of haemoglobin
0	0
100	0.7
200	2.4
300	4.1
400	5.9
500	6.5
600	7.2

(i) Plot the results on the grid.

(4)

(ii) Draw the curve of best fit.

(1)

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4. Question 5 · part b.ii · 1 mark

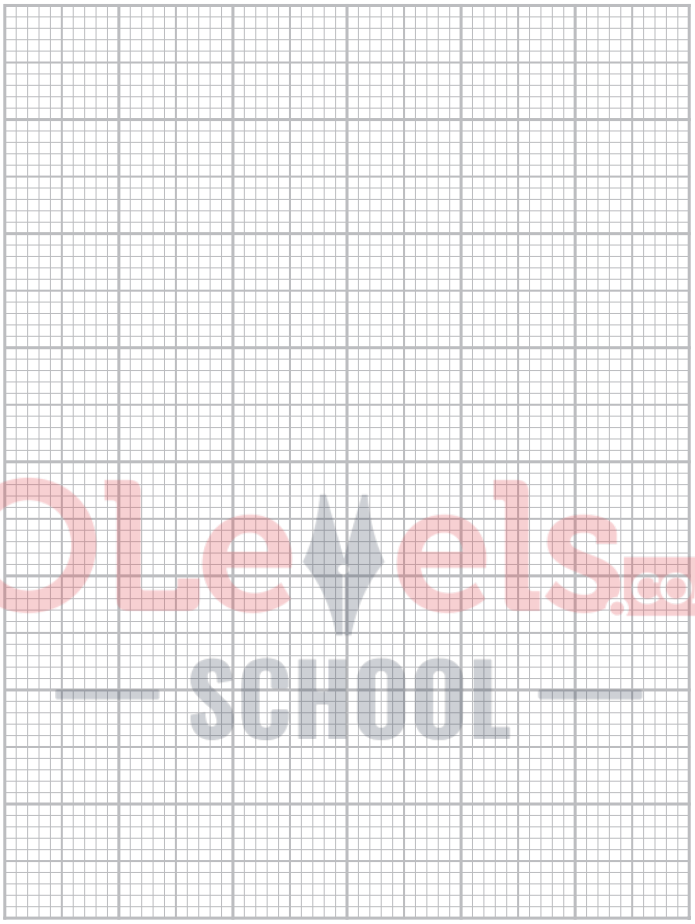
January 2023 Paper 2

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4. Question 5 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(iii) Discuss how the percentage change in mass of haemoglobin, after training at altitude, could result in an increase in performance during exercise.

(5)

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(Total for Question 5 = 14 marks)

4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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5. Question 4 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

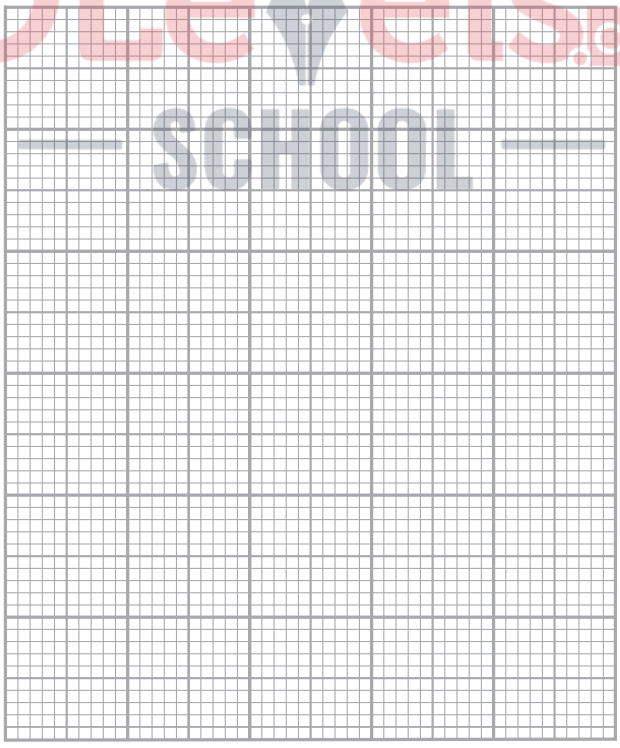
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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5. Question 4 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

(vi) Name a part of the body where starch digestion takes place.

(1)

(b) Describe a test for starch in a sample of food.

(2)

(Total for Question 4 = 13 marks)

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5. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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6. Question 4 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

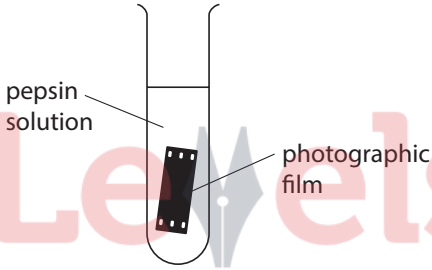
4 Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin



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The table shows the student's results.

Percentage pepsin concentration (%)	Time taken for film to become clear in seconds
1	112
2	102
3	93
4	84
5	84
6	84

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6. Question 4 · part a.ii · 1 mark

June 2025 Paper 1R

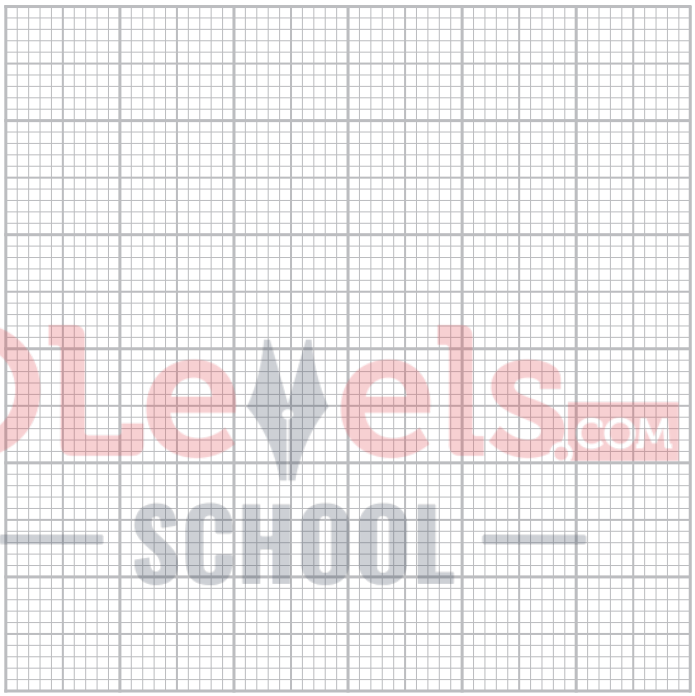
Further practice · Attempt part a.ii only.

(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



(b) (i) Explain why the photographic film becomes clear.

(3)

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6. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Calculate a percentage change or percentage difference

Two values of one quantity, from text, a table or a graph, compared as a percentage of a stated starting or reference value.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PERCENTAGE-CHANGE

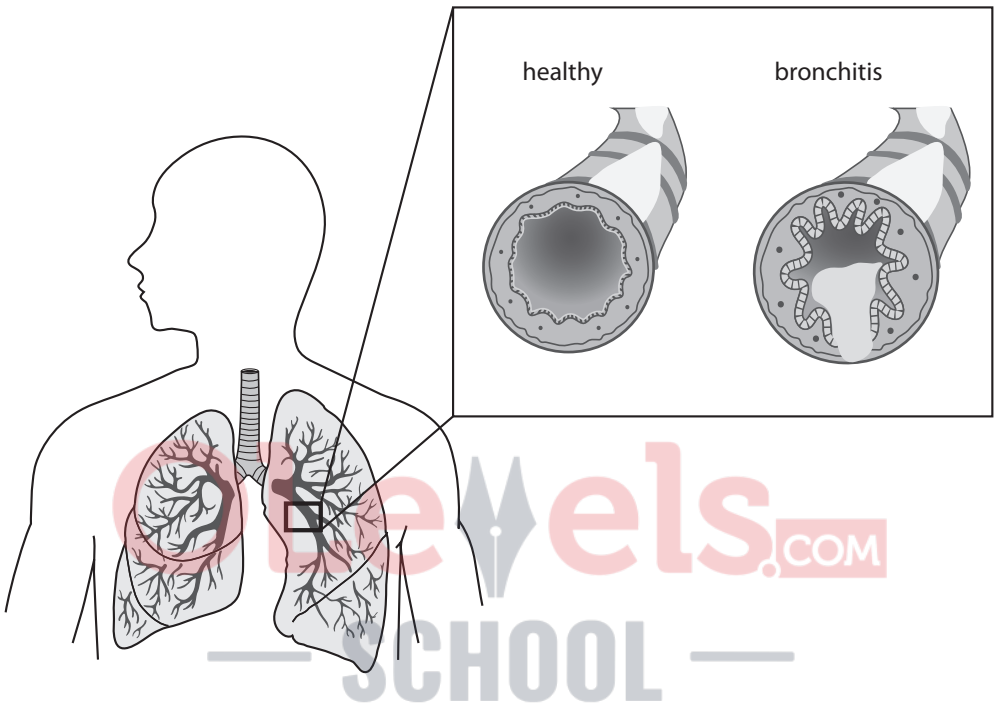
1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

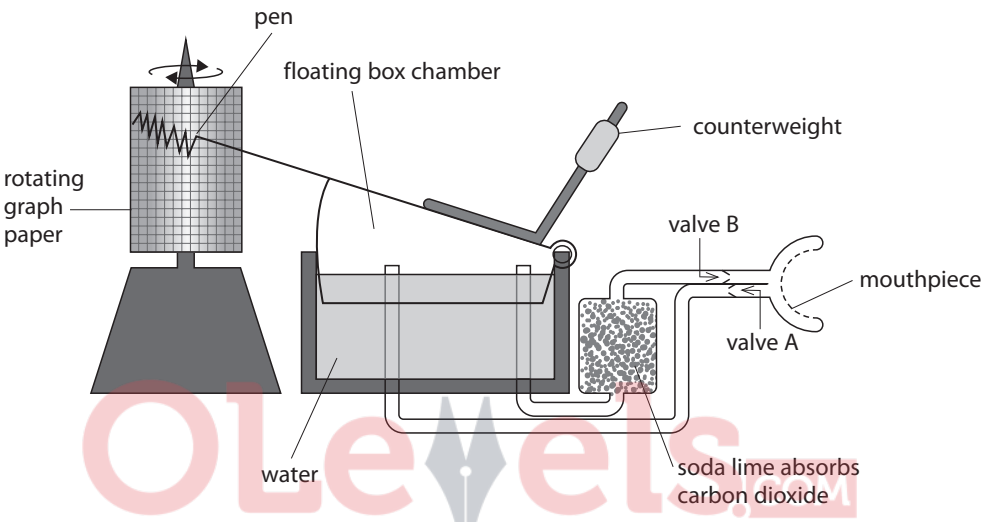
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

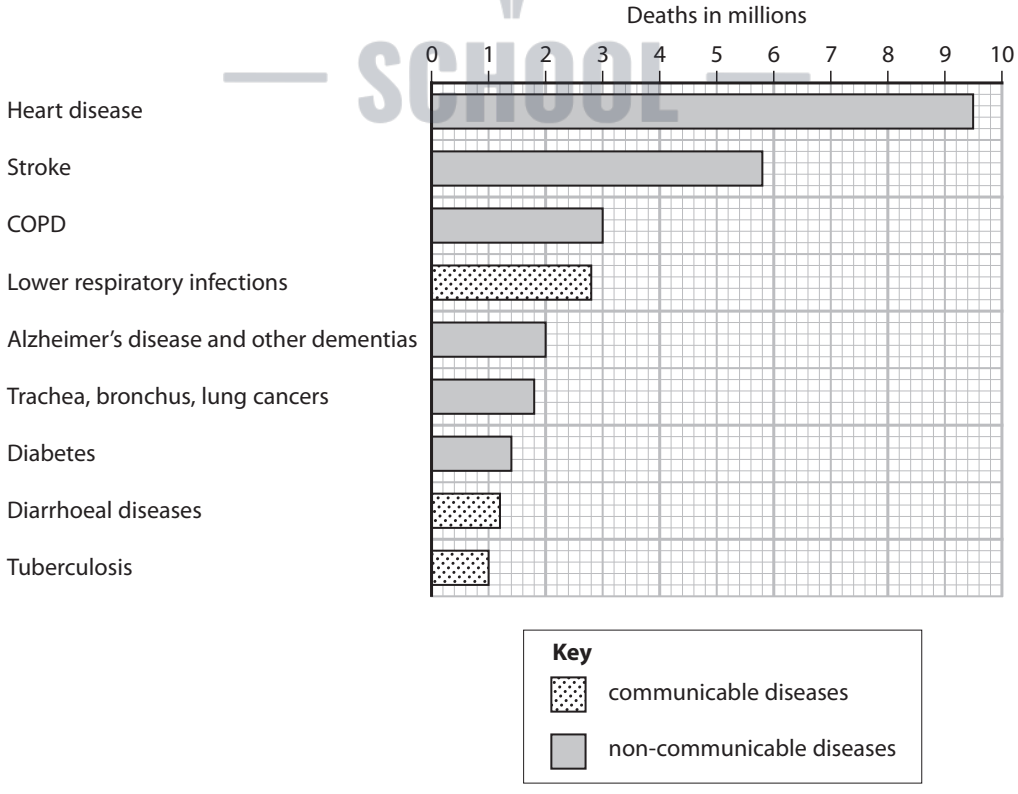
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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1. Question 6 · part c.ii · 3 marks

January 2020 Paper 1

Worked-example retry · Attempt part c.ii only.

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(i) More deaths are caused by non-communicable diseases than by communicable diseases.

Suggest what is meant by the term **non-communicable disease**.

(1)

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(ii) In 2000, the number of deaths from heart disease was 7.0 million.

Calculate the percentage change in the number of deaths from heart disease from 2000 to 2016.

(3)

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percentage change = ...%

(iii) Suggest a reason for the change in the number of deaths from heart disease from 2000 to 2016.

(1)

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(Total for Question 6 = 13 marks)

1. Question 6 · part c.ii · 3 marks · Working and answer

Working and answer

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2. Question 5 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

5 Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

(2)

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

(5)

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2. Question 5 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

	Concentration of sucrose solution in mol per dm³				
	0.2	**0.4**	**0.6**	**0.8**	**1.0**
Initial mass in g	8.4	7.9	8.7	8.3	8.4
Mass after soaking in g	8.7	8.0	8.2	7.5	7.3
Change in mass in g	+ 0.3	+ 0.1	– 0.5	– 0.8	– 1.1
Percentage change in mass	+ 3.6		– 5.8	– 9.6	– 13.1

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

(2)

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

(2)

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2. Question 5 · part c.i · 2 marks · Working and answer

Working and answer

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3. Question 9 · part a.i · 3 marks

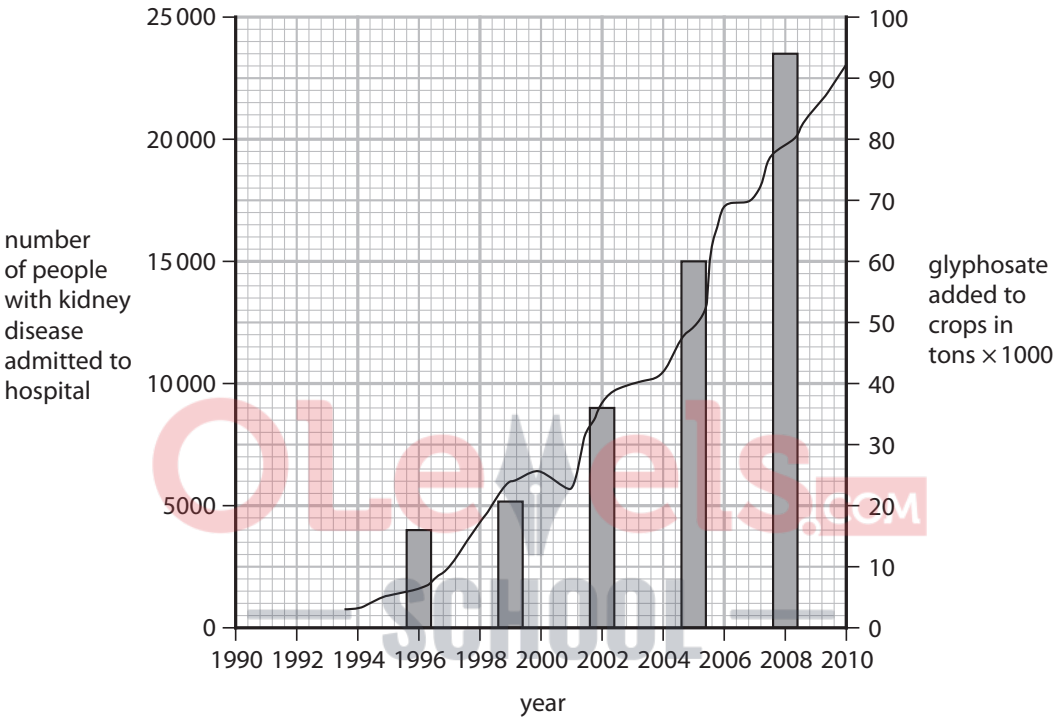
November 2020 Paper 1R

Further practice · Attempt part a.i only.

9 (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.



(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

(3)

percentage increase = ... %

20

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3. Question 9 · part a.i · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

(1)

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(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

(4)

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3. Question 9 · part a.i · 3 marks · Working and answer

Working and answer

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4. Question 5 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

atropine injected →

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4. Question 5 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

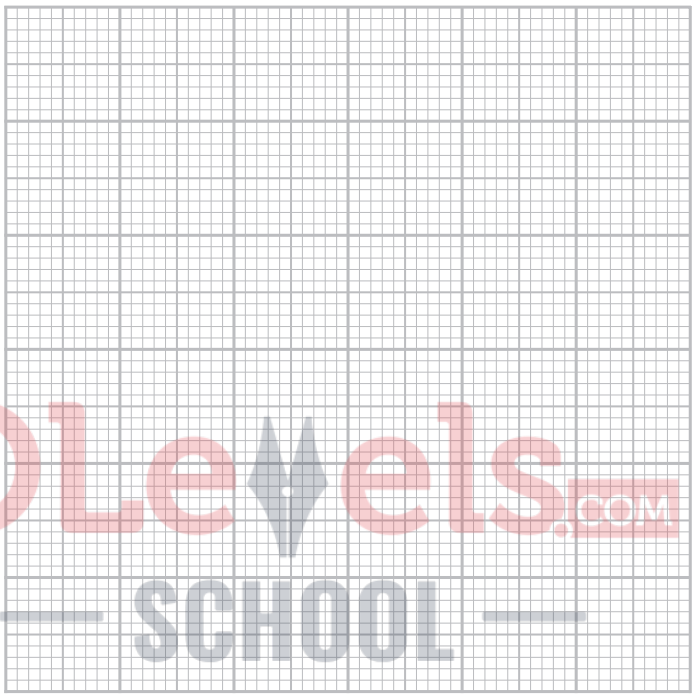
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

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(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

4. Question 5 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

(ii) Explain why this percentage change is not achieved as soon as atropine is injected.

(2)

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(c) Suggest the function of the vagus nerve.

(2)

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(Total for Question 5 = 14 marks)

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Working and answer

5. Question 6 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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5. Question 6 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

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(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

5. Question 6 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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5. Question 6 · part b.ii · 3 marks · Working and answer

Working and answer

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6. Question 4 · part c.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part c.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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6. Question 4 · part c.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part c.i only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

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6. Question 4 · part c.i · 3 marks · Working and answer

Working and answer

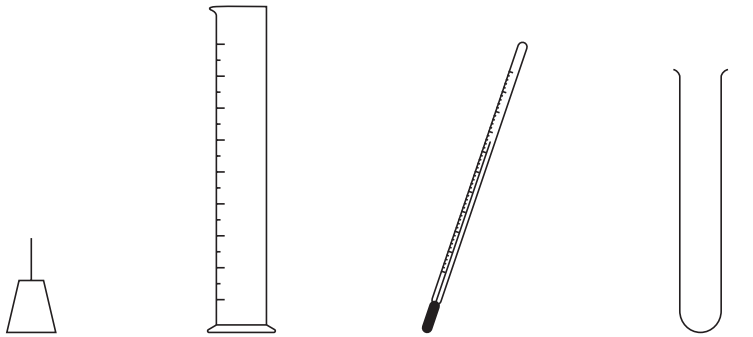
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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

3 (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

(5)

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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Name two variables that need to be controlled in the investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Give two safety precautions needed for this investigation.

(2)

1 .....

.....

2 .....

.....

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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

Daily Energy Requirement in kcal

Age of adults in years	Males	Females
19–24	2772	2175
25–34	2749	2175
35–44	2629	2103
45–54	2581	2103
55–64	2581	2079
65–74	2342	1912
75 and over	2294	1840

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

(i) Describe the patterns in the energy requirements for males and for females.

(2)

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

(1)

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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7. Question 3 · part b.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part b.iii only.

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(iii) Calculate the percentage difference in the energy requirement of males compared with females in the age range of 35–44 years.

(3)

percentage difference = ... %

(Total for Question 3 = 15 marks)

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Working and answer

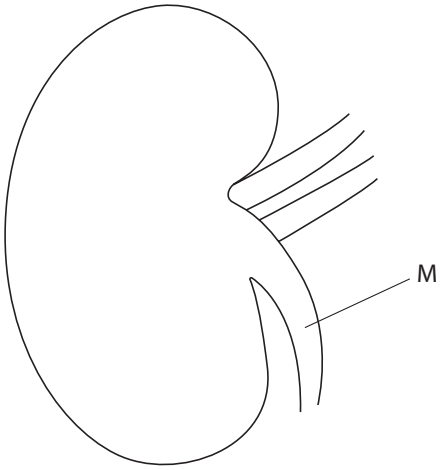
8. Question 7 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

8. Question 7 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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8. Question 7 · part c.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part c.i only.

(ii) Which is the main nitrogen containing excretory product produced from the breakdown of protein in the diet?

Use the information in the table to explain your answer.

(2)

(iii) The breakdown of muscle tissue continues at a constant rate and is not affected by diet.

Discuss whether the information in the table supports this statement.

(4)

(Total for Question 7 = 14 marks)

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8. Question 7 · part c.i · 3 marks · Working and answer

Working and answer

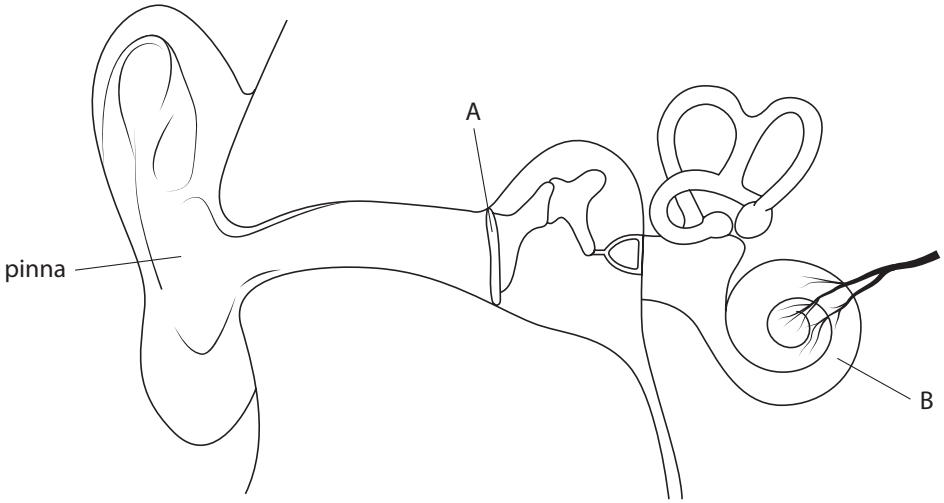
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9. Question 5 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

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9. Question 5 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

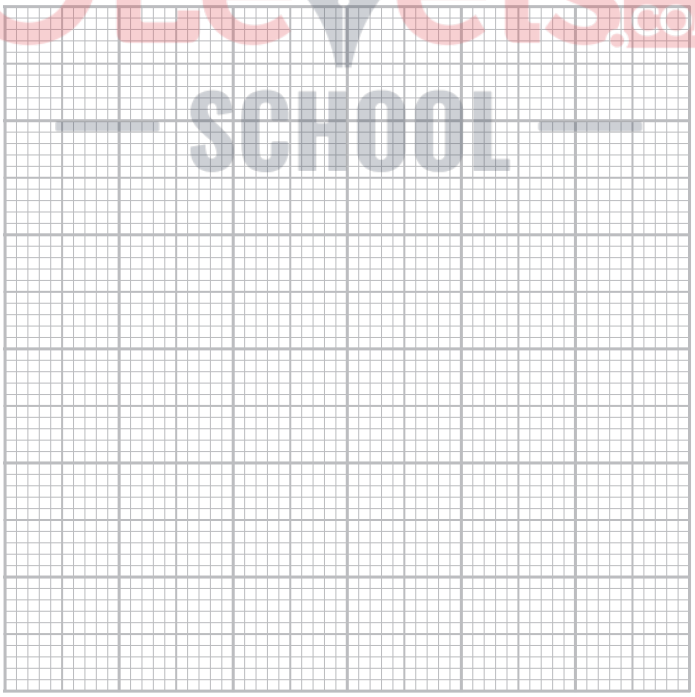
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)



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9. Question 5 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

(3)

percentage change = ... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

(2)

1 .....

.....

2 .....

.....

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(ii) Describe how the reliability of the results could be improved.

(2)

.....

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(Total for Question 5 = 15 marks)

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9. Question 5 · part b.ii · 3 marks · Working and answer

Working and answer

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10. Question 8 · part b.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

8 (a) (i) Describe the structure of a virus.

(2)

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

(5)

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10. Question 8 · part b.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(b) In 2010, AIDS-related illnesses caused the death of 1.4 million people.

In 2021, AIDS-related illnesses caused the death of 650 000 people.

(i) Name the microorganism that causes AIDS.

(1)

.....

(ii) Calculate the percentage decrease in the number of people dying from AIDS-related illnesses between 2010 and 2020.

(2)

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percentage decrease = ... %

(Total for Question 8 = 10 marks)

Working and answer

11. Question 3 · part a.ii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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11. Question 3 · part a.ii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

(Total for Question 3 = 15 marks)

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11. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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Practice workbook · Calculate an amount from a percentage, fraction or probability

A total population, volume or mass is given with a percentage, fraction or probability; the matching number, or the new value after a percentage rise or fall, is wanted.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-AMOUNT-FROM-PERCENTAGE

1. Question 7 · part c.i · 2 marks

June 2022 Paper 1

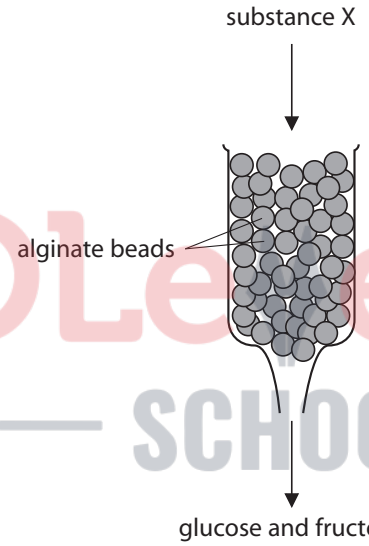
Worked-example retry · Attempt part c.i only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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1. Question 7 · part c.i · 2 marks

June 2022 Paper 1

Worked-example retry · Attempt part c.i only.

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(b) Which process is involved in the conversion of glucose into glycogen in the human body?

(1)

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

(2)

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number of people = ... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

38	62	51	96	49	78	82	91	65

Determine the median number of people with diabetes.

(2)

median = ...

(Total for Question 7 = 10 marks)

1. Question 7 · part c.i · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

June 2019 Paper 2

Further practice · Attempt part a only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

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number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

.....

.....

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 4 · part c.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part c.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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3. Question 4 · part c.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part c.ii only.

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(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

3. Question 4 · part c.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part c.ii only.

(d) Diabetes causes a narrowing of the arteries supplying blood to the lower limbs.

Suggest why this can lead to the need to amputate the lower limbs.

(3)

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(Total for Question 4 = 14 marks)

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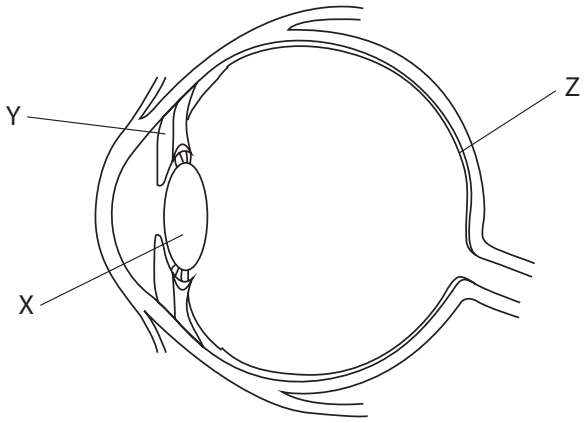
Working and answer

4. Question 1 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
	transmits electrical impulses to the brain
structure X	refracts light rays
	protects the eye from pathogens
structure Y	
	detects light rays
structure Z	controls the shape of the lens
	controls the size of the pupil

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4. Question 1 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

.....

.....

.....

(ii) State one treatment for cataracts.

(1)

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

4. Question 1 · part b.iii · 2 marks · Working and answer

Working and answer

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5. Question 5 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

5 (a) Describe the possible effects of a DNA mutation.

(3)

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

Age of mother in years	15–19	20–24	25–29	30–34	35–39	40–44
Probability of chromosome aberration	0.0010	0.0010	0.0010	0.0015	0.0045	0.0140

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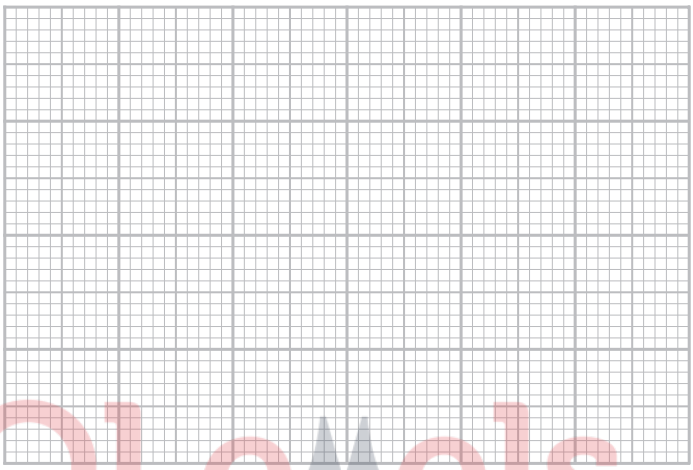
5. Question 5 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(i) Plot a bar chart of the results.

(5)



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(ii) In one year, 43 000 children are born to mothers aged 30–34.

Calculate the probable number of these children who will have a chromosome aberration.

Give your answer to two significant figures.

(3)

number of children = ...

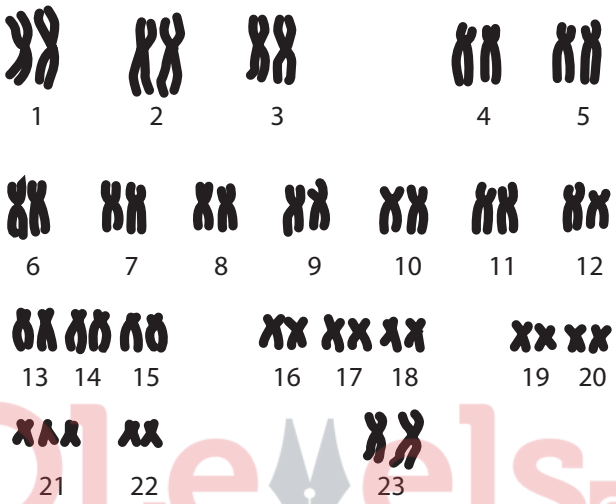
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5. Question 5 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(c) The diagram shows the full diploid number of chromosomes in a person.



Justify the statement that this diagram shows the genotype of a female with a chromosome aberration.

(4)

(Total for Question 5 = 15 marks)

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5. Question 5 · part b.ii · 3 marks · Working and answer

Working and answer

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6. Question 6 · part a.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

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6 (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

Region	Total number of people tested	Percentage of colour blind males	Percentage of colour blind females
A	150	10.00	1.00
B	16 000	6.60	0.40
C	5000	0.80	0.01

(i) Calculate the number of colour blind males in the test group in region C.

(3)

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number of males = ...

(ii) State why the results from region A are not considered to be reliable.

(1)

.....

6. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

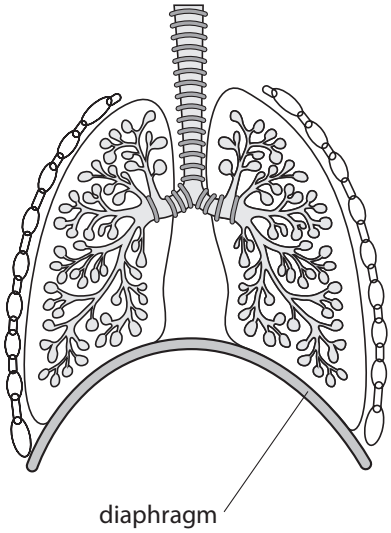
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7. Question 4 · part b.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

4 (a) The diagram shows the human lungs.



(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

1 .....

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2 .....

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3 .....

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4 .....

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7. Question 4 · part b.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

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(b) A spirometer can be used to test lung function.

(i) Describe how a spirometer is used to obtain values for tidal volume.

(4)

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(ii) Asthma is a condition that causes the tubes in the lungs to narrow.

A person without asthma has a vital capacity of 4600 cm^3.

People with severe asthma have a 30% lower vital capacity compared with a healthy person.

Calculate the vital capacity of a person with severe asthma.

(2)

vital capacity of a person with severe asthma = ... cm^3

(Total for Question 4 = 13 marks)

7. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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8. Question 5 · part b.ii · 2 marks

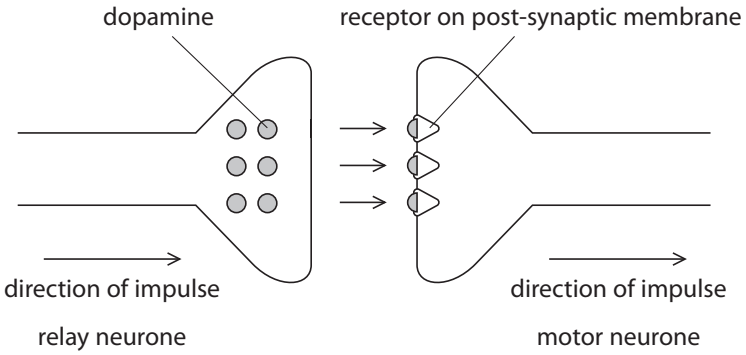
June 2023 Paper 2R

Further practice · Attempt part b.ii only.

5 (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.



(i) Describe what is meant by a neurotransmitter.

(2)

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

(2)

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P 7 2 5 8 8 A 0 1 4 2 4

8. Question 5 · part b.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Suggest how an excess of dopamine will affect bodily functions.

(2)

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.....

.....

.....

(b) In 2017 in the United Kingdom, 145 000 people were diagnosed with Parkinson's disease.

This figure is expected to have an 18% increase by 2025.

(i) Give two symptoms of Parkinson's disease.

(2)

1 .....

.....

2 .....

.....

(ii) Calculate the number of people expected to be diagnosed with Parkinson's disease in 2025.

(2)

number of people = ...

(Total for Question 5 = 10 marks)

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8. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 4 · part a.iii · 3 marks

June 2025 Paper 2

Further practice · Attempt part a.iii only.

4 A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

1.00 au = −7 %	1.50 au = −10 %	0.00 au (distilled water) = +8 %
	0.25 au = +4 %	0.50 au = −1 %

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

(4)

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P 7 8 7 6 9 A 0 8 2 0

9. Question 4 · part a.iii · 3 marks

June 2025 Paper 2

Further practice · Attempt part a.iii only.

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(ii) Determine the median value for sugar concentration.

(1)

median value = ...

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

(3)

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mass of cylinder = ... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

(4)

concentration of sugar solution = ... au

9. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

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10. Question 2 · part c · 3 marks

January 2022 Paper 2

Further practice · Attempt part c only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 dm^3 of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

(3)

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10. Question 2 · part c · 3 marks

January 2022 Paper 2

Further practice · Attempt part c only.

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(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

(1)

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

(1)

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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10. Question 2 · part c · 3 marks

January 2022 Paper 2

Further practice · Attempt part c only.

(c) Calculate the average volume of oxygen absorbed by the lungs during one minute of breathing, at rest. (lines 13 to 15)

(3)

volume of oxygen absorbed = ... dm^3

(d) Explain why pregnant women are advised not to smoke cigarettes.

(4)

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(Total for Question 2 = 14 marks)

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10. Question 2 · part c · 3 marks · Working and answer

Working and answer

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11. Question 7 · part a · 2 marks

June 2019 Paper 2

Further practice · Attempt part a only.

7 The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 cm^3 of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 cm^3 of this wine.



250 cm^3 glass of Ruby Red wine

(a) Calculate the number of units of alcohol in the glass of wine.

(2)

number of units = ...

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11. Question 7 · part a · 2 marks

June 2019 Paper 2

Further practice · Attempt part a only.

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(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

(2)

time = ... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

(2)

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11. Question 7 · part a · 2 marks · Working and answer

Working and answer

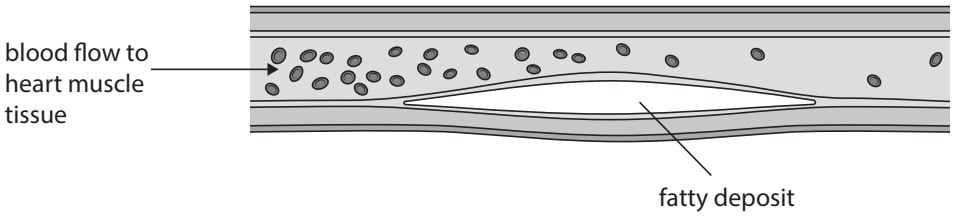
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12. Question 7 · part b.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

7 (a) The diagram shows a fatty deposit in an artery carrying blood to the heart muscle.



(Source: adapted from https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis)

Explain how a build-up of a fatty deposit in an artery carrying blood to the heart muscle can cause heart disease.

(4)

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12. Question 7 · part b.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

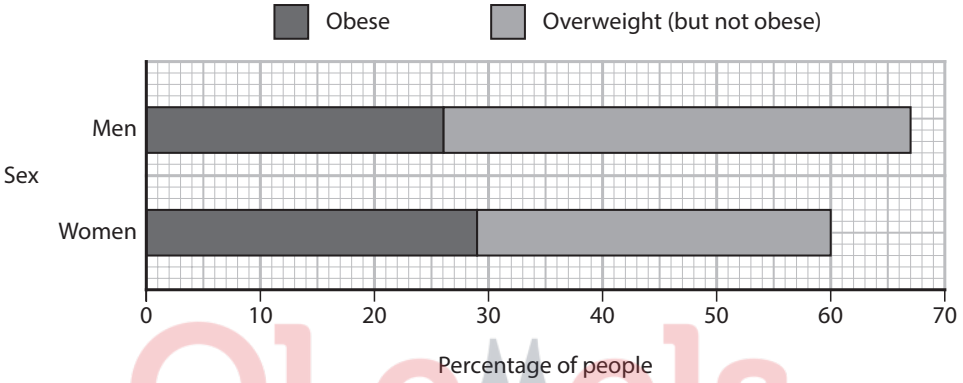
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(b) People with obesity are more at risk of heart disease than those who are overweight.

The chart shows the percentage of men and women in the United Kingdom who were either overweight or obese in 2018.



(Source: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2018/summary)

(i) In 2018 there were 66.5 million people living in the UK. Of these people, 33 million were men.

Calculate how many more women than men were classified as obese.

(3)

extra number of women = ...

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P 7 4 6 3 0 A 0 1 9 2 4

Turn over ▶

12. Question 7 · part b.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

(ii) Describe a calculation that could be used to determine if a person is obese.

(3)

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12. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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13. Question 3 · part b.i · 3 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

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.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

13. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

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14. Question 3 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

.....

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.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

14. Question 3 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

(1)

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

(2)

(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

(3)

(Total for Question 3 = 13 marks)

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14. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Describe a trend or compare data sets and conclude

Describing the pattern in one or more data series from a graph, table or chart, comparing groups or conditions, and drawing a conclusion the data support.

27 question parts: 1 worked-example retry and 26 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-DESCRIBE-COMPARE-TRENDS

1. Question 4 · part a.iv · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.iv only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

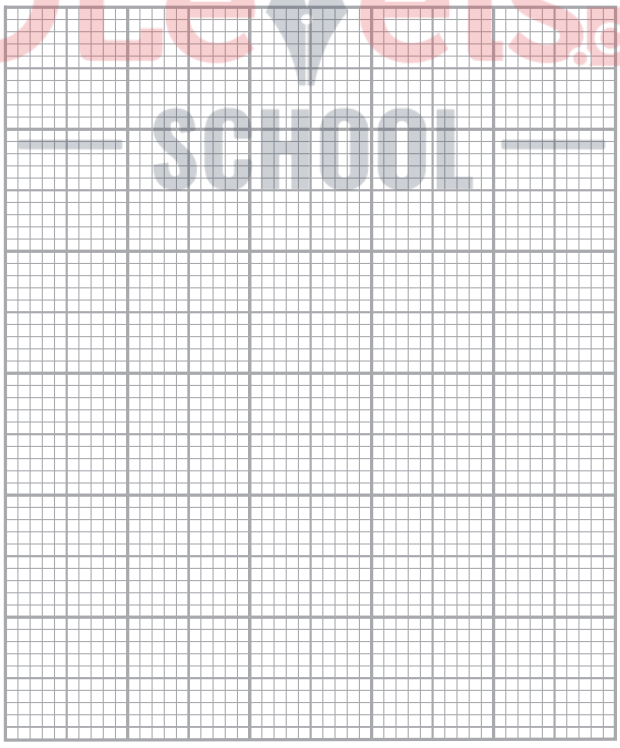
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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1. Question 4 · part a.iv · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.iv only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

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(vi) Name a part of the body where starch digestion takes place.

(1)

(b) Describe a test for starch in a sample of food.

(2)

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(Total for Question 4 = 13 marks)

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1. Question 4 · part a.iv · 2 marks · Working and answer

Working and answer

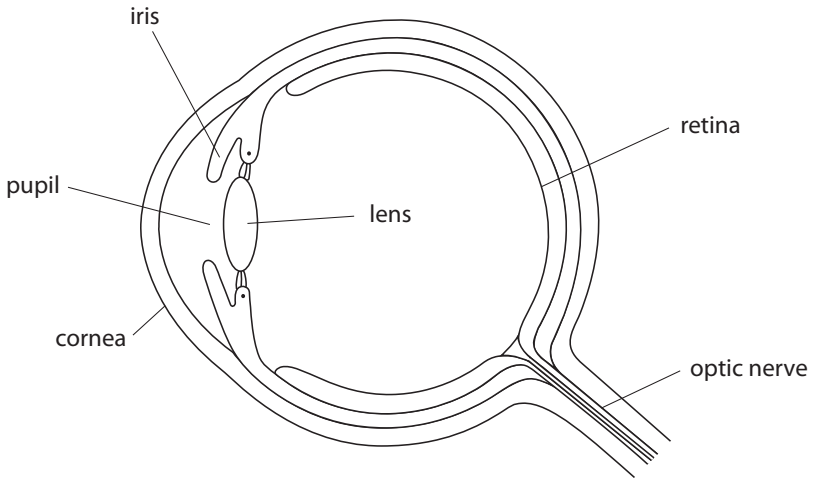
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2. Question 3 · part b.iv · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.iv only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part b.iv · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.iv only.

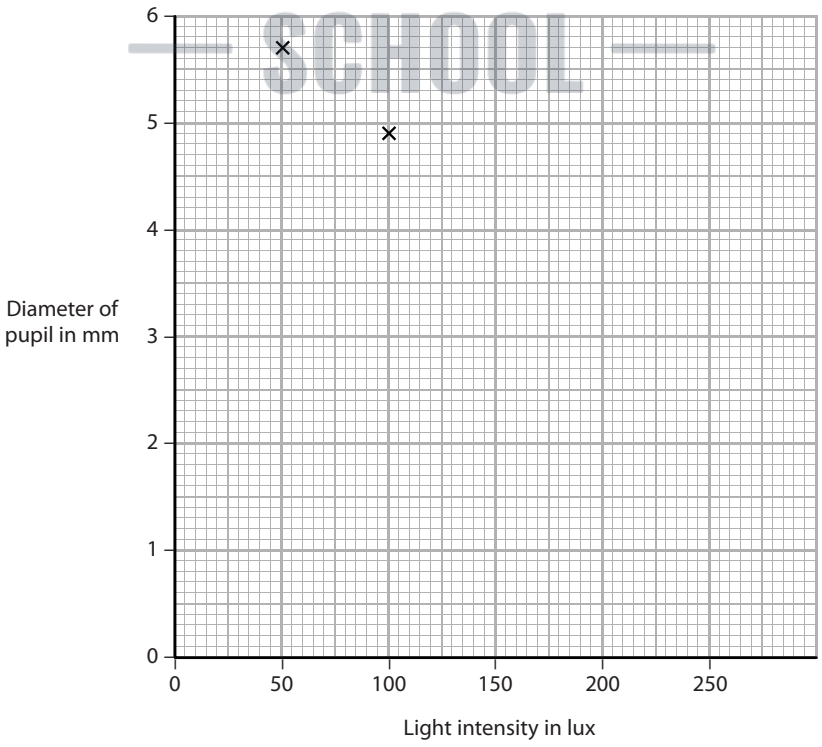
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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2. Question 3 · part b.iv · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.iv only.

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(iii) Explain which result is anomalous.

(2)

(iv) State the effect of light intensity on the diameter of the pupil.

(1)

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

(2)

(Total for Question 3 = 8 marks)

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Working and answer

3. Question 6 · part a.i · 1 mark

June 2019 Paper 1

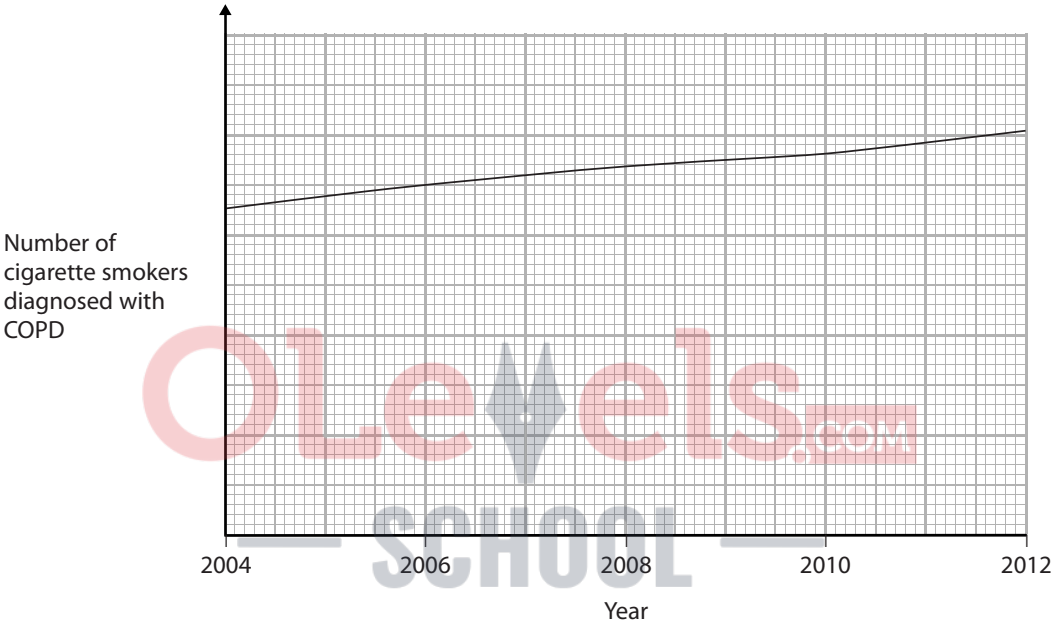
Further practice · Attempt part a.i only.

6 (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.



(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

(1)

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

(2)

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3. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 3 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

3 There are many factors that affect the height of a human.

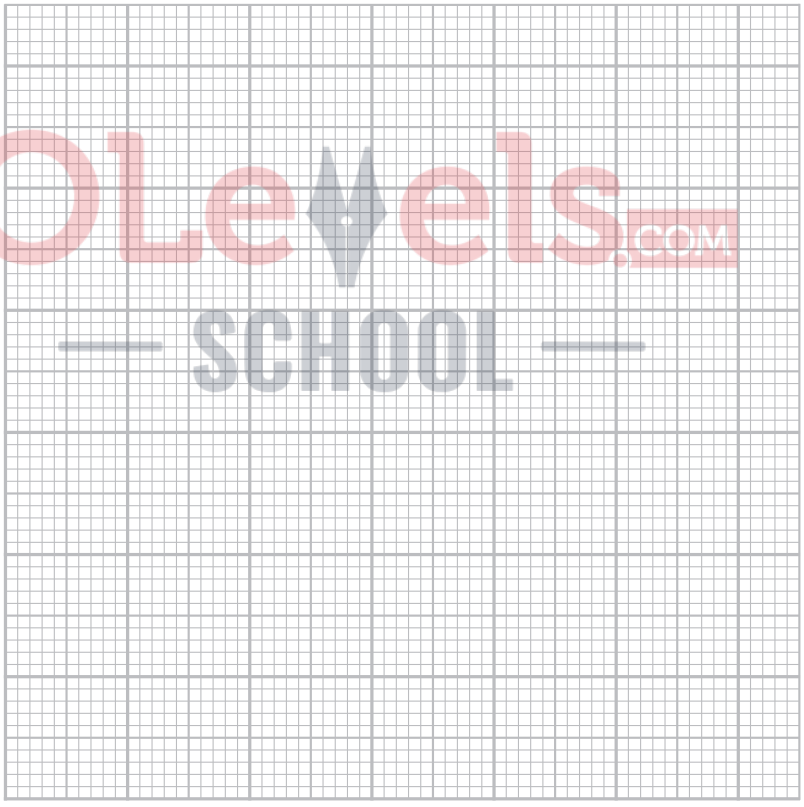
(a) The heights of a group of students are measured.

The table shows the results.

Height in cm	150–151	152–153	154–155	156–157	158–159	> 160
Number of students	4	14	25	20	10	4

(i) Plot a bar chart of the results.

(5)



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4. Question 3 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Describe the trend shown in this bar chart.

(2)

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(b) State three factors that could affect the heights of the students.

(3)

1 .....

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2 .....

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3 .....

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(Total for Question 3 = 10 marks)

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Working and answer

5. Question 3 · part c.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.i only.

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3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

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2 .....

.....

(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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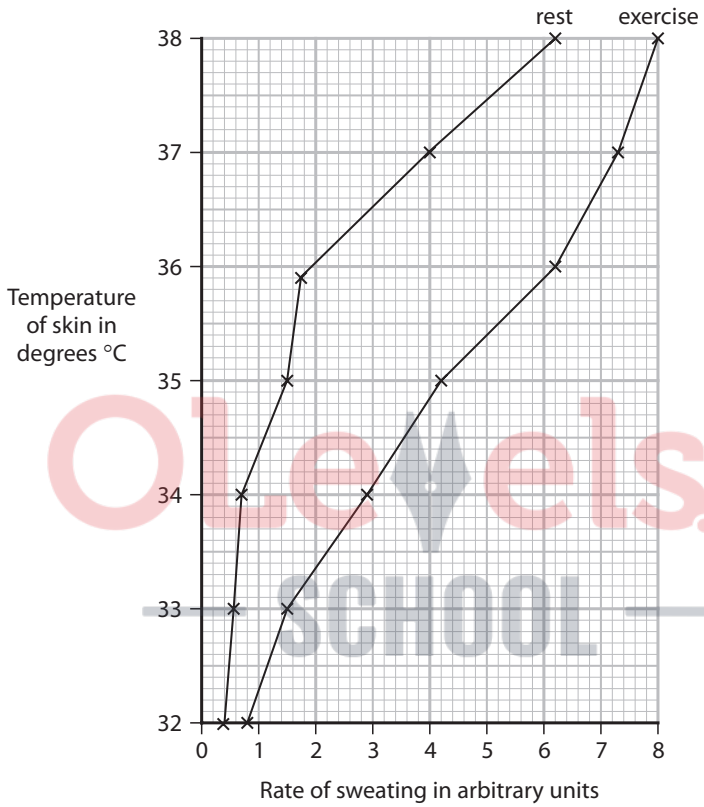
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5. Question 3 · part c.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.i only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



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5. Question 3 · part c.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.i only.

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(i) Describe how the skin temperature varies as the rate of sweating changes for the person at rest.

(2)

(ii) Suggest a method that could be used to measure the rate of sweating.

(2)

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(d) Explain the importance of maintaining a constant body temperature.

(3)

(Total for Question 3 = 13 marks)

5. Question 3 · part c.i · 2 marks · Working and answer

Working and answer

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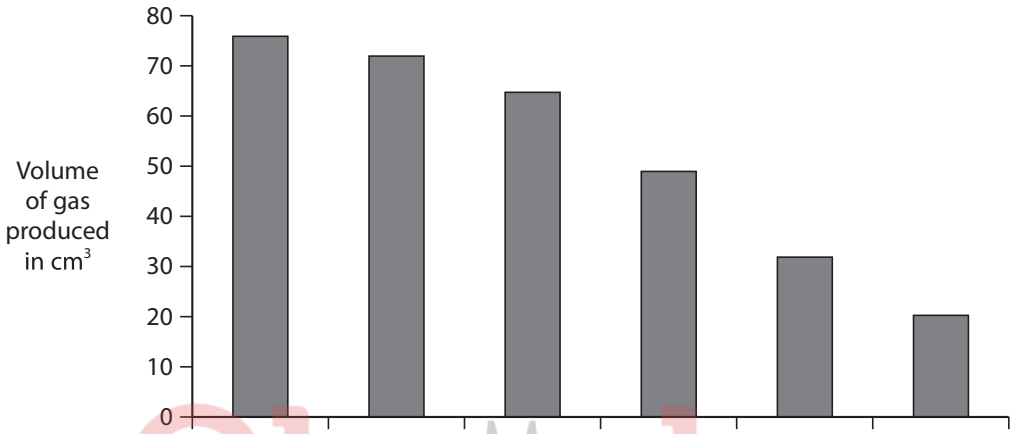
6. Question 3 · part a.iii · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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P 7 2 5 8 5 A 0 6 2 0

6. Question 3 · part a.iii · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(iv) Explain the student's results obtained in the investigation.

(3)

(b) Explain the effect on the results if the student had added a competitive inhibitor at the start of the investigation.

(2)

(Total for Question 3 = 12 marks)

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6. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

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7. Question 4 · part b.i · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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7. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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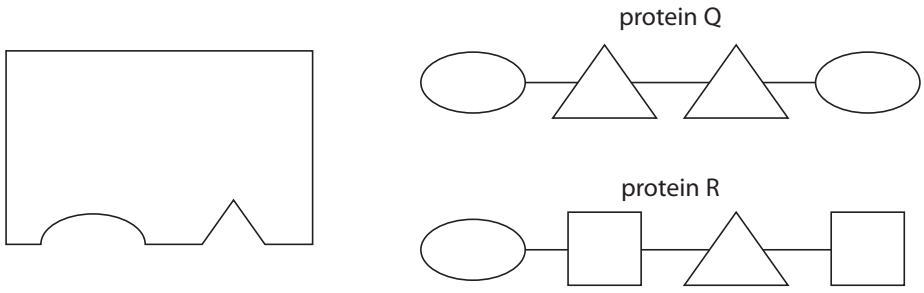
8. Question 2 · part b.i · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

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2 (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.



Enzyme P **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

(3)

amino acids	amylases	fatty acids	proteases
protein Q	protein R	starch	sugar

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

(1)

.....

(iii) Name the molecule that codes for the production of enzyme P.

(1)

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8. Question 2 · part b.i · 3 marks

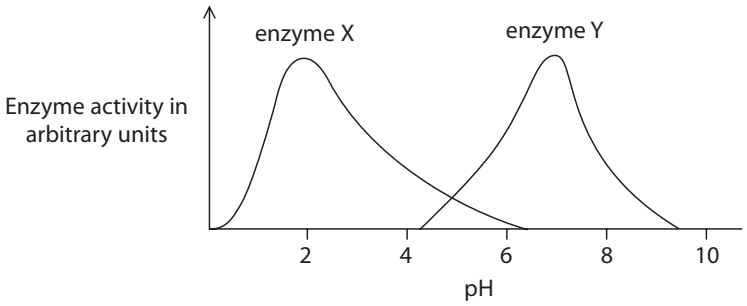
June 2023 Paper 2R

Further practice · Attempt part b.i only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.



(i) Describe how the activity of enzyme X changes with pH.

(3)

(ii) Explain why no activity is shown for enzyme Y at pH2.

(2)

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8. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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9. Question 6 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

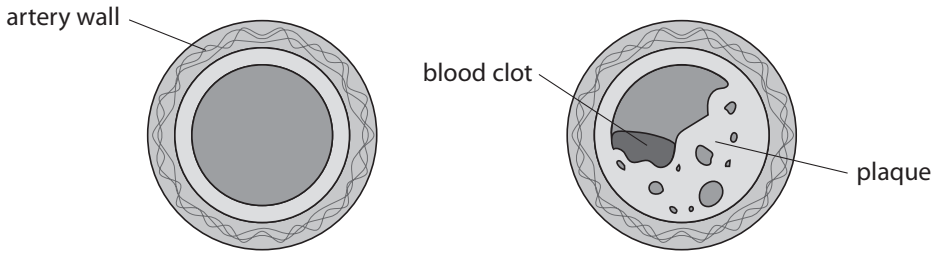
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6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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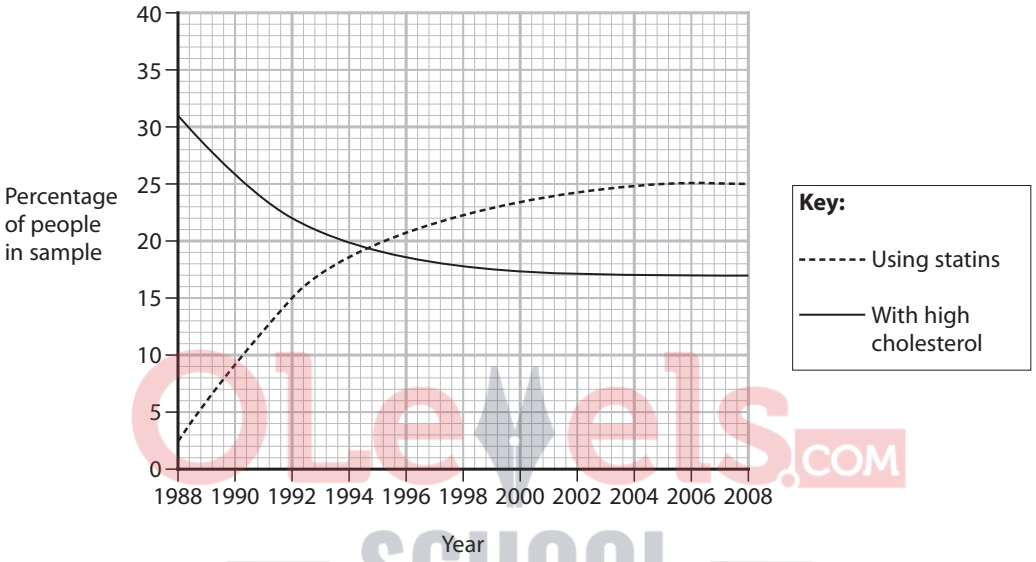
9. Question 6 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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P 7 4 4 9 5 A 0 1 8 2 8

9. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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10. Question 1 · part c.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

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1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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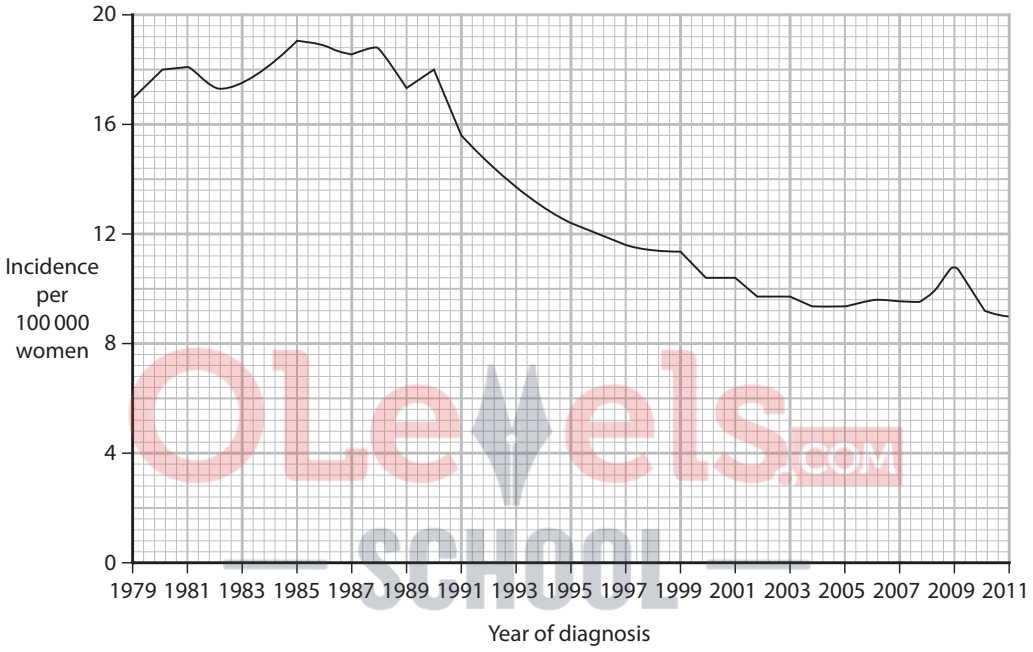
10. Question 1 · part c.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.



(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

(2)

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10. Question 1 · part c.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

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(ii) Calculate the difference in the incidence per 100 000 of women with cervical cancer between 1979 and 2011.

(2)

difference = ... per 100 000

(d) Cervical cancer that is caused by a virus can now be prevented by vaccination.

The table gives information about vaccination.

Which row gives correct information about vaccination?

(1)

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	Contents of vaccination	Response by the body	How the pathogen is destroyed
☒ **A**	antibodies	antigens produced	antigens destroy pathogen
☒ **B**	antibodies	white blood cells produced	white blood cells destroy pathogen
☒ **C**	antigens	antibodies produced	antibodies destroy pathogen
☒ **D**	antigens	platelets produced	platelets destroy pathogen

(Total for Question 1 = 12 marks)

Working and answer

11. Question 2 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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11. Question 2 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

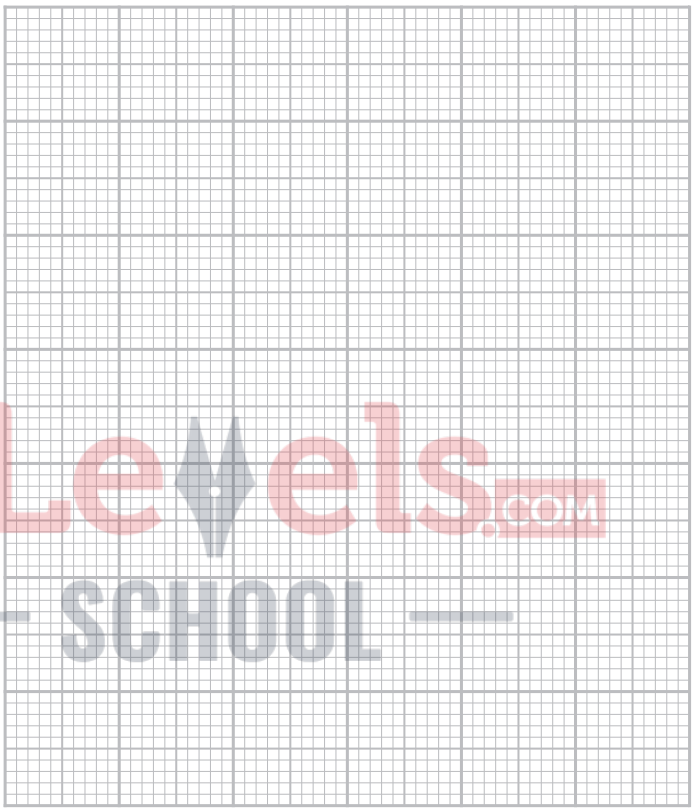
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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

.....

.....

11. Question 2 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

(3)

(c) Name the independent variable in this investigation.

(1)

(d) Give two variables that the student should control in this investigation.

(2)

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

(1)

(Total for Question 2 = 12 marks)

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11. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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12. Question 4 · part a.i · 2 marks

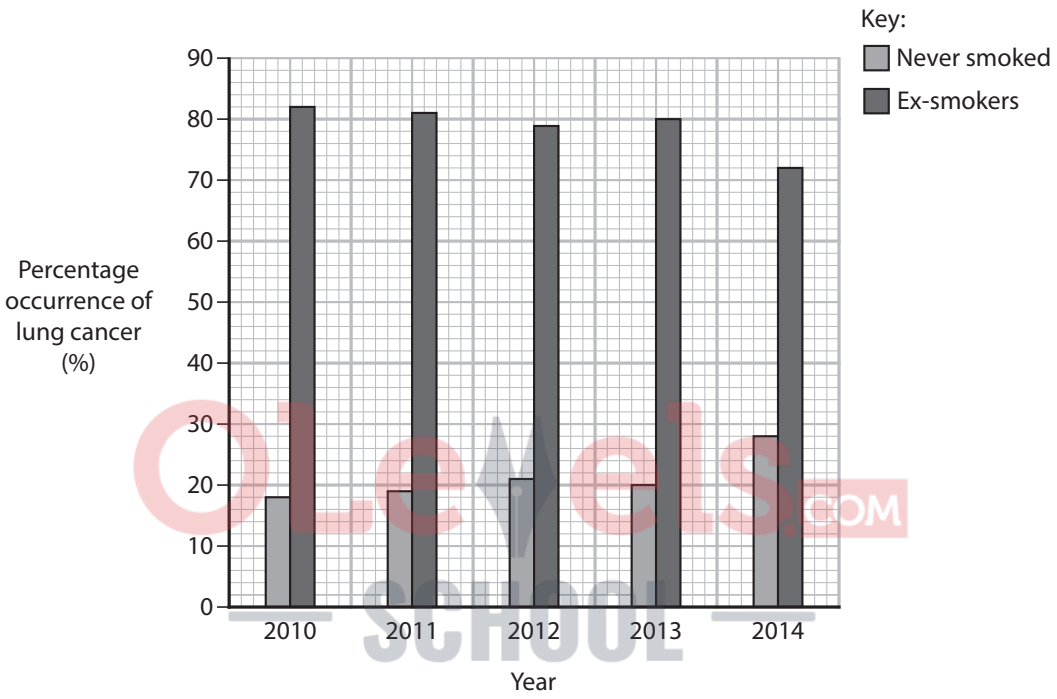
June 2025 Paper 2R

Further practice · Attempt part a.i only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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12. Question 4 · part a.i · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.i only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

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(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

12. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

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9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

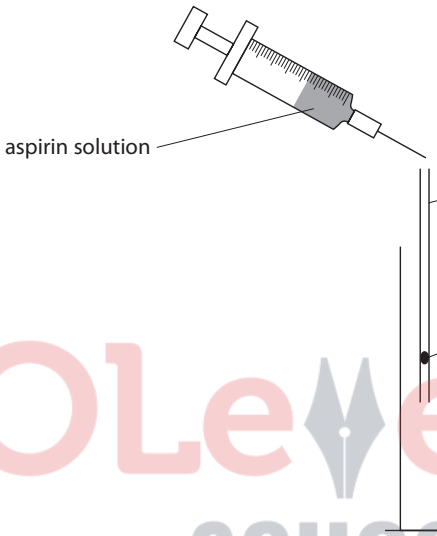
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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.



The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

(1)

(ii) Describe a suitable control test for this investigation.

(1)

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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

The table shows the student's results.

Time in minutes	Total volume of aspirin solution in the measuring cylinder in cm^3		
	1 aspirin tablet	2 aspirin tablets	3 aspirin tablets
5	0	0	0
10	0	0	0
15	0	0	1
20	0	1	2
25	1	1	3
30	1	2	4
35	2	4	6
40	5	7	9
45	10	12	16
50	17	19	22
55	25	27	31
60	35	37	39

(iii) Explain the pattern shown by the results.

(3)

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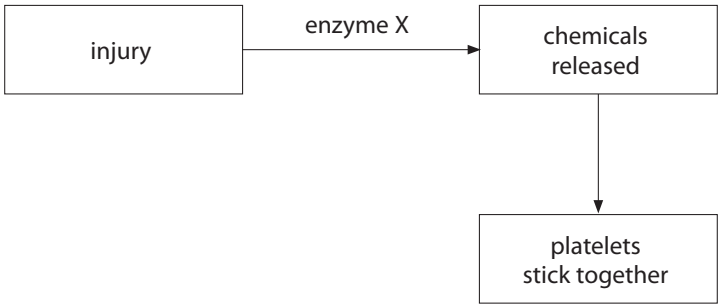
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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

(c) The diagram shows a pathway involved in blood clotting.



Aspirin is an inhibitor of enzyme X.

Explain how the inhibition of enzyme X reduces the formation of blood clots.

(4)

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(Total for Question 9 = 15 marks)

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13. Question 9 · part b.iii · 3 marks · Working and answer

Working and answer

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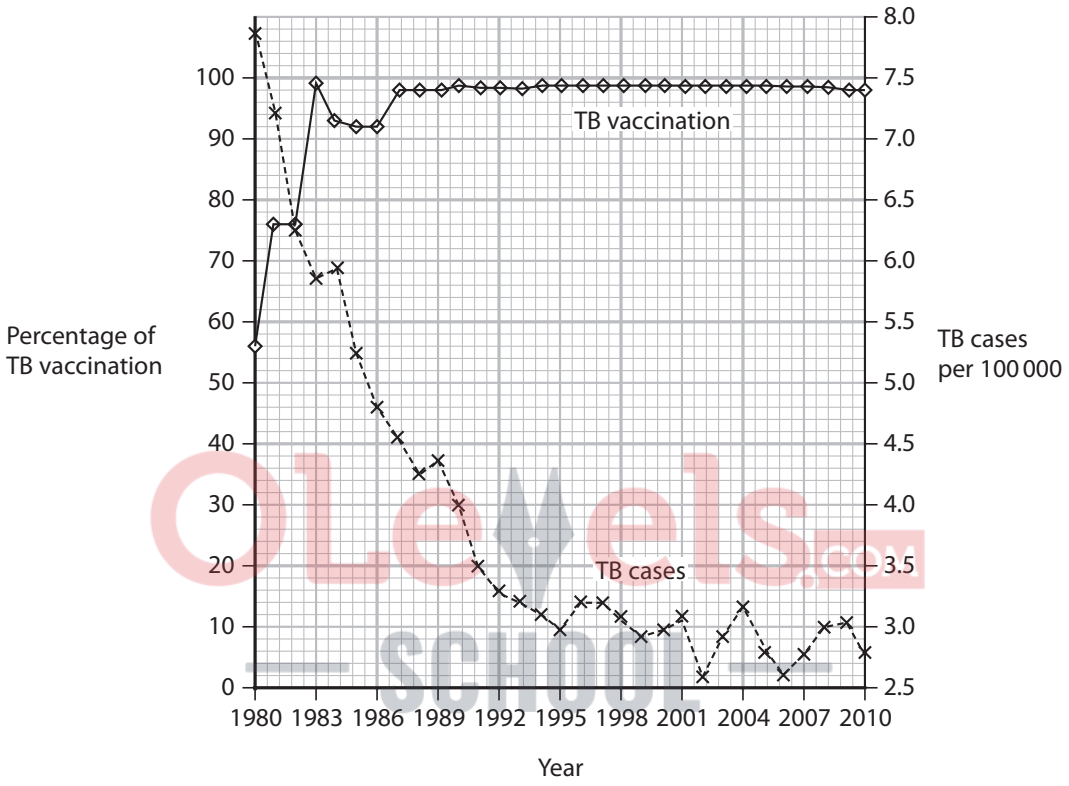
14. Question 4 · part a.ii · 4 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

4 The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.



(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

(1)

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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14. Question 4 · part a.ii · 4 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

(4)

(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

(3)

number of cases = ...

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14. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

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(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

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(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

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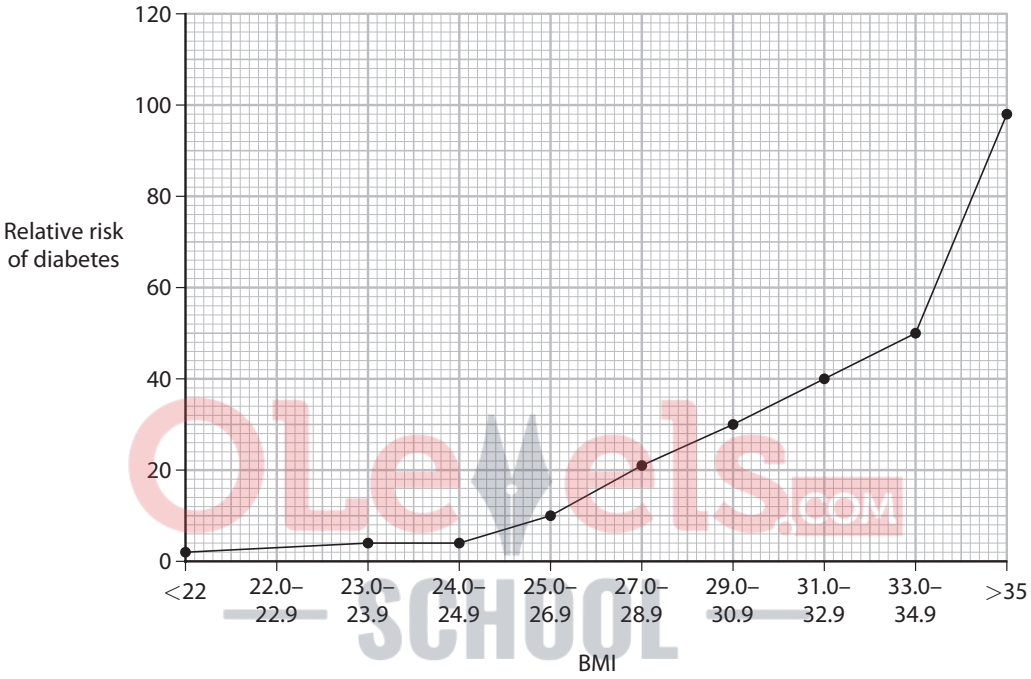
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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

(c) The graph shows the relative risk of developing a form of diabetes called type 2 for different BMI values.



(Source: adapted from Colditz GA, et al., *Ann Intern Med.*, 1995; 122:481–486)

(i) Describe the trend shown by the graph.

(2)

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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

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(ii) Give some dietary advice that could help an obese person lower their BMI.

(2)

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(Total for Question 3 = 11 marks)

Working and answer

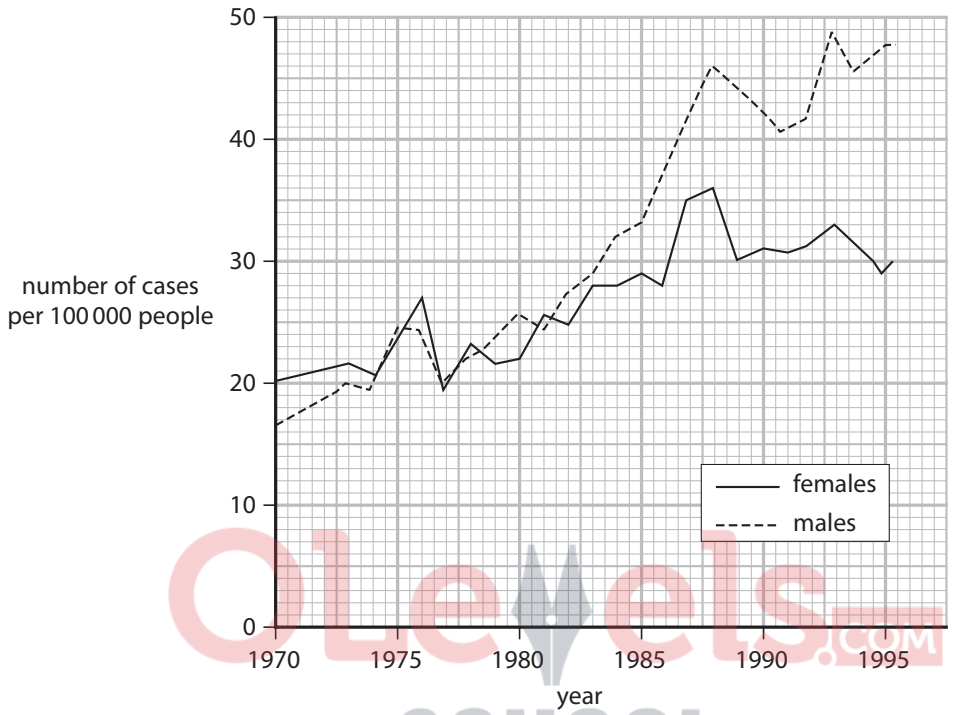
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16. Question 6 · part a.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

6 A form of skin cancer called melanoma has been increasing in both males and females.



(a) (i) Describe the trends shown by the graph.

(3)

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P 6 7 0 6 0 R R A 0 1 4 2 0

16. Question 6 · part a.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

(ii) Suggest why cancer can occur in the skin.

(2)

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

(3)

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mean increase per 100 000 males per year = ...

(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

(3)

number of females with melanoma = ...

(Total for Question 6 = 11 marks)

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16. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

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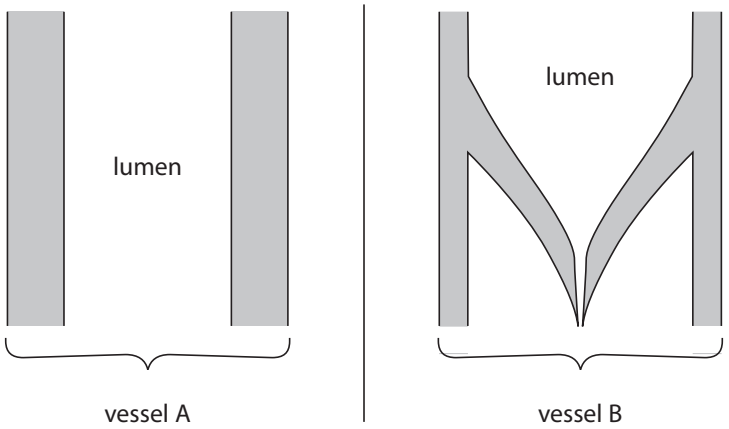
17. Question 3 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

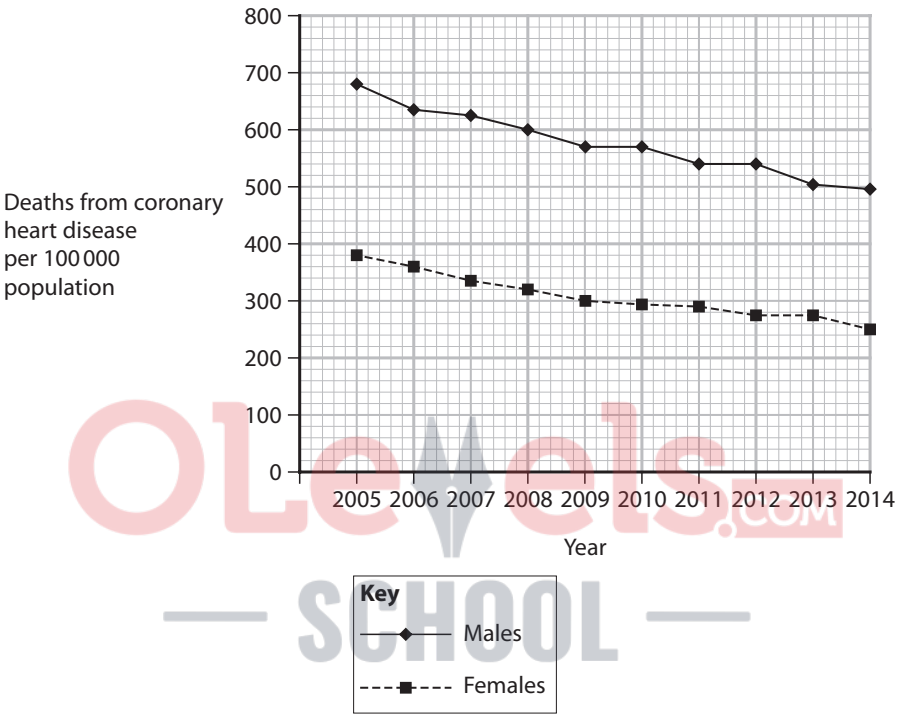
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17. Question 3 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

(b) Coronary heart disease kills many people each year. The graph shows the number of deaths per 100 000 of the whole population in a part of the world.



(i) Describe the trends shown by the graph.

(3)

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17. Question 3 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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(ii) The part of the world for which the data was collected had a population of 60 million people in 2006.

Calculate the number of men who died from coronary heart disease in 2006.

(3)

number of men who died = ...

(Total for Question 3 = 12 marks)

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Working and answer

18. Question 4 · part c.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part c.ii only.

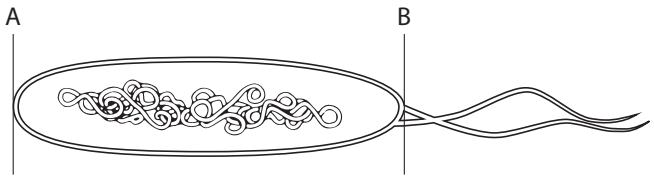
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4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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18. Question 4 · part c.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part c.ii only.

(c) The effectiveness of a vaccine in preventing disease was studied over six months.

Two different groups of people were used. One group of people were aged between 45 and 64 years and the second group were aged over 65 years.

The results are shown in the table.

	Percentage effectiveness against disease (%)	
Time since vaccination in months	**45 to 64 years**	**over 65 years**
1	90	80
2	85	78
3	80	75
4	68	58
5	60	50
6	50	40

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18. Question 4 · part c.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part c.ii only.

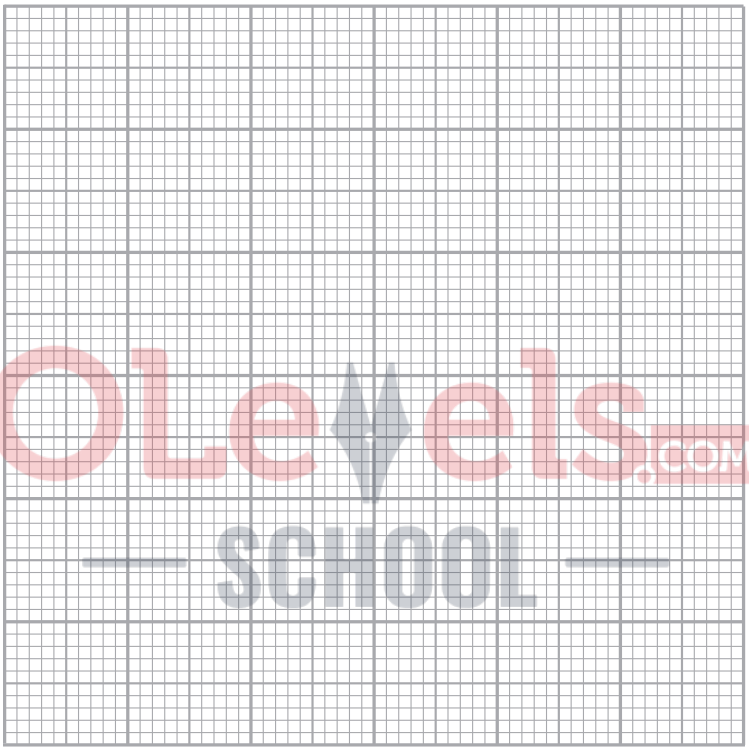
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(i) Plot a graph of the results. Join the points with straight lines.

(5)



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(ii) Describe the trends shown by this graph.

(3)

(Total for Question 4 = 15 marks)

18. Question 4 · part c.ii · 3 marks · Working and answer

Working and answer

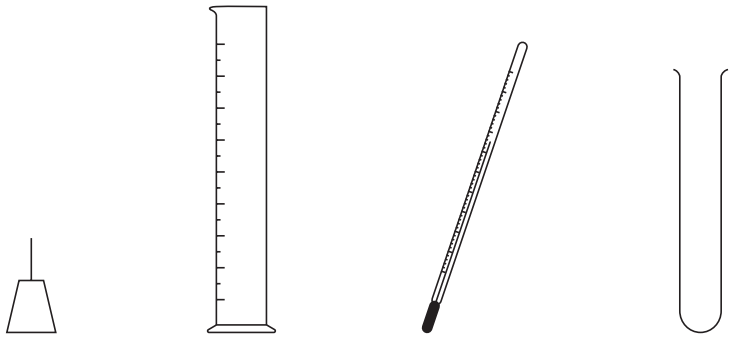
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19. Question 3 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

3 (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

(5)

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19. Question 3 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

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(ii) Name two variables that need to be controlled in the investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Give two safety precautions needed for this investigation.

(2)

1 .....

.....

2 .....

.....

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19. Question 3 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

Daily Energy Requirement in kcal

Age of adults in years	Males	Females
19–24	2772	2175
25–34	2749	2175
35–44	2629	2103
45–54	2581	2103
55–64	2581	2079
65–74	2342	1912
75 and over	2294	1840

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

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(i) Describe the patterns in the energy requirements for males and for females.

(2)

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

(1)

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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19. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

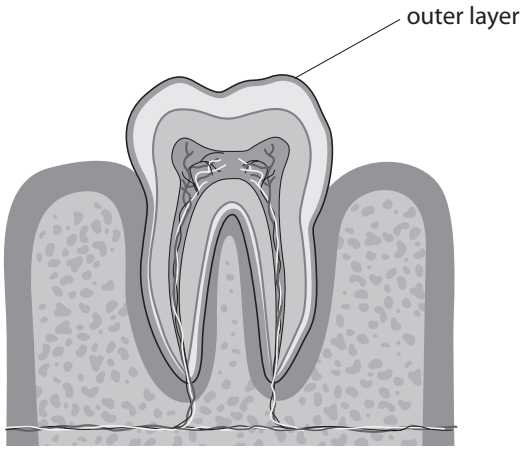
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20. Question 1 · part b.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.i only.

1 (a) The diagram shows a section through one type of tooth.



(i) State the name of this type of tooth.

(1)

(ii) Which substance forms the outer layer of the tooth?

(1)

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

(3)

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20. Question 1 · part b.i · 2 marks

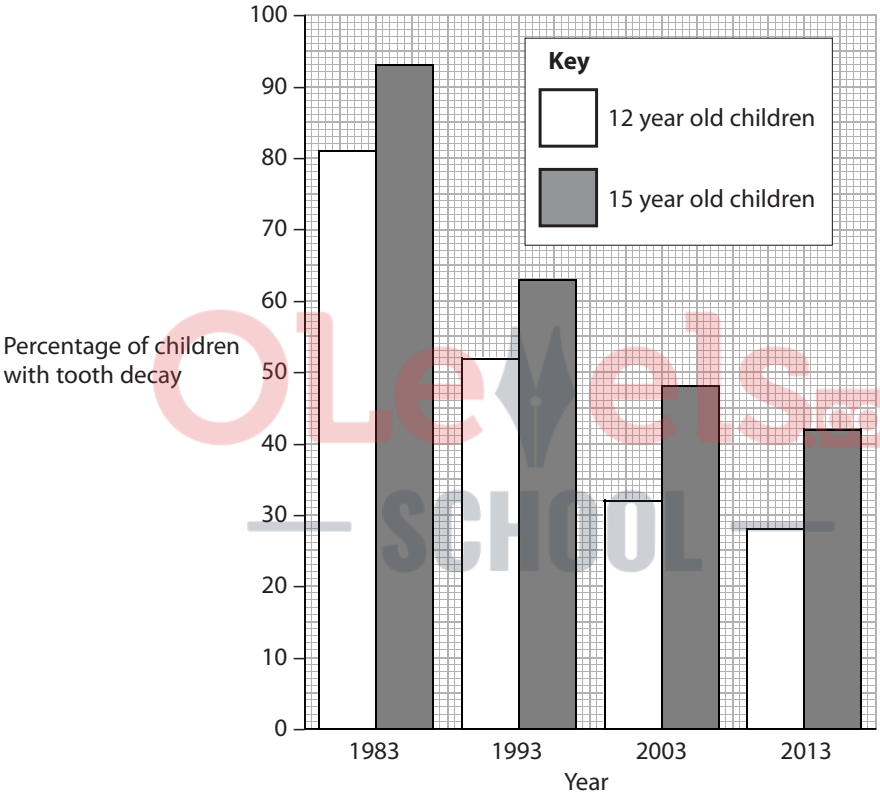
June 2019 Paper 2

Further practice · Attempt part b.i only.

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(b) The bar chart shows the percentage of children with tooth decay.

It gives this information for 12 and 15 year old children over a period of 30 years.



(i) Give two conclusions that can be made from the data in the bar chart.

(2)

1 .....

.....

2 .....

.....

(ii) Suggest a reason for the change in the percentage of children with tooth decay over the 30 year period.

(1)

.....

.....

(Total for Question 1 = 8 marks)

20. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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21. Question 7 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

.....

2 .....

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(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

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2 .....

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21. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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22. Question 6 · part a.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	MRSA	*S. aureus*	MRSA
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

1 .....

.....

.....

2 .....

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22. Question 6 · part a.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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22. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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23. Question 2 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

2 To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

Time in minutes	Blood glucose levels in mg per 100 cm^3	
	person without diabetes	person with diabetes
0	65	110
30	120	215
60	130	230
90	95	200
120	80	190
150	65	180
180	65	170

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23. Question 2 · part a.iii · 3 marks

November 2023 Paper 1

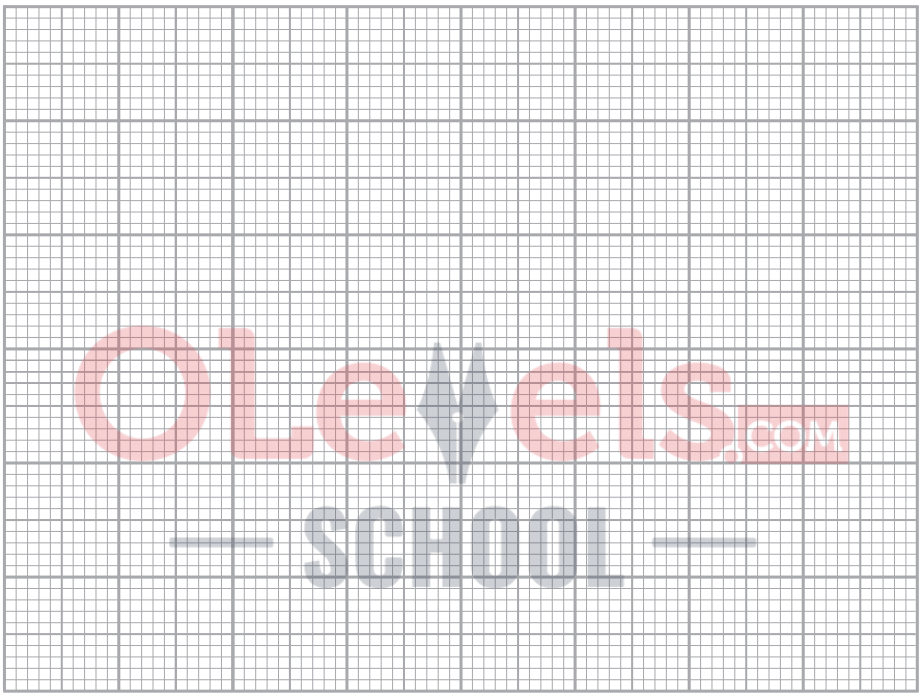
Further practice · Attempt part a.iii only.

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(a) (i) Plot the results on the grid.

Join the points with straight lines.

(6)



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

(2)

without diabetes = ... mg per 100 cm^3

with diabetes = ... mg per 100 cm^3

23. Question 2 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

(iii) Give **three** differences between the glucose levels for the two people during the three hours.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(b) One symptom of diabetes is glucose in the urine.

Describe how a sample of urine could be tested safely for glucose.

(5)

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(Total for Question 2 = 16 marks)

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23. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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24. Question 3 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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24. Question 3 · part b.i · 3 marks

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Further practice · Attempt part b.i only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

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(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

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(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

(Total for Question 3 = 15 marks)

24. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

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25. Question 3 · part e · 2 marks

November 2023 Paper 2

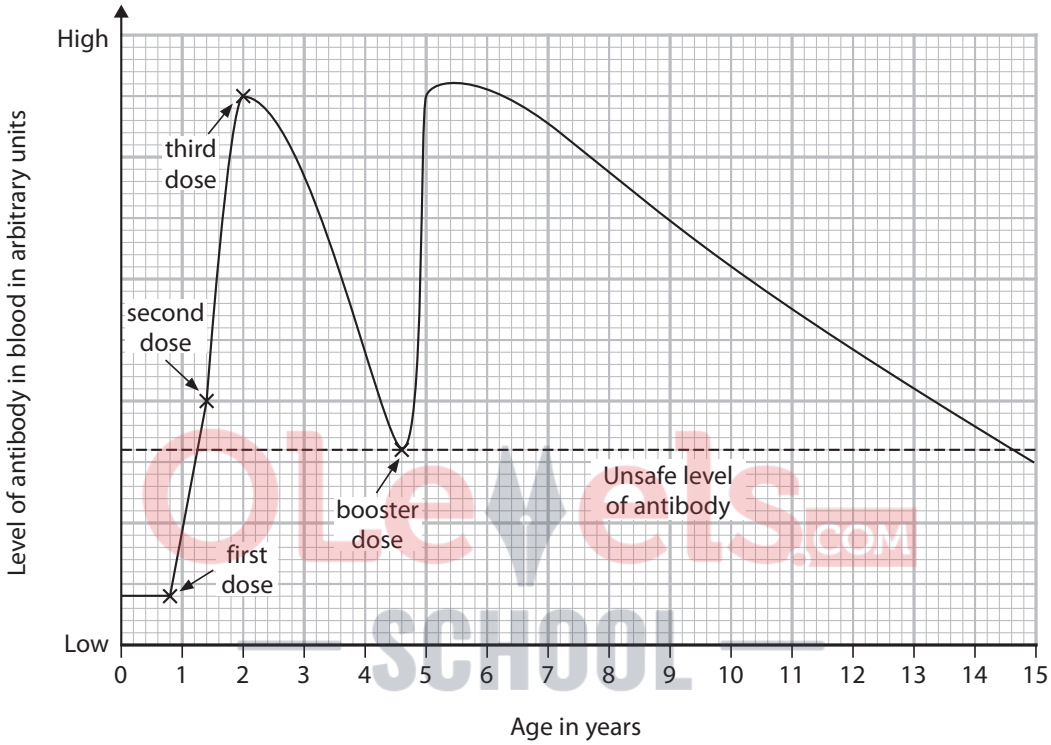
Further practice · Attempt part e only.

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3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

.....

.....

P 7 4 4 9 4 A 0 5 2 0

25. Question 3 · part e · 2 marks

November 2023 Paper 2

Further practice · Attempt part e only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

(2)

1 .....

2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

(2)

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(d) Explain why a booster dose is necessary.

(5)

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P 7 4 4 9 4 A 0 6 2 0

25. Question 3 · part e · 2 marks

November 2023 Paper 2

Further practice · Attempt part e only.

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(e) Give two differences between the effects of the first three doses and the effect of the booster dose.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 15 marks)

Working and answer

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26. Question 3 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

3 A woman who smokes when she is pregnant risks harming her developing fetus.

The table lists the birth masses of 310 children born to non-smoking mothers or to mothers who smoke.

	Birth masses of children in kg					
	Less than 2.4	**2.4 – 3.1**	**3.2 – 3.5**	**3.6 – 3.9**	**More than 3.9**	**Total number of births in sample**
Number of children born to non-smoking mothers	4	42	70	60	34	210
Number of children born to mothers who smoke	3	24	44	23	6	100

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(a) (i) What is the most common birth mass in kg? **(1)**

- ☒ **A** 2.4 – 3.1
- ☒ **B** 3.2 – 3.5
- ☒ **C** 3.6 – 3.9
- ☒ **D** More than 3.9

(ii) Calculate the percentage of children with a birth mass of 3.6 kg or more in the sample of non-smoking mothers. **(2)**

percentage = ...

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P 6 0 1 8 7 A 0 8 2 4

26. Question 3 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

(b) Use data from the table to deduce what can be concluded about the effect of smoking on birth mass.

(2)

(c) Suggest why smoking affects birth mass.

(5)

(Total for Question 3 = 10 marks)

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26. Question 3 · part b · 2 marks · Working and answer

Working and answer

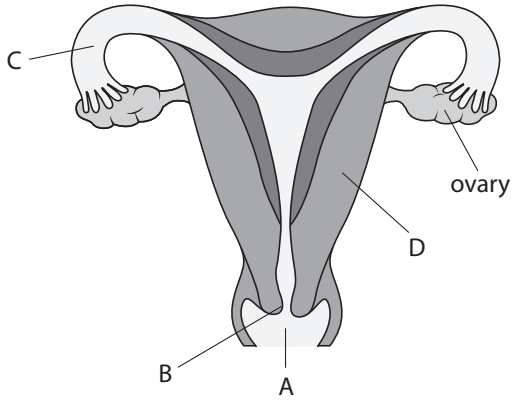
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27. Question 2 · part d.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part d.ii only.

2 (a) The diagram shows some structures of the female reproductive system.



(i) Where are sperm deposited during sexual intercourse?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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27. Question 2 · part d.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part d.ii only.

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(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

FSH	wall	fertilisation	oestrogen	ovulation
lining	ADH	follicles	menstruation	

Use words from the box to complete the passage about the ovaries.

(3)

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

(1)

.....

.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

(1)

.....

.....

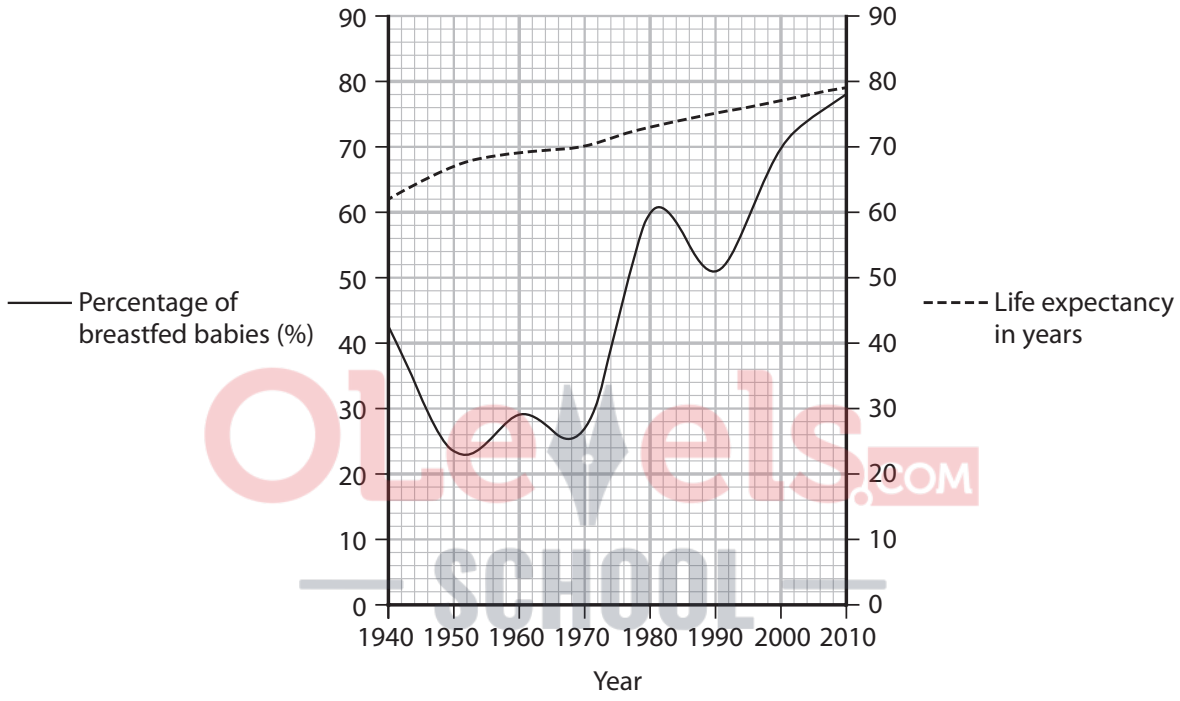
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27. Question 2 · part d.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part d.ii only.

(d) The graph shows the percentage of babies that are breastfed and the life expectancy of breastfed babies from 1940 to 2010 in the UK.



(i) Compare the percentage of babies that were breastfed in 1940 with the percentage of babies that were breastfed in 2010.

(1)

.....

.....

(ii) The graph suggests that breastfeeding has no effect on life expectancy.

Describe how the graph shows this.

(2)

.....

.....

.....

.....

(Total for Question 2 = 10 marks)

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27. Question 2 · part d.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Judge whether data support a claim and spot their limits

Health and investigation data where the student decides if evidence supports a claim, why an estimate is uncertain, which result is anomalous, or why data are unreliable or incomplete.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INTERPRET-HEALTH-DATA-CLAIMS

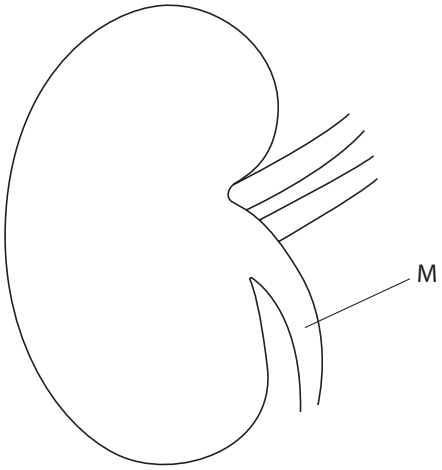
1. Question 7 · part c.iii · 4 marks

June 2022 Paper 2R

Worked-example retry · Attempt part c.iii only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

1. Question 7 · part c.iii · 4 marks

June 2022 Paper 2R

Worked-example retry · Attempt part c.iii only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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1. Question 7 · part c.iii · 4 marks

June 2022 Paper 2R

Worked-example retry · Attempt part c.iii only.

(ii) Which is the main nitrogen containing excretory product produced from the breakdown of protein in the diet?

Use the information in the table to explain your answer.

(2)

(iii) The breakdown of muscle tissue continues at a constant rate and is not affected by diet.

Discuss whether the information in the table supports this statement.

(4)

(Total for Question 7 = 14 marks)

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1. Question 7 · part c.iii · 4 marks · Working and answer

Working and answer

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2. Question 9 · part a.ii · 1 mark

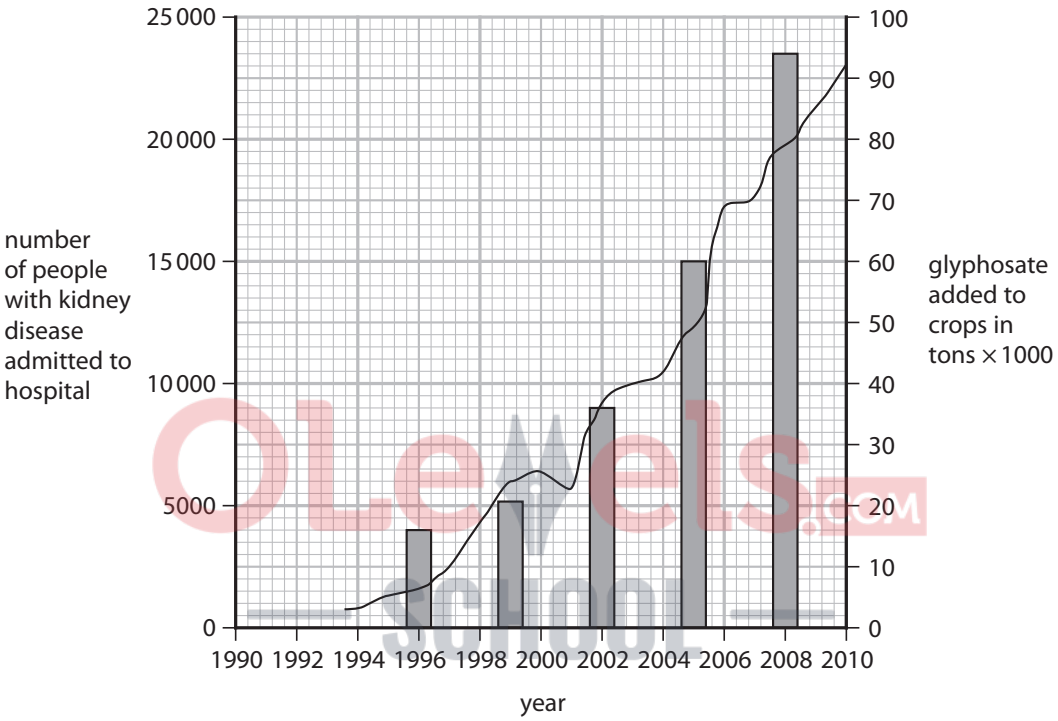
November 2020 Paper 1R

Further practice · Attempt part a.ii only.

9 (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.



(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

(3)

percentage increase = ... %

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P 6 2 0 5 2 A 0 2 0 2 4

2. Question 9 · part a.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

(1)

.....

.....

(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

(4)

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2. Question 9 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

7 A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 cm^3 per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GFR in cm^3 per minute	80	75	79	79	67	60	50	45	35

(a) (i) Plot a line graph of the data on the grid.

(4)



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3. Question 7 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

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(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

(2)

(iii) A person needs dialysis if the GFR falls to 15 cm^3 per minute.

Use the data to estimate when the person will probably need to start dialysis.

(2)

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(iv) Discuss the accuracy of this estimate.

(3)

3. Question 7 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

(b) During the early stages when a person's kidneys are not working correctly, they are advised to control the amount of protein in their diet.

Explain why the protein in their diet should be controlled.

(3)

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(Total for Question 7 = 14 marks)

3. Question 7 · part a.iv · 3 marks · Working and answer

Working and answer

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4. Question 6 · part a.ii · 2 marks

June 2019 Paper 1

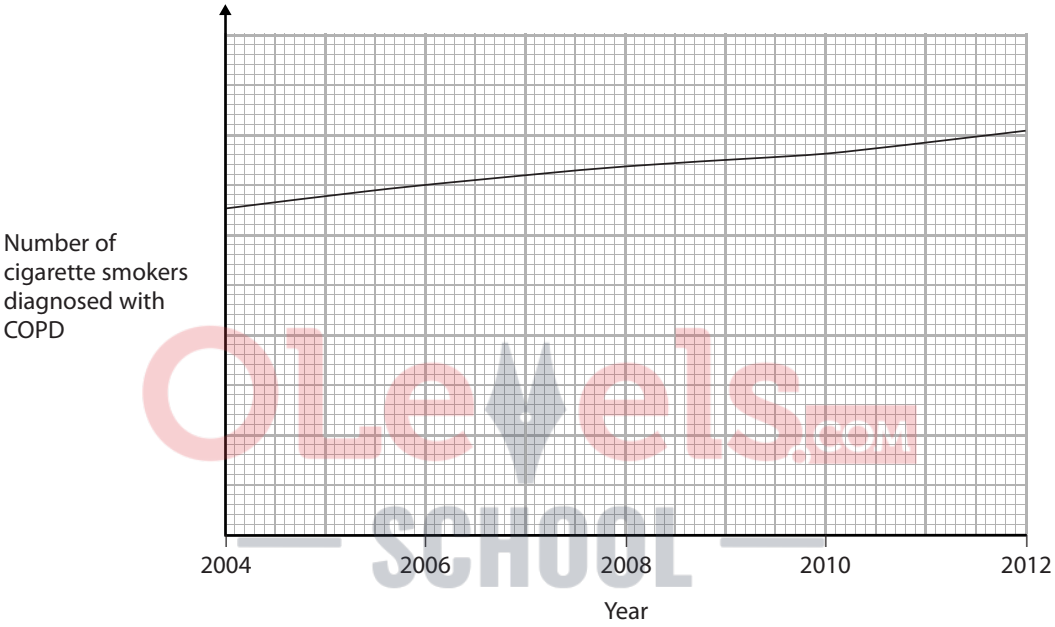
Further practice · Attempt part a.ii only.

6 (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.



(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

(1)

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

(2)

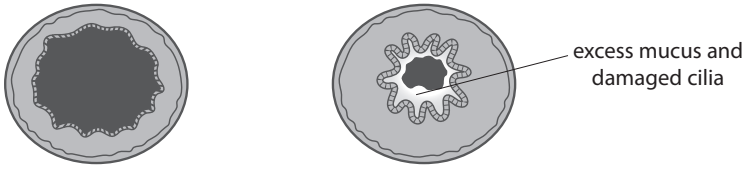
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4. Question 6 · part a.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.ii only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.



healthy airway

chronic bronchitis

Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

(2)

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4. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 7 · part a.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

.....

2 .....

.....

(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

.....

2 .....

.....

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5. Question 7 · part a.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

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(b) (i) Individuals who are overweight or obese have an increased risk of diabetes.

People with diabetes can have high blood glucose levels.

Explain how blood glucose levels are reduced in a person who does not have diabetes.

(3)

(ii) Suggest why one symptom of diabetes is frequent urination.

(2)

(Total for Question 7 = 11 marks)

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5. Question 7 · part a.iii · 2 marks · Working and answer

Working and answer

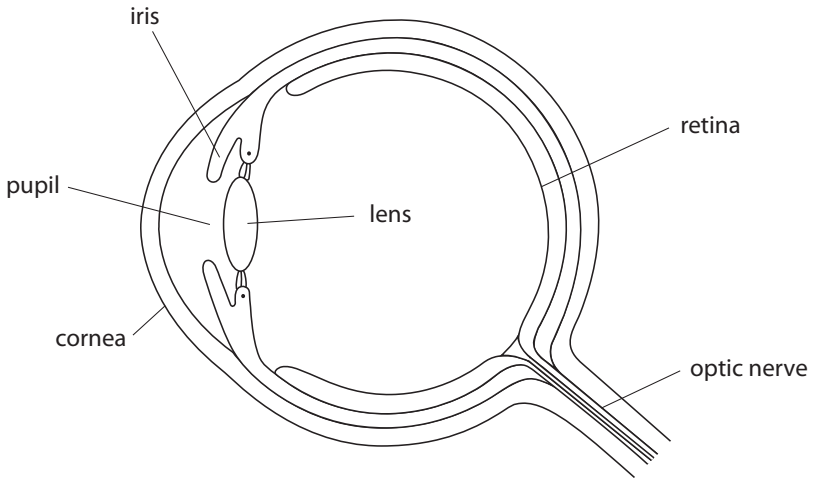
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6. Question 3 · part b.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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6. Question 3 · part b.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

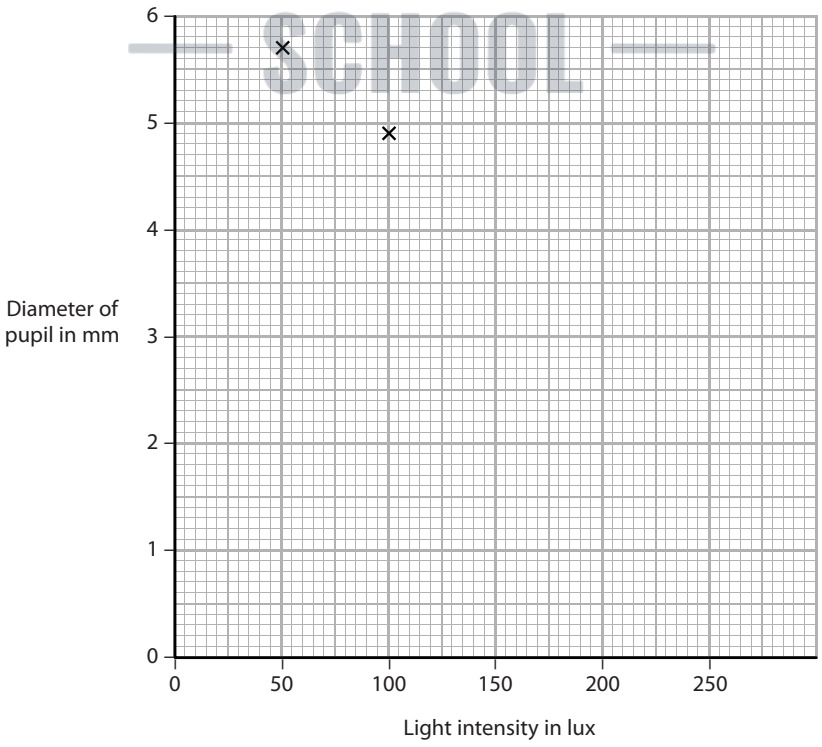
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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6. Question 3 · part b.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

(iii) Explain which result is anomalous.

(2)

(iv) State the effect of light intensity on the diameter of the pupil.

(1)

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

(2)

(Total for Question 3 = 8 marks)

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6. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

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7. Question 7 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

7 A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 cm^3 per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GFR in cm^3 per minute	80	75	79	79	67	60	50	45	35

(a) (i) Plot a line graph of the data on the grid.

(4)



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7. Question 7 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

(2)

(iii) A person needs dialysis if the GFR falls to 15 cm^3 per minute.

Use the data to estimate when the person will probably need to start dialysis.

(2)

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(iv) Discuss the accuracy of this estimate.

(3)

P 7 2 5 8 5 A 0 1 9 2 0

7. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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8. Question 6 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

6 (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

Region	Total number of people tested	Percentage of colour blind males	Percentage of colour blind females
A	150	10.00	1.00
B	16 000	6.60	0.40
C	5000	0.80	0.01

(i) Calculate the number of colour blind males in the test group in region C.

(3)

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number of males = ...

(ii) State why the results from region A are not considered to be reliable.

(1)

.....

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8. Question 6 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

(iii) Justify the statement that colour blindness is a recessive sex-linked condition.

(5)

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8. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Scale per-unit or reference quantities and combine them

A content per 100 g, per kg, per unit or per pot is given; amounts for another mass are found, combined, or used to find how much is needed.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-REFERENCE-RATIO-SCALING

1. Question 2 · part a · 3 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

2 The table gives the nutritional information for a pot of yoghurt.

	Mass per 100 g of yoghurt	Mass per 85 g pot
fat	2.7 g	2.3 g
carbohydrate	12.6 g	10.7 g
protein	6.1 g	
calcium	149 mg	127 mg
vitamin D	0.9 µg	0.8 µg

(a) Calculate the mass of protein in one pot of yoghurt.

Give your answer to two significant figures.

(3)

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mass of protein = ... g

(b) Describe how the yoghurt could be tested to show that protein is present.

(3)

.....

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1. Question 2 · part a · 3 marks · Working and answer

Working and answer

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2. Question 8 · part a · 4 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

8 The table shows the rate of blood flow through different organs when a person is at rest. The table also shows the masses of these organs.

Organ	Mass in kg	Rate of blood flow in cm^3 per minute	
		through the whole organ	per 100g of the organ
kidneys	0.3	1200	
skeletal muscles	25.0		3
liver	1.5	1500	100

(a) Calculate the missing values and put them in the table.

(4)

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(b) Explain why the liver has a high rate of blood flow.

(4)

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2. Question 8 · part a · 4 marks · Working and answer

Working and answer

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3. Question 3 · part a.v · 4 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
MALES					
15 to 18 years old	12 600	59	0.9	50	12
25 to 50 years old	12 200	63	0.9	60	10
FEMALES					
15 to 18 years old	9200	46	0.7	50	15
25 to 50 years old	9200	50	0.7	60	15

(i) Describe what is meant by a balanced diet.

(2)

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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3. Question 3 · part a.v · 4 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

(2)

.....

.....

.....

.....

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

(1)

.....

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

(4)

extra mass = ... mg

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3. Question 3 · part a.v · 4 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

(b) In the 19th century more than two million sailors on long ocean voyages died from a nutritional disease called scurvy.

(i) Give a symptom of scurvy.

(1)

.....

.....

(ii) Suggest why scurvy is an uncommon nutritional disease in the modern world.

(1)

.....

.....

(Total for Question 3 = 14 marks)

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Working and answer

4. Question 5 · part e · 3 marks

June 2024 Paper 2R

Further practice · Attempt part e only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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4. Question 5 · part e · 3 marks

June 2024 Paper 2R

Further practice · Attempt part e only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

(1)

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

(1)

.....

(ii) Name a component of the blood that is involved in blood clotting.

(1)

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

(3)

number of red blood cells = ...

(Total for Question 5 = 10 marks)

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4. Question 5 · part e · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 7 · part a.iii · 3 marks

June 2022 Paper 2

Further practice · Attempt part a.iii only.

7 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

(2)

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

(2)

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5. Question 7 · part a.iii · 3 marks

June 2022 Paper 2

Further practice · Attempt part a.iii only.

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(iii) Calculate the mass, in grams, of boiled broccoli that should be eaten to provide the recommended daily intake of vitamin C. (lines 12 to 14)

(3)

mass = ... g

(b) DCPIP is a chemical used to determine the amount of vitamin C in food.

Describe a method, using DCPIP, to compare the amount of vitamin C in fresh orange juice with that in boiled orange juice.

(5)

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(Total for Question 7 = 12 marks)

5. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

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6. Question 2 · part b.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

2 (a) Describe what is meant by a balanced diet.

(2)

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

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6. Question 2 · part b.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

(3)

mass = ... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

(2)

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number of slices = ...

6. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 2 · part c · 2 marks

June 2023 Paper 2

Further practice · Attempt part c only.

2 The table gives the nutritional information for a pot of yoghurt.

	Mass per 100 g of yoghurt	Mass per 85 g pot
fat	2.7 g	2.3 g
carbohydrate	12.6 g	10.7 g
protein	6.1 g	
calcium	149 mg	127 mg
vitamin D	0.9 µg	0.8 µg

(a) Calculate the mass of protein in one pot of yoghurt.

Give your answer to two significant figures.

(3)

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mass of protein = ... g

(b) Describe how the yoghurt could be tested to show that protein is present.

(3)

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7. Question 2 · part c · 2 marks

June 2023 Paper 2

Further practice · Attempt part c only.

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(c) A pot of yoghurt provides 16% of a person's daily requirement of calcium.

Calculate how many pots of yoghurt a person would need to eat to obtain their daily requirement of calcium.

(2)

number of pots = ...

(d) Explain why yoghurt is good for bone development.

(5)

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(Total for Question 2 = 13 marks)

7. Question 2 · part c · 2 marks · Working and answer

Working and answer

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8. Question 2 · part b.iii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

2 (a) Describe what is meant by a balanced diet.

(2)

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

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P 7 5 9 5 9 A 0 4 2 4

8. Question 2 · part b.iii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

(3)

mass = ... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

(2)

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number of slices = ...

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8. Question 2 · part b.iii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

(iv) When meat is rubbed on a piece of paper it leaves a greasy mark due to the presence of fat in the meat.

Describe another test to show the presence of fat in the meat.

(5)

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(Total for Question 2 = 13 marks)

8. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

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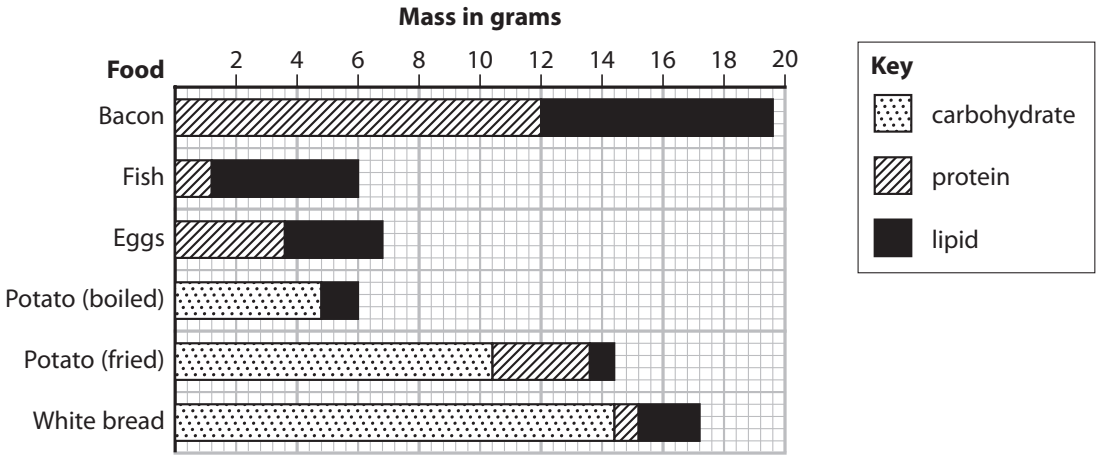
9. Question 7 · part a.iv · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iv only.

7 Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.



(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

(1)

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

(1)

(iii) Deduce the mass of lipid in 26 g of bacon.

(2)

mass of lipid = ... g

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P 6 7 0 5 9 A 0 2 0 2 4

9. Question 7 · part a.iv · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iv only.

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(iv) The energy in one gram of protein is 17 kJ and the energy in one gram of lipid is 37 kJ.

Calculate the total energy in 26 g of fish.

(3)

total energy = ... kJ

(b) Describe how to test for the presence of lipid in the fish.

(3)

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(Total for Question 7 = 10 marks)

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9. Question 7 · part a.iv · 3 marks · Working and answer

Working and answer

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10. Question 6 · part b.i · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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10. Question 6 · part b.i · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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10. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

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Practice workbook · Substitute into a given equation, rearranging and converting units

A physical-style relationship such as speed, distance and time, energy from heating water or force times distance, cardiac output or reaction time is given or implied.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-SUBSTITUTE-FORMULA

1. Question 3 · part b.iv · 3 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b.iv only.

3 (a) Describe what is meant by a specialised cell.

(2)

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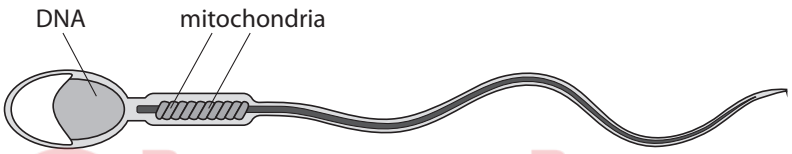
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE



(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

(2)

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

(1)

.....

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1. Question 3 · part b.iv · 3 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b.iv only.

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(iii) Explain why the sperm cell has many mitochondria.

(2)

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(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

(3)

time taken = ... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

(2)

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(Total for Question 3 = 12 marks)

1. Question 3 · part b.iv · 3 marks · Working and answer

Working and answer

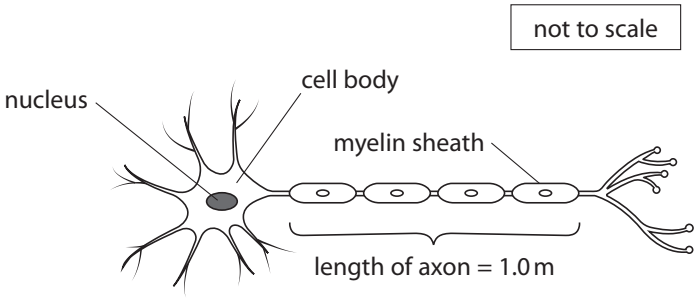
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2. Question 5 · part b.ii · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

5 The diagram shows a motor neurone.



(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

(2)

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diameter = ... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

(1)

.....

.....

(b) Nerve impulses travel at an average speed of 1.2×10^8 µm per second.

Hormones travel at an average speed of 1.5×10^6 µm per second.

(i) Explain this difference in speed.

(2)

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2. Question 5 · part b.ii · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Calculate the time taken for a nerve impulse to travel along the axon of the motor neurone.

(3)

time taken = ... s

(Total for Question 5 = 8 marks)

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Working and answer

3. Question 3 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

3 (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

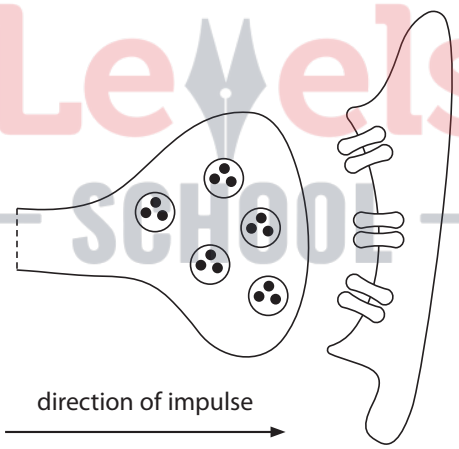
Complete the passage about the nervous system by giving the correct words.

(4)

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system.

The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.



(i) State the name of the junction between two neurones.

(1)

.....

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3. Question 3 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(ii) Describe how an impulse reaching the end of one neurone produces an impulse in the next neurone.

(2)

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(c) A nerve impulse travels at a speed of 120 metres per second.

Calculate the distance that this nerve impulse will travel in 5.0 minutes.

(2)

distance = ... m

(Total for Question 3 = 9 marks)

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3. Question 3 · part c · 2 marks · Working and answer

Working and answer

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4. Question 7 · part c · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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4. Question 7 · part c · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

(4)

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(iii) Describe the function of a motor neurone in the nervous system.

(2)

4. Question 7 · part c · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c only.

(c) Neurotransmitters allow different neurones to communicate with each other.

The distance between two neurones is 100 nm.

Neurotransmitters diffuse across the gap between neurones at a speed of 4.0×10^6 nm/s.

Calculate the time taken, in milliseconds, for a neurotransmitter to cross the gap between two neurones.

Give your answer in standard form.

(4)

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time taken = ... ms

(Total for Question 7 = 14 marks)

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4. Question 7 · part c · 4 marks · Working and answer

Working and answer

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5. Question 1 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

1 (a) The sentences are about the heart.

Complete the sentences by using words or numbers from the box.

(8)

80	100	120	atria	body	diastolic	elastic tissue
lungs	muscle	systolic	tendon	vena cava	ventricles	

The heart consists of four chambers.

The two upper chambers are called ..... .

The left upper chamber receives blood from the ..... whilst the right upper chamber receives blood from the body through the ..... .

The two lower chambers are called ..... .

The left lower chamber pumps blood to the ..... .

This left lower chamber has a very thick wall made of ..... .

When the chamber contracts, it generates a pressure known as the ..... pressure.

The average value for this pressure is about ..... mmHg.

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5. Question 1 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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(b) Cardiac output is the volume of blood pushed out of the heart in one minute.

Cardiac output is calculated using this formula.

cardiac output = heart rate × stroke volume

The stroke volume is the volume, in dm^3, of blood pushed out during one heartbeat. The heart rate is the number of beats per minute.

Calculate the stroke volume for a person with a heart rate of 68 beats per minute and a cardiac output of 6.1 dm^3.
Give your answer in cm^3.

(3)

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stroke volume = ... cm^3

(Total for Question 1 = 11 marks)

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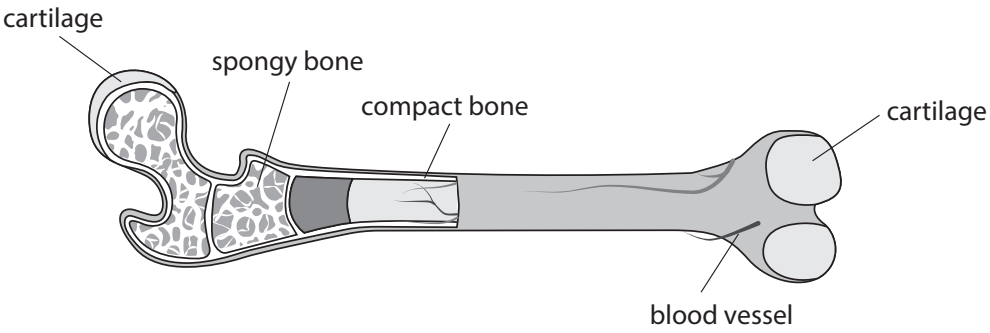
Working and answer

6. Question 1 · part b.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

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6. Question 1 · part b.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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(b) Bones are moved by muscles.

(i) Use the words in the box to complete the passage about bones and muscles.

(4)

cartilage	contracts	ligaments	lowers	pulls
pushes	raises	relaxes	tendons	

Muscles are attached to bones by ..... .

To raise the lower arm, the biceps ..... and the triceps ..... .

The biceps ..... on the radius, which raises the lower arm.

(ii) Energy is transferred by the biceps muscle when raising the lower arm.

The biceps muscle applies a force of 125 N to raise the lower arm a distance of 0.50 m.

Use this formula to calculate the energy transferred by the biceps muscle when raising the lower arm 0.50 m.

energy transferred (J) = force (N) × distance (m)

(2)

energy transferred = ... J

(Total for Question 1 = 12 marks)

6. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iv only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

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(ii) Describe the function of a motor neurone.

(2)

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7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

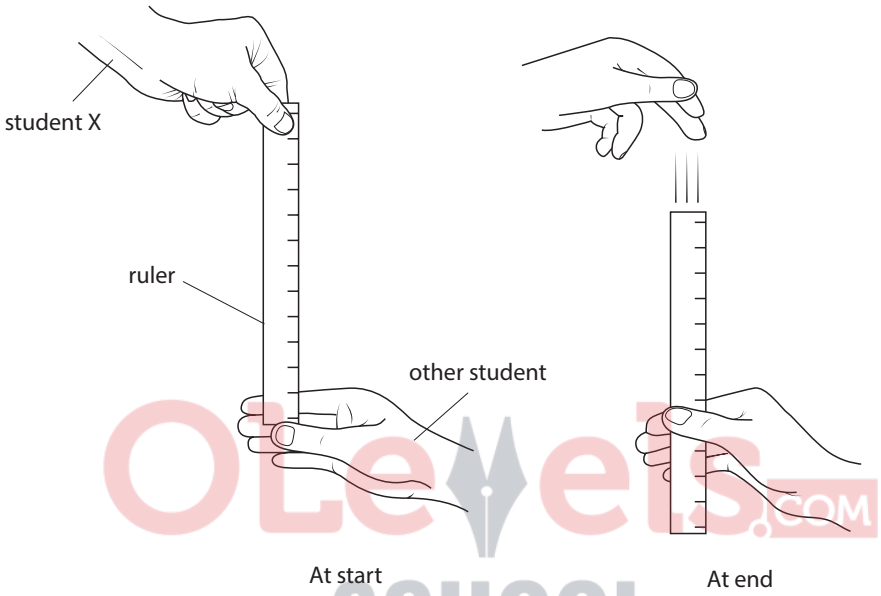
Further practice · Attempt part b.iv only.

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(b) Student X uses this equipment to investigate the reaction time of another student.



Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iv only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

(6)

(ii) Name the dependent variable in this investigation.

(1)

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

(1)

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7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iv only.

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(iv) The reading on the ruler for one of the students is 0.15 m.

Calculate the reaction time of this student.

Use this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

(2)

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reaction time = ... s

(Total for Question 3 = 16 marks)

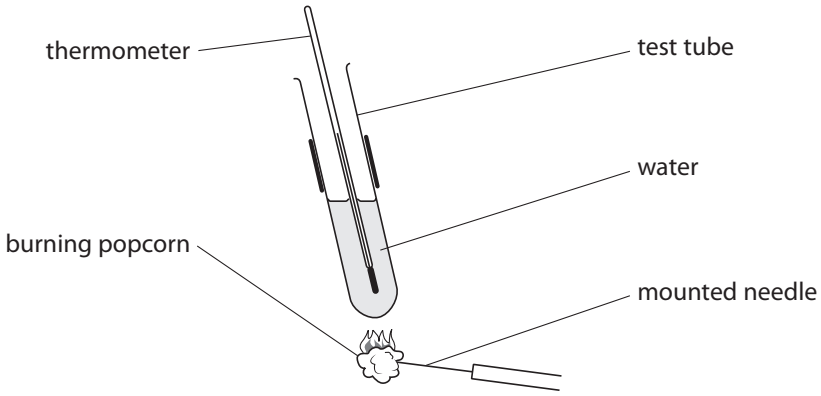
Working and answer

8. Question 2 · part b.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

2 A student uses this apparatus to investigate the energy contained in popcorn.



This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

(1)

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(ii) Explain one safety precaution that the student should take during the investigation.

(2)

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8. Question 2 · part b.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

(3)

energy released = ... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

(3)

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8. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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9. Question 7 · part b.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.i only.

7 The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 cm^3 of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 cm^3 of this wine.



250 cm^3 glass of Ruby Red wine

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(a) Calculate the number of units of alcohol in the glass of wine.

(2)

number of units = ...

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9. Question 7 · part b.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.i only.

(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

(2)

time = ... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

(2)

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9. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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